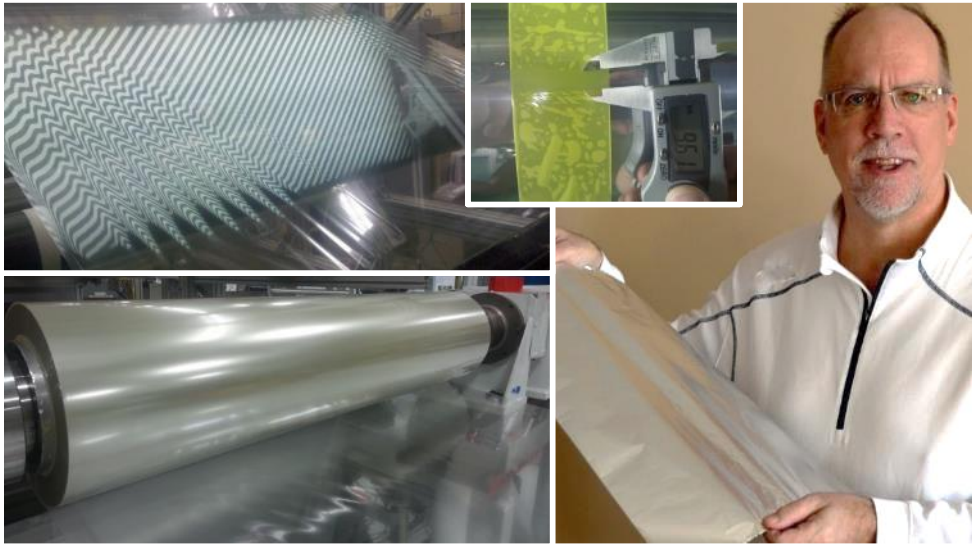

6.1 Winding Better Rolls

Winding Process, Roll Structure, Winder Design

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WH Fundamentals

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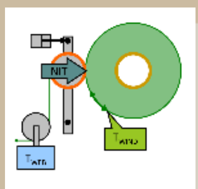
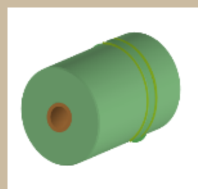
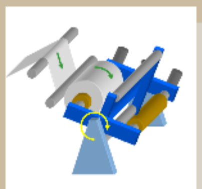
6: Winding: Process, Roll Quality, Equipment Design



Winding Process



Roll Quality



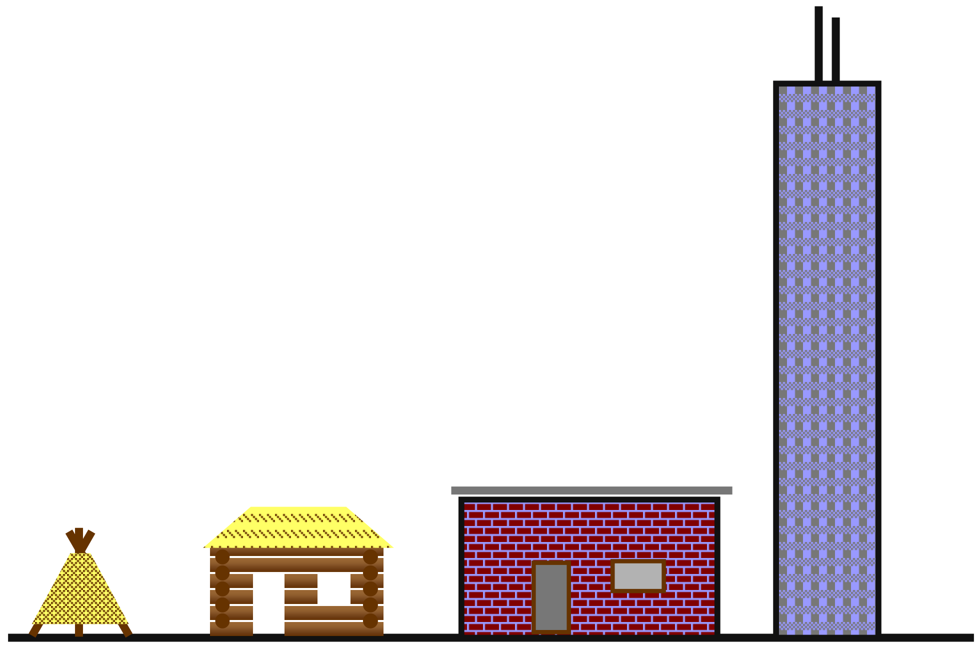
Winder Design

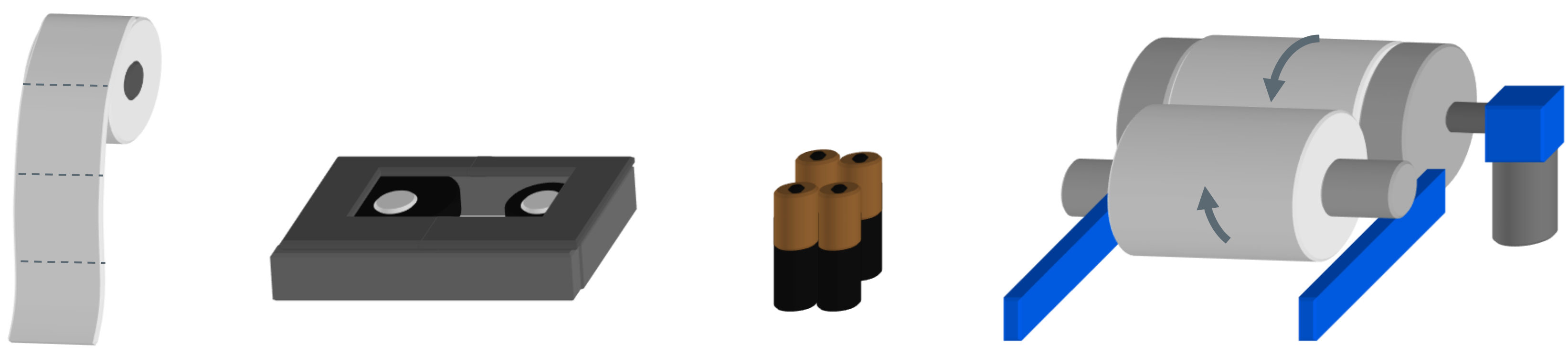
+ Winding, Rolls, Winders
  - ✓ What winder design is best for a product?
  - ✓ What determines the pressure inside a wound roll?
  - ✓ What causes roll and web defects?

Wound Roll Construction

Think of building wound rolls as construction.

- Buildings can be made from various materials.
- Rolls can be made from various materials.

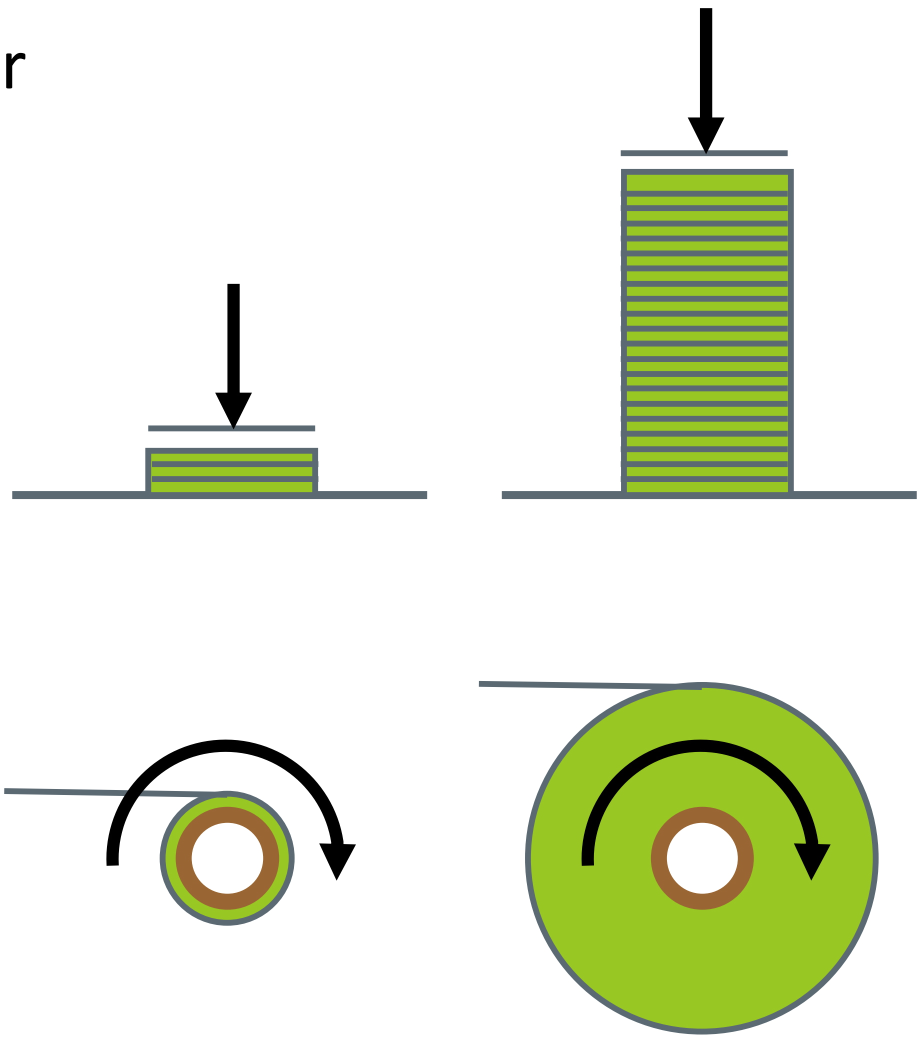




Wound Roll Structure

What makes one building or roll differ from another?

- Buildings differ in their foundation, footprint, height, and profile.
- Rolls differ in their core stiffness, core size, buildup ratio, and tensioning profile.



Wound Roll Structure

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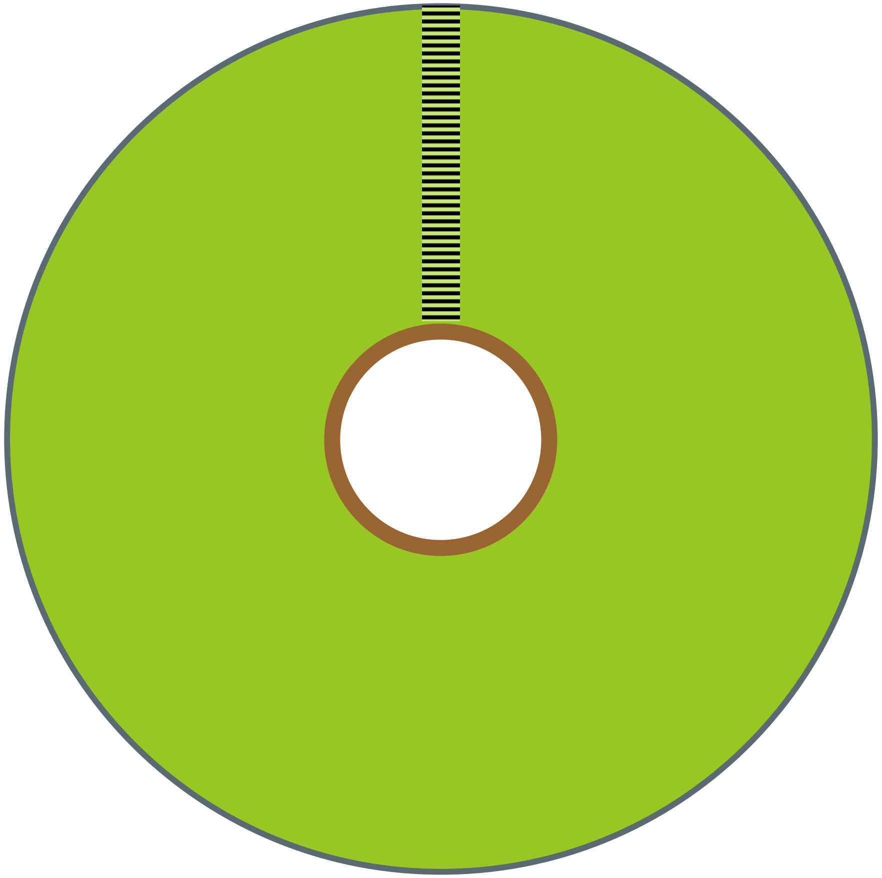
How many layers are in a roll?

$$n = \frac{(r_F - r_C)}{t}$$

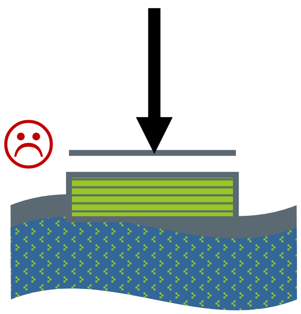
Example:

R_F = 0.5m (20-in)
R_C = 0.05m (2-in)
t = 50 micron (0.002-in)

n = (0.5-0.05)/0.00005
= 9000 layers

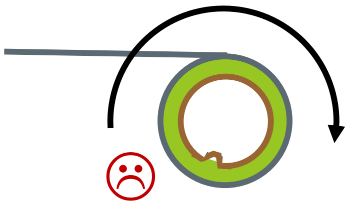


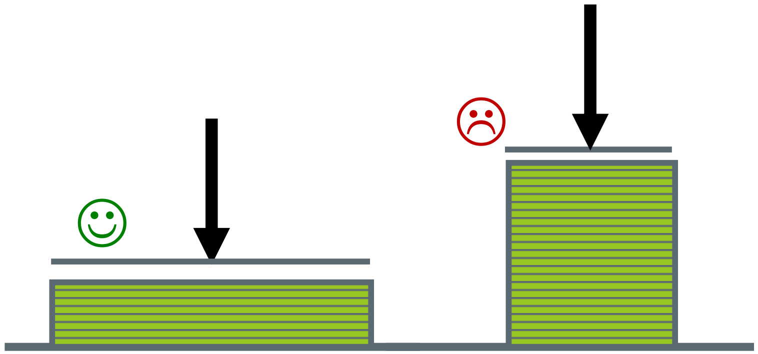
Wound Roll Structure



Don’t build on swampland.

Don’t wind on weak cores.

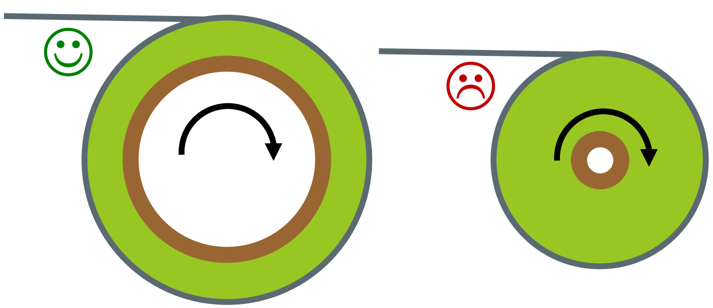


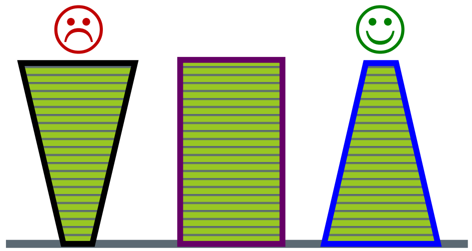


Low aspect ratios are more stable…

…whether height/width…

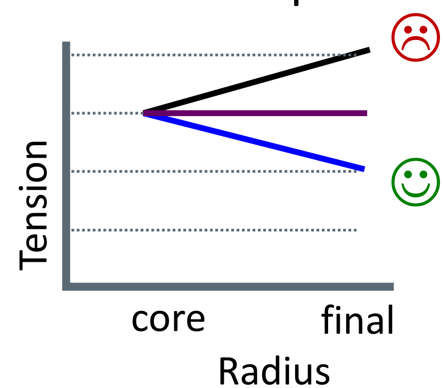
…or final/core diameter.





Some profiles are more stable than other.

Rolls are profiled using tension taper.



The Goals of Winding are Simple

1. Create a package with a structure to withstand winding, shipping, storage, and unwinding.
2. Efficiently end one roll and begin another with minimal waste.
3. Upon unwinding, the web quality from the roll should be equal or better than before winding.

Unfortunately, winding rolls can be quite complex, making these simple goals difficult.

Winding Better Rolls is Complicated

- Web (and core) material property are diverse,
- Important process variables are under-appreciated,
- Winding equipment has many alternate designs,
- Wound roll internal structure is difficult to detect,
- Roll and web quality can be challenging to quantify.

6.0 Winding

6.1. Winding Success

6.2. Material Properties (Web and Core)

6.3. Winding a Roll

- Starting a Roll: Roll Transfers
- Managing Initial Contact
- Tension and Torque
- Winder Tensioning
- Co-Axial Winding

6.4. Roll Structure and Web Quality

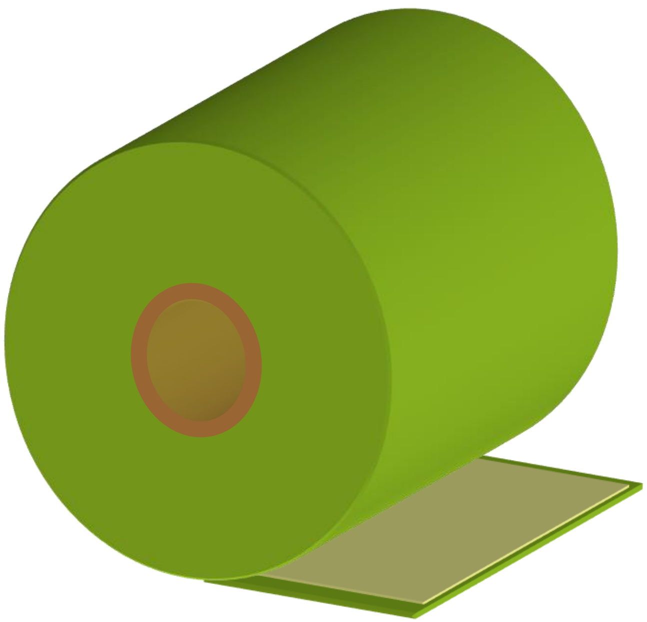
- Wound-In Stresses and Bagginess
- Lateral Errors
- Cross Machine Direction Variations (Gauge Bands and Bagginess)
- In-Roll Buckling

6.6. Wind-Ability, Winding Optimization

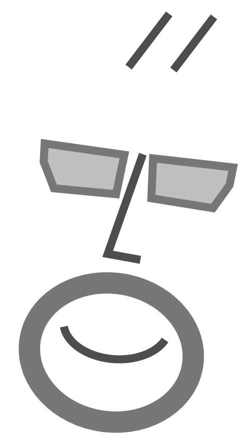
1. Measuring Winding Success

1.1. What is a Good Roll?

1.1. Winding-Related Waste



What is the purpose of winding?
What is a good roll?
What is roll quality?



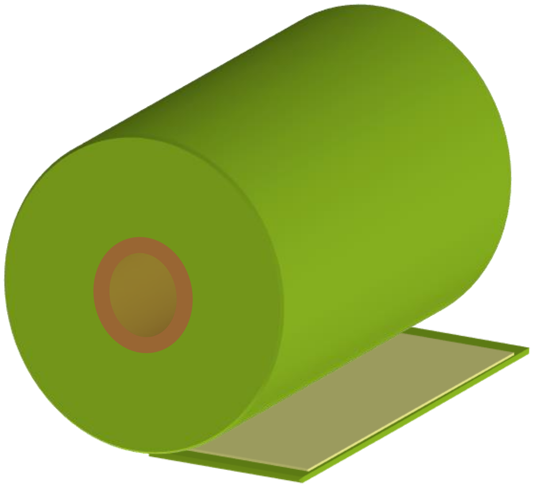
Characteristics of a Good Roll

Good Structure:

- Transport-able
- Store-able
- Unwind-able

Good Geometry:

- Correct Length
- Correct Width
- Correct Wind Direction
- Core Inner Diameter



Good Roll Appearance

Good Web Appearance and Function

Quality Out = Quality In

Quantity Out = Quantity In

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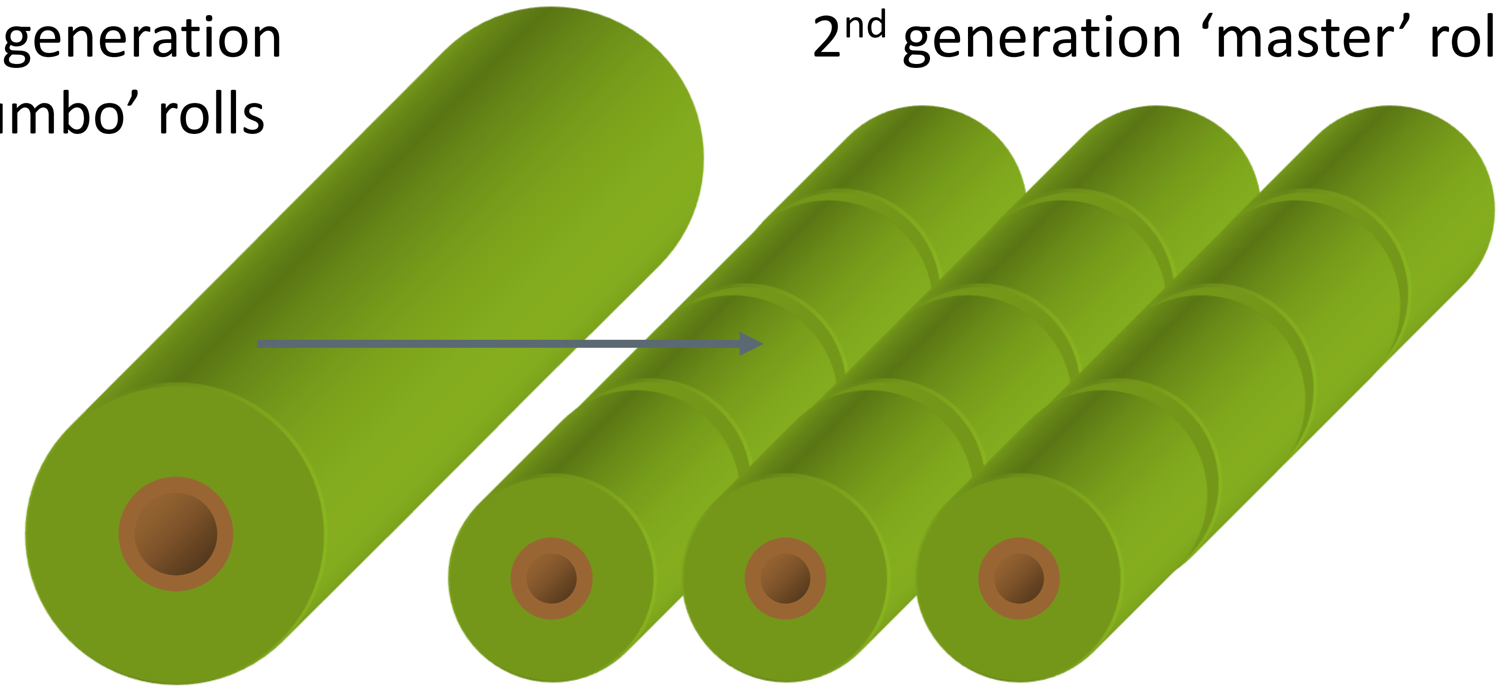
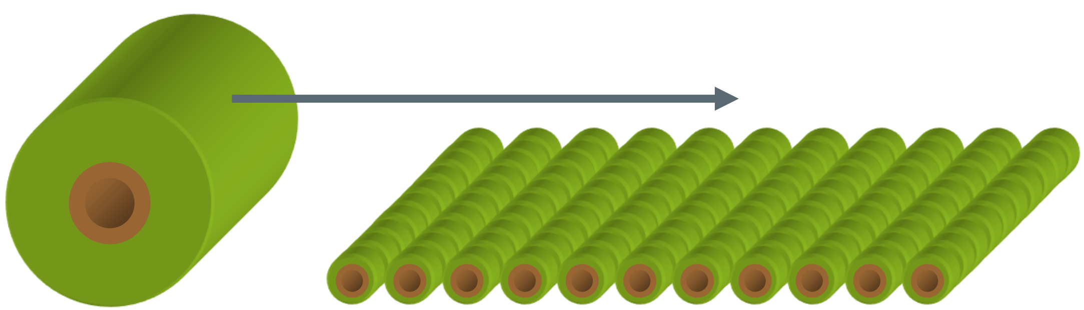
Roll Genealogy

1st generation 'jumbo' rolls

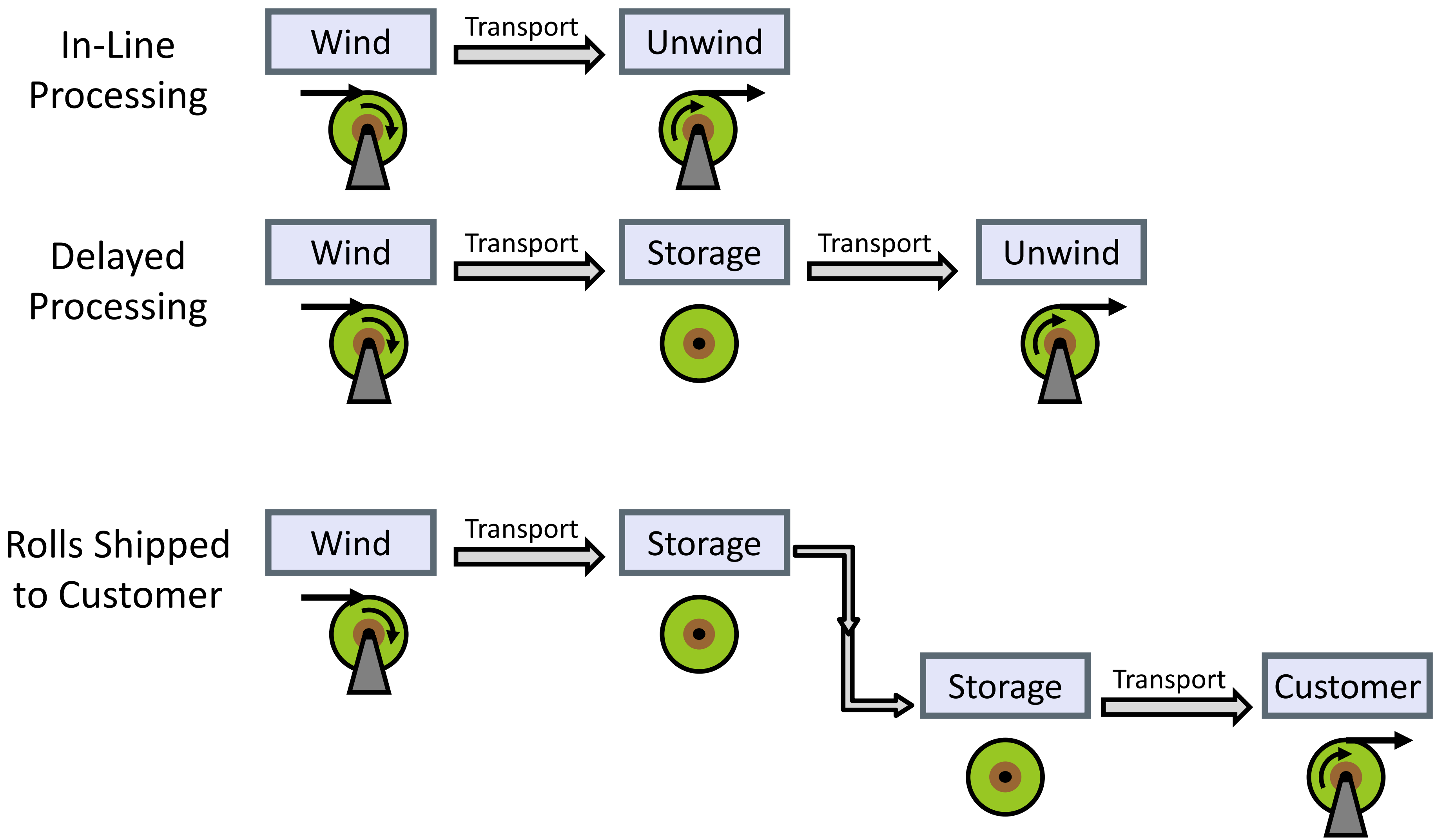
2nd generation 'master' rolls.





Master rolls are converted into 3rd generation 'slit' rolls.

Wind – Transport – Store – Transport - Unwind



The Winding Process

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Start Winding — Attached Web to Core

Winding — Build a Roll

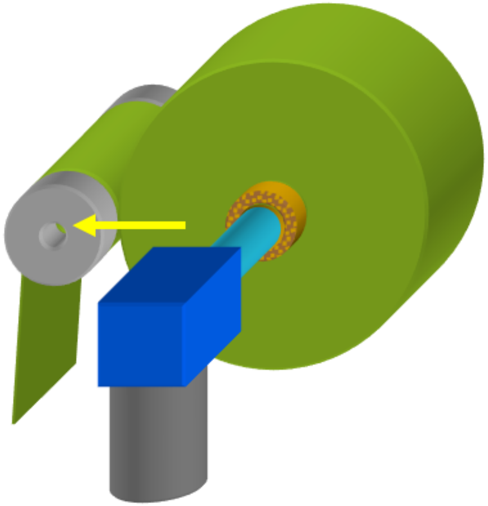
Finish Winding — Cut Web and Unload Roll

Control Pre-Winding Tension

Align Web to Core

Hold Core Position

Transmit Center Torque without Internal Roll Slippage



Control Air Lubrication and Entrainment

No Wrinkles

Winding-Related Waste Has Many Forms

- Roll shape defects include any non-cylindrical aspect, including undesired variation in lateral position or width error, roundness, or diameter profile.
- Web defects include lay-flat imperfections (curl, bagginess, skew) and surface imperfections (scratches, impressions, dimples.
- Roll pressure-related defects including, cinching / slippage, excessive peel force or blocking, thickness loss, and core damage.

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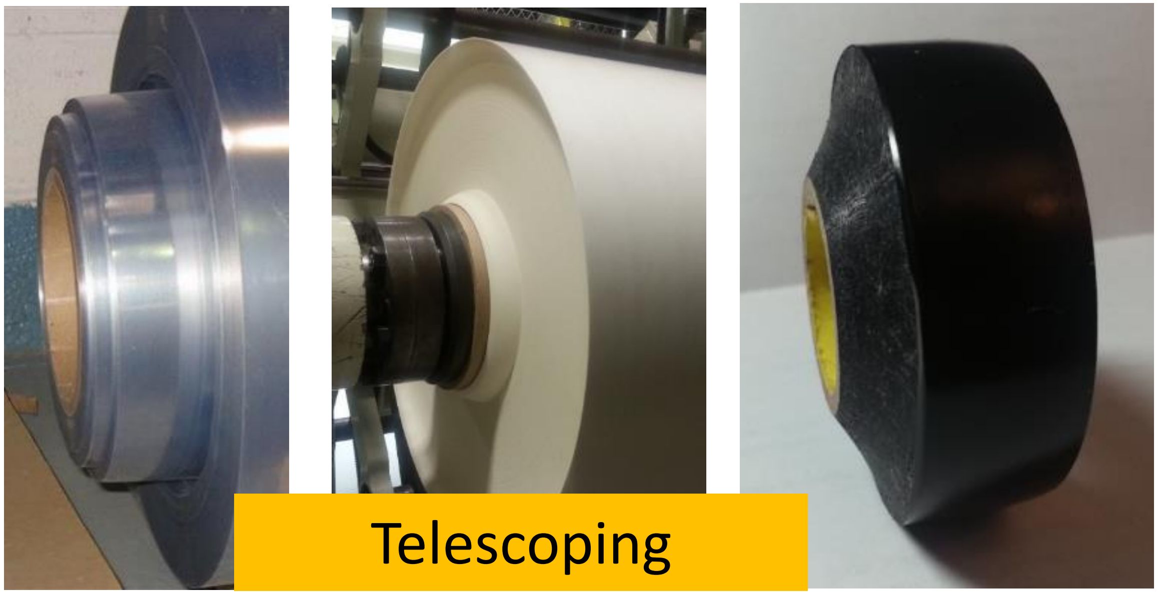
Wound Roll Defects



Telescoping

Crushed Core

Gauge Bands

Starring

CD Buckles

MD Buckles

Winding Value Add or Value Subtract?

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Winding creates waste!

WASTE!

1. Roll defects
2. Web defects
3. Core scrap
4. Outer wraps
5. Trim waste

If possible, integrate processes to avoid wind-unwind waste.

6.0 Winding

6.2 Material Properties (Web and Core)

Mechanical Properties

- Tensile-Elongation
- Stack Compressibility
- Poisson's Ratio
- Visco-Elasticity
- Temperature
- Moisture

Web Geometry

- Thickness
- Thickness Profile
- Width
- Laminates

Surface Properties

- Coefficient of Friction
- Roughness, Porosity

Roll Geometry

- Core Size
- Length
- Final Diameter

Cores

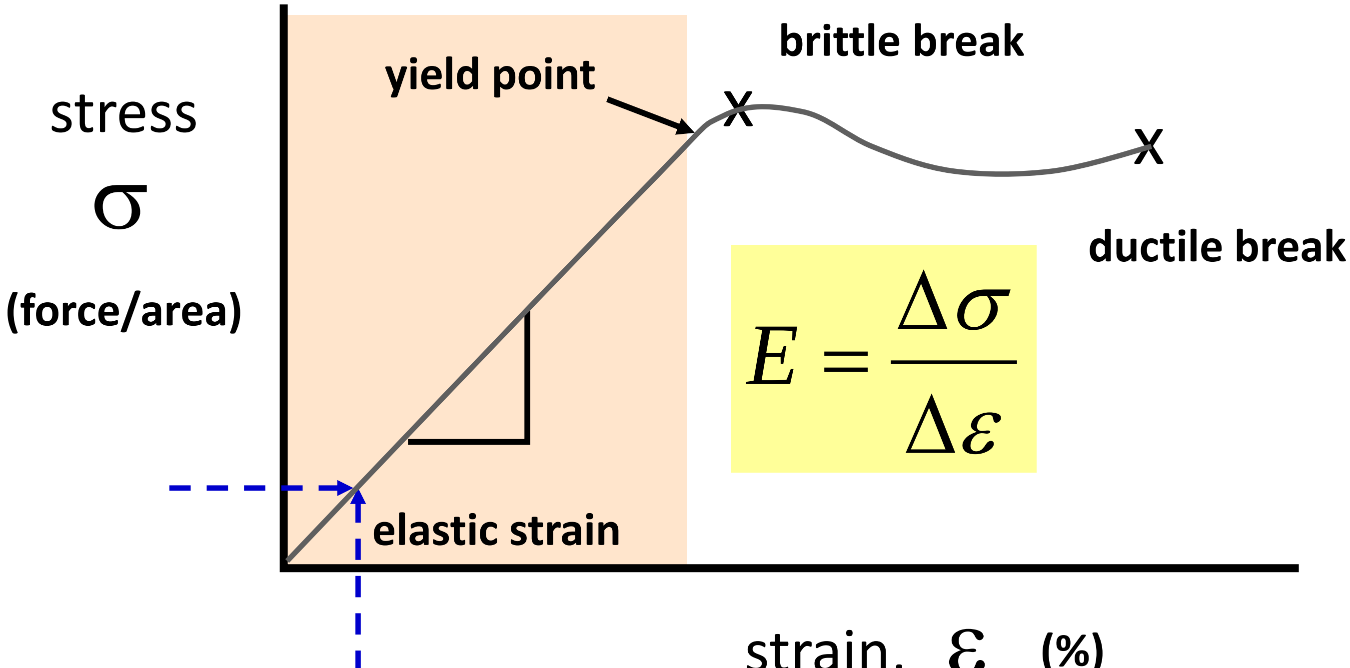
- Core Materials
- Core Compression
- Core Collapse

Materials of Winding

- Winding is highly dependent on material properties.
- Web thickness, Young’s modulus, stack compressibility, and coefficient of friction are the strongest material property variables.
- Cores geometry and materials mostly affect near core defects, such as impressions, blocking, and cinching-induced telescoping.

Tensile-Elongation or Stress-Strain

Typical web handling tensions are 10 to 20% of a web's yield or break stress.



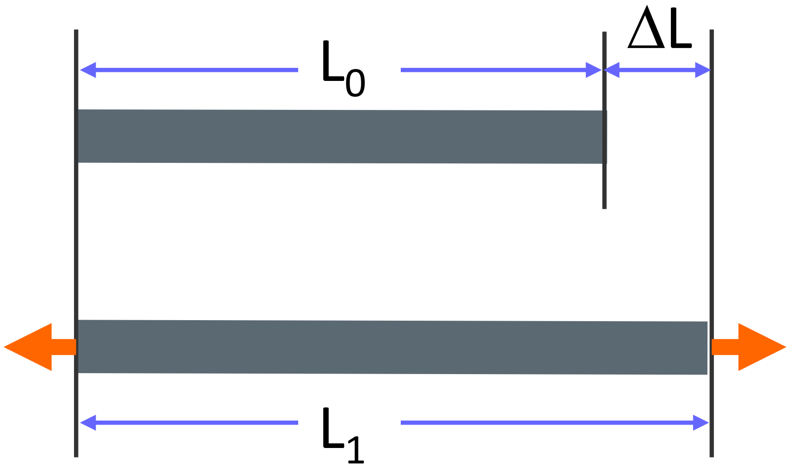
Tension set point should be 10-20% of yield or break point.

Strain Defined

Strain, ε, is the ratio of the change in a dimension over the untensioned dimension. For tensioning, strain is the change in length over the untensioned length.

L_0 = Dimension of zero tension web

L_1 = Dimension of tensioned web



$$\varepsilon = \frac{L_1 - L_0}{L_0} = \frac{\Delta L}{L_0}$$

The Key to Winding is Strain

- Most views of web handling and winding emphasize the **T**ension, **N**ip load, and **T**orque (The TNTs of Winding).
- These forces and torques should be viewed in terms of stresses and strains in the web.
- Without strain, there is no tension and no pressure holding a roll together (except when layers are bonded by adhesion).

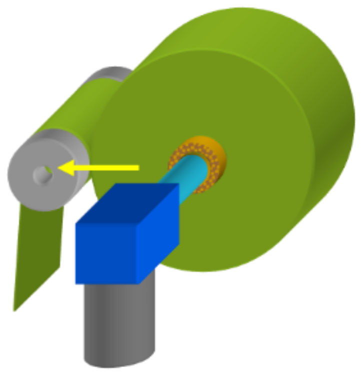
The Key to Winding is Strain

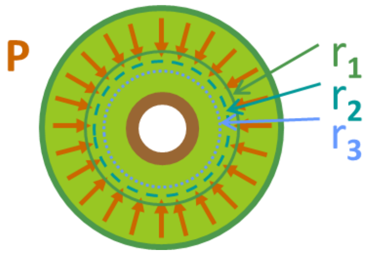
- Winder settings create the initial strain of the web as it enters the roll.
- Strains are usually less than 1%. Changes within a roll that are equal or greater than winding strains are significant.
- Compression and other dimensional changes within a roll can reduce the strains to near zero.
- Larger dimensional changes can lead to negative or compressive strain and stresses, causing buckling within the roll.

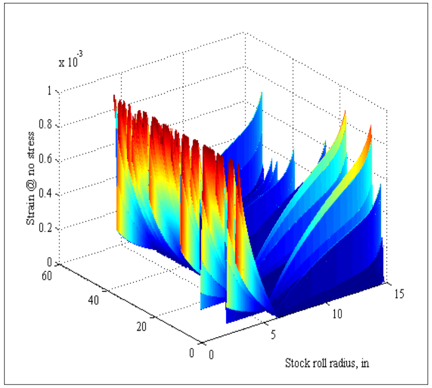
The Strains of Winding

Roll stresses are determined by:

- Web strain at first contact with the winding roll as function of diameter, tension, nip, torque, and speed.
- Strain losses from dimensional changes of web, coating, entrained air, and core.
- Strain variations from thickness profile building to internal cross-roll diameter.



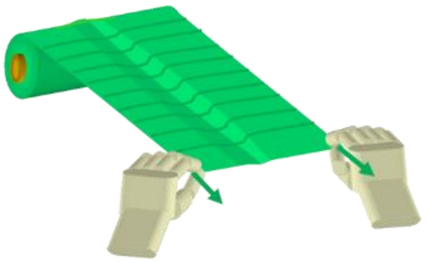




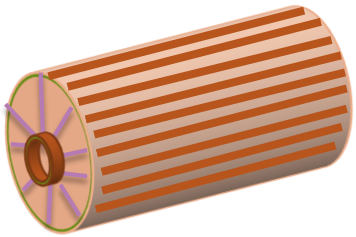
The Strains of Winding

Roll defects are determined by:

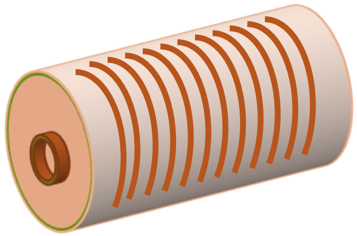
- Residual strain variations create bagginess and camber.



- Cross machine direction buckles form when in-roll machine direction (MD) strains are compressive.

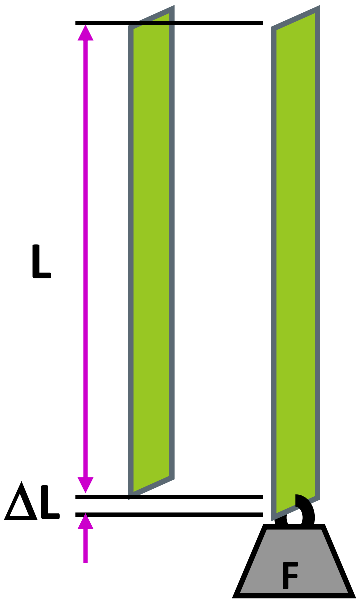


- Machine direction buckles when in-roll radial compression and machine direction strain loss releases Poisson's contraction, creating cross machine (CD) compressive stresses.

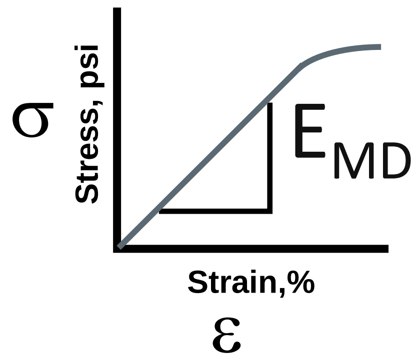


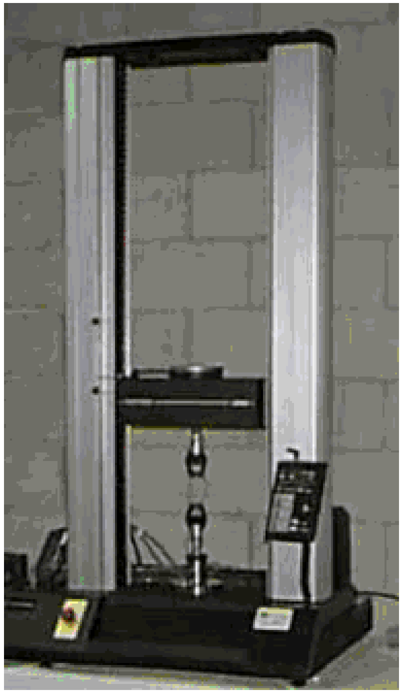
Modulus of Elasticity

Young's modulus of elasticity is found from tension-elongation tests
(a relatively simple measurement).



Young's Modulus





$$\sigma = \frac{T}{t}$$

$$\varepsilon = \frac{\Delta L}{L}$$

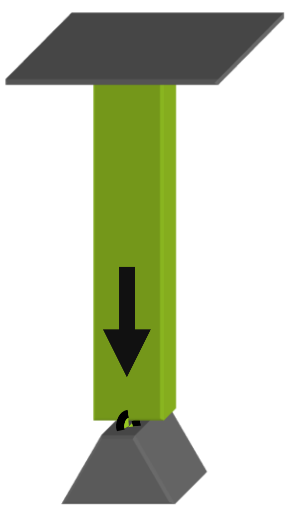
$$E_{MD} = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{TL}{(\Delta L)t}$$

Roll Formation Properties

Pressure buildup within a roll (a.k.a. ***the tourniquet effect***) is determined by material properties and winding conditions.

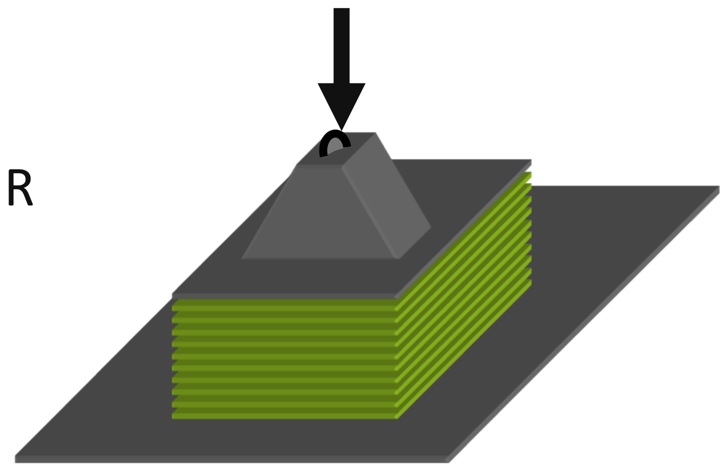
A key property of a winding roll is the ratio of **machine direction** and **radial** modulii

E_{MD}



MD Modulus

E_R



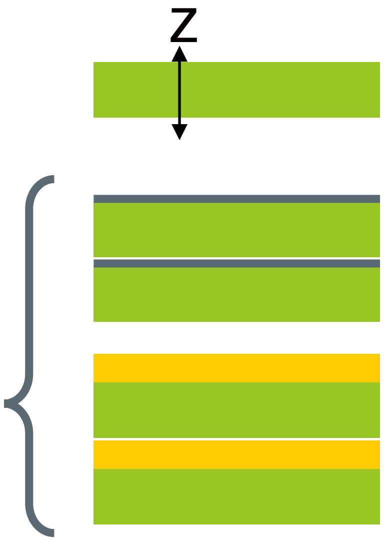
Radial Modulus

Radial Modulus

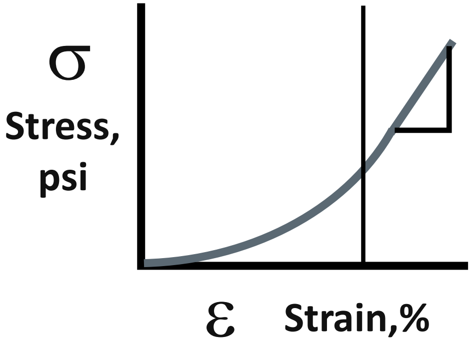
Why does radial modulus differ from MD modulus?

What's in the stack?

Most webs and rolls are anisotropic, differing properties in X, Y, and Z directions.



Many rolls have laminate, coating, or interleave layers.



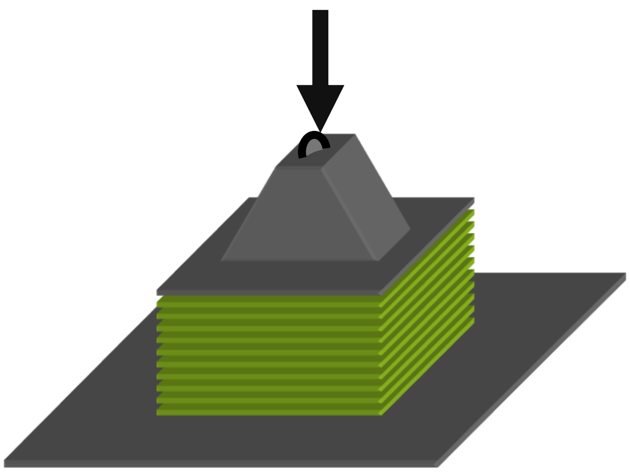
Surface asperities allow air between layers and act as small springs when loaded.

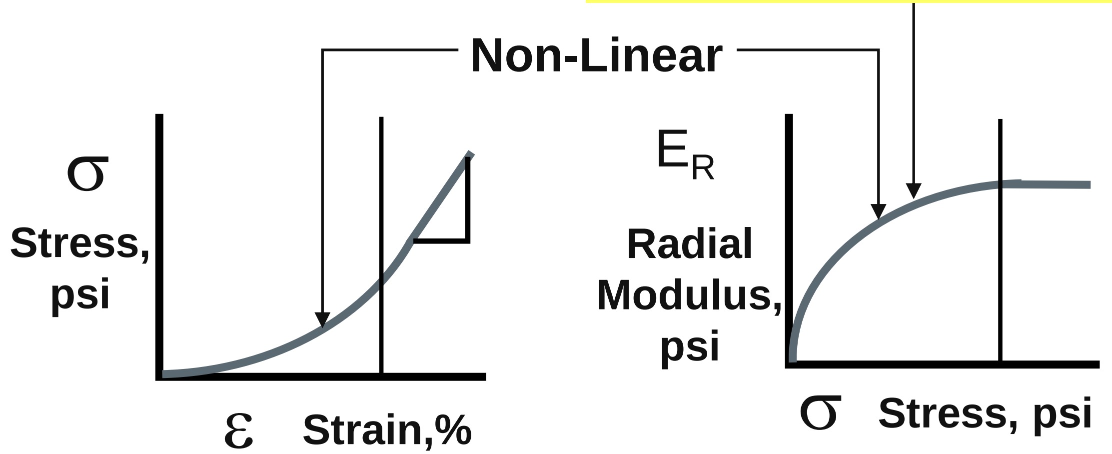
Radial Modulus

Radial modulus (a.k.a. stack modulus) is found from stack compression tests
(a more difficult, high load measurement).

$$E_r(P) = K_2(P + K_1)$$

$$E_r(P) = k_1 + k_2 P + k_3 P^2$$





Radial Modulus

(a.k.a. Stack Modulus)

$$\sigma = \frac{F}{A}$$

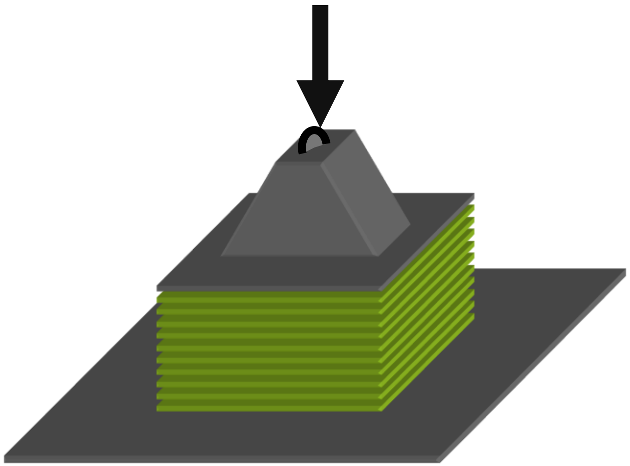
$$\varepsilon = \frac{\delta h}{h}$$

$$E_R = \frac{\delta\sigma}{\delta\varepsilon} = \frac{Fh}{(\delta h)A}$$

Radial Modulus

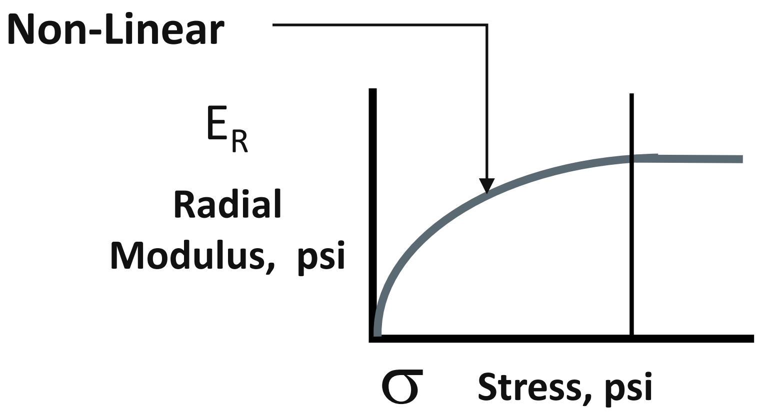
Radial modulus is a function of the stack pressure.

- Rolls wound at higher tension will have more tourniquet effect.
- This makes rolls non-linear – doubling tension will more than double internal roll pressures.



Radial Modulus

(a.k.a. Stack Modulus)

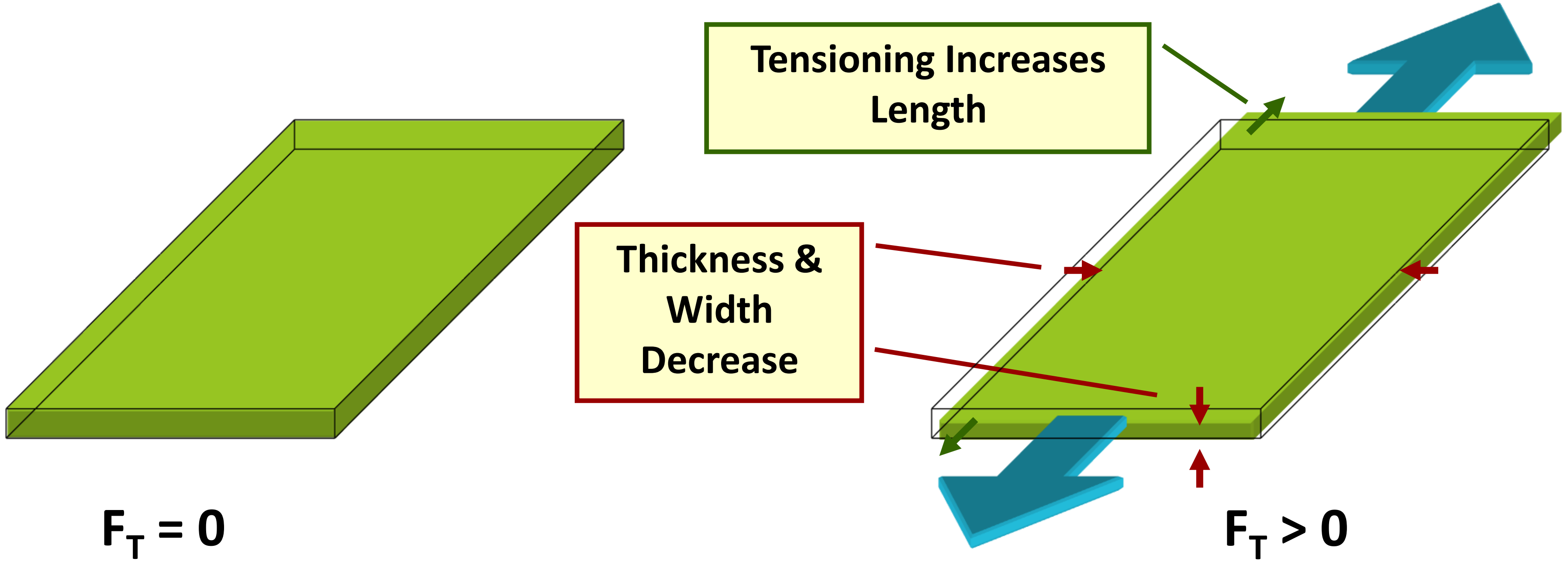


$$E_R = \frac{\delta\sigma}{\delta\varepsilon} = \frac{Fh}{(\delta h)A}$$

Strain = Dimensional Change

For solid materials, tensile and compressive stresses do not significantly change density.

Increases in length (MD strain) is offset by decreases in the width and thickness.

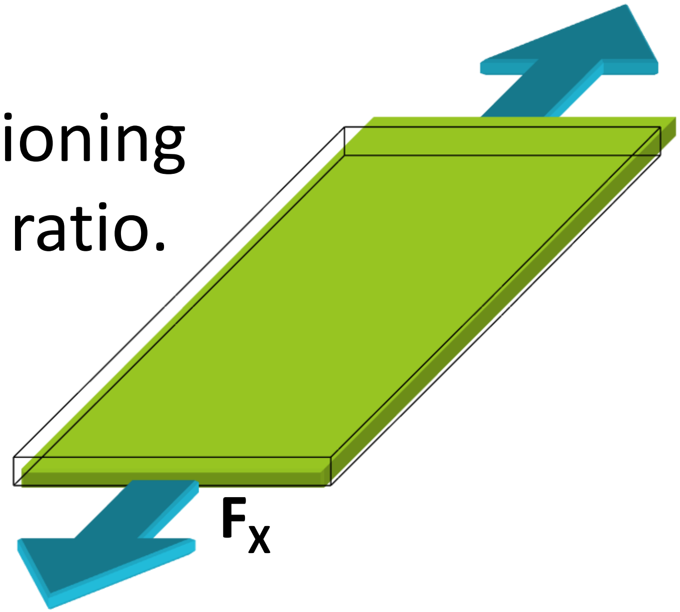


Poisson's Ratio and Width Contraction

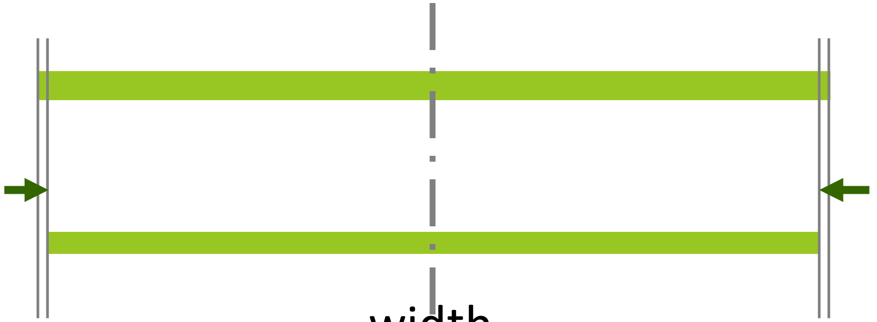
Tension will increase web length and decrease width and thickness.

The ratio of width contraction to tensioning elongation is defined as the Poisson's ratio.

$$\varepsilon_Y = -\nu\varepsilon_X = -\nu\left(\frac{F_X}{twE}\right)$$

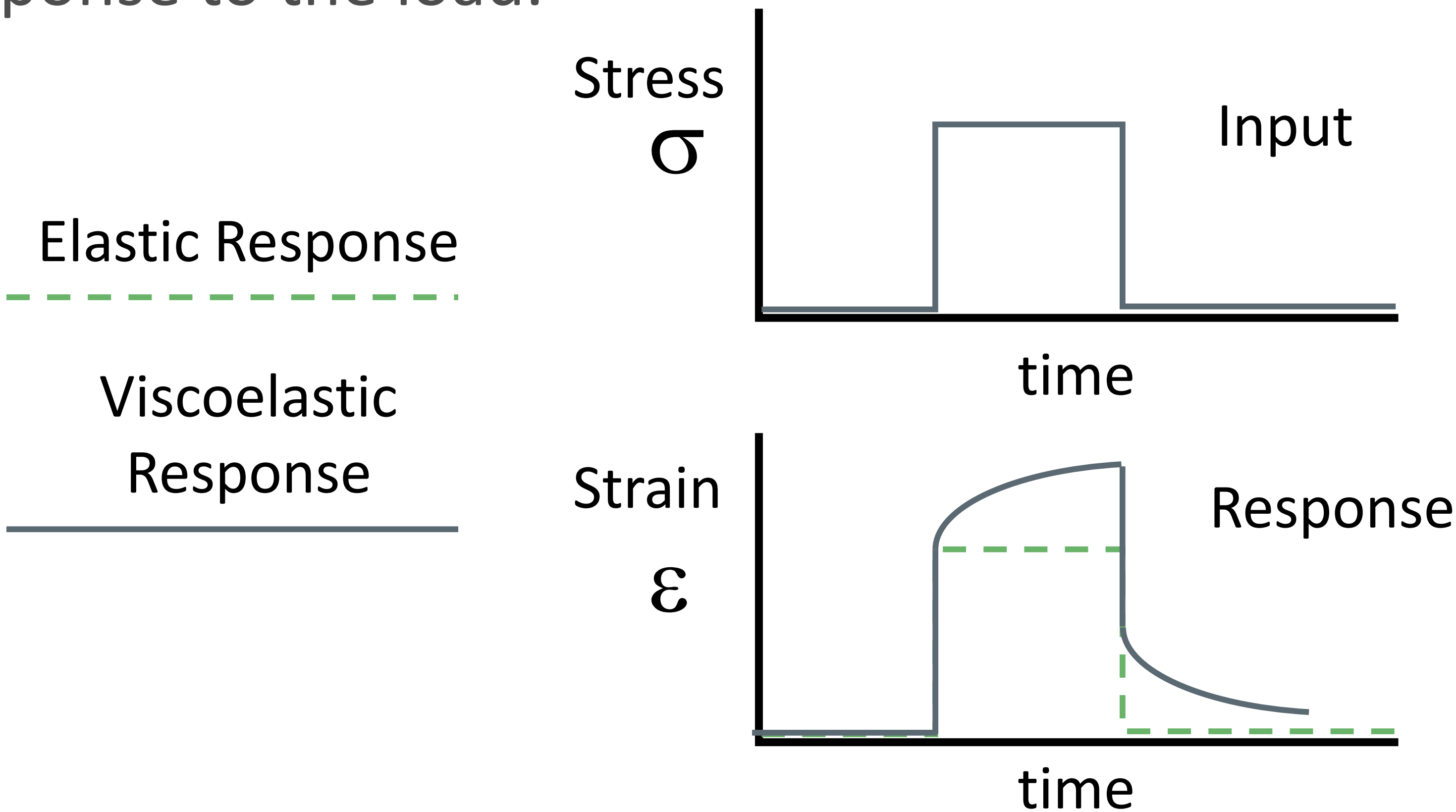


$$\delta w_T = w_0\left(-\nu\varepsilon_X\right)$$



Viscoelastic Behavior - Creep

The creep test applies a constant load to material and measure the instant (elastic) and time-dependent (viscous) response to the load.



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Moisture and Temperature

Webs change thickness, width, and length with changes in moisture content and temperature.

Dimensional changes of a magnitude similar or larger than web strain will significantly change roll structure.

Wind, store, and unwind roll at stable humidity and temperature conditions, optimally 40-60% relative humidity and 65-75^{o}F.

Courtesy of Machine Design magazine

Buckling defects from hygroscopic expansion of paper rolls.

Courtesy of RTape

Moisture and Temperature of Winding

Winding Guideline:

- Winding, storage, and unwinding should be controlled to the same temperature and humidity conditions.

- Many material change their properties as a function of moisture and temperature
  - Hydrophilic materials, such as paper and BOPA (Nylon)
  - Thermoplastic materials (nearly all polymer films)
- Hygroscopic and thermal expansion (and contraction) are the most important material property dynamics that can greatly change roll tightness or create in-roll buckling.
- Thermoplastic web may creep easier at elevated temperatures, creating more profile-induce bagginess if wound or stored at elevated temperatures.

Moisture and Temperature of Winding

- Some processes require heating rolls over time for curing or heat relaxation.
- Wound rolls layers are usually good thermal insulators and moisture barriers, requiring long time periods (over 24 hrs) to react to new environmental conditions.
- Increased moisture of hydrophilic rolls is a common cause of outside layer buckling and web edge bagginess.
- Since moisture changes dramatically by geography and season, many hydrophilic rolls are wrapped with moisture barrier packaging during transport and storage.

6.2 Material Properties (Web and Core)

Mechanical Properties
- Tensile-Elongation
- Stack Compressibility
- Poisson's Ratio
- Visco-Elasticity
- Temperature
- Moisture

Web Geometry
- Thickness
- Thickness Profile
- Width
- Laminates

Surface Properties
- Coefficient of Friction
- Roughness, Porosity

Roll Geometry
- Core Size
- Length
- Final Diameter

Cores
- Core Materials
- Core Compression
- Core Collapse

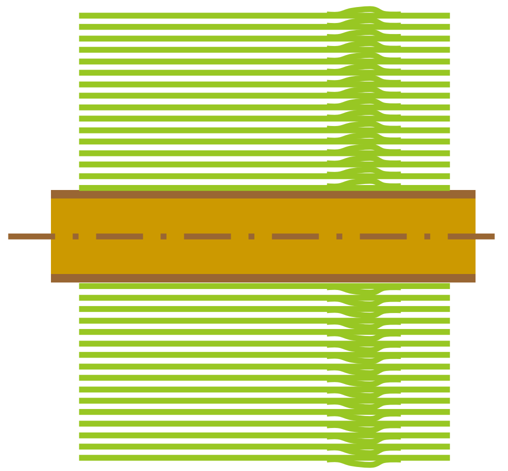
Thickness, Thickness Profile

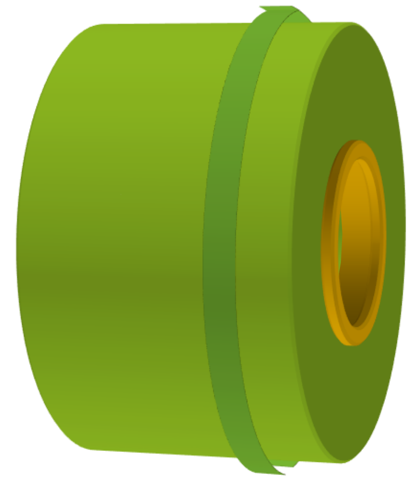
What is a web's thickness profile?

- Thickness mapped vs. width.
- Does the web have thick edges? Lanes?
- Is the thickness profile consistent over time?
- Does the product have intentional thickness profile (including coatings or cutouts)?
- How is thickness profile measured? (Besides in winding defects)

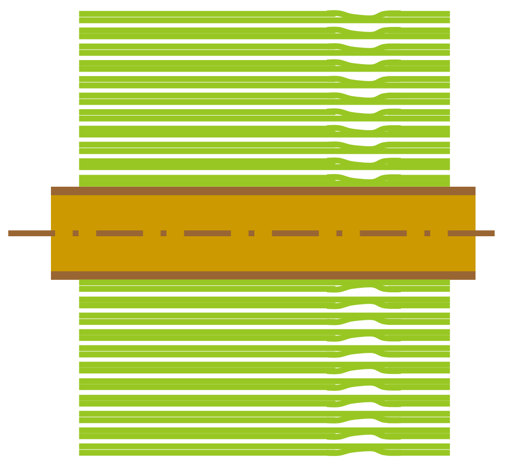
Softbands and Hardbands

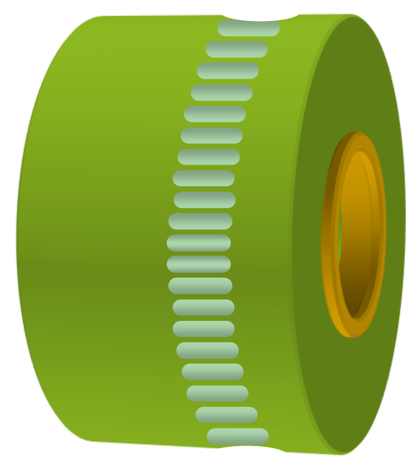
Hardbands are magnifications of thick web or coating thickness variations.





Softbands are magnifications of thin web or coating thickness variations.





Winding and Web Width

Wide Rolls

- More likely to have significant thickness profile variations.
- More core and nip roller deflection.

Narrow Rolls

- Widths below 100mm may see side air leakage effects in winding.
- Rolls of high aspect ratio (buildup / width) may collapse under in-roll pressures.

6.2 Material Properties (Web and Core)

Mechanical Properties
- Tensile-Elongation
- Stack Compressibility
- Poisson's Ratio
- Visco-Elasticity
- Temperature
- Moisture

Web Geometry
- Thickness
- Thickness Profile
- Width
- Laminates

Surface Properties
- Coefficient of Friction
- Roughness, Porosity

Roll Geometry
- Core Size
- Length
- Final Diameter

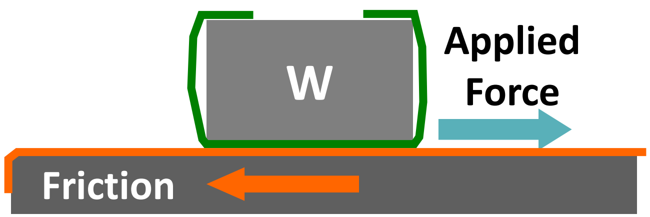
Cores
- Core Materials
- Core Compression
- Core Collapse

Friction Coefficient

The friction coefficient, μ, is the ratio of frictional force relative to normal load. Two common friction coefficient measurement methods are:

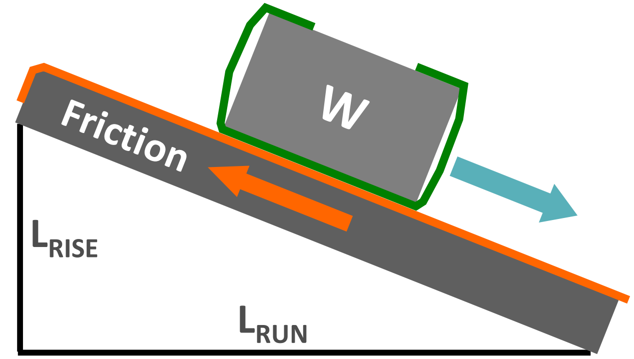
Sliding Block Test

$$\mu = \frac{F}{W}$$



Incline Plane Slide Test

$$\mu = \frac{L_{RISE}}{L_{RUN}}$$



Winding and Material Properties: Friction, Porosity

- Nip-induced tension (NIT) is directly proportional to product side A to B coefficient of friction.
  - Lower friction webs will have less added tightness from nipping.
  - High friction webs will have more tightness from nipping.
- For non-porous webs, air entrainment is a function of radius, speed, and tension, but porous webs can absorb air within their structure (paper, nonwoven, fabrics)

Winding and Material Properties: Roughness

- The volume of air entrained at winding is not a function of roughness, but roughness (side A and B) determines what amount of air is required to lubricate the roll’s outer layers.
- Roughness and thickness variations will also determine how much air can remain within a roll and how much will escape over time.
- Many polymer films have slip additives to modify the film-film coefficient of friction.

6.2 Material Properties (Web and Core)

Mechanical Properties

- Tensile-Elongation
- Stack Compressibility
- Poisson's Ratio
- Visco-Elasticity
- Temperature
- Moisture

Web Geometry

- Thickness
- Thickness Profile
- Width
- Laminates

Surface Properties

- Coefficient of Friction
- Roughness, Porosity

Roll Geometry

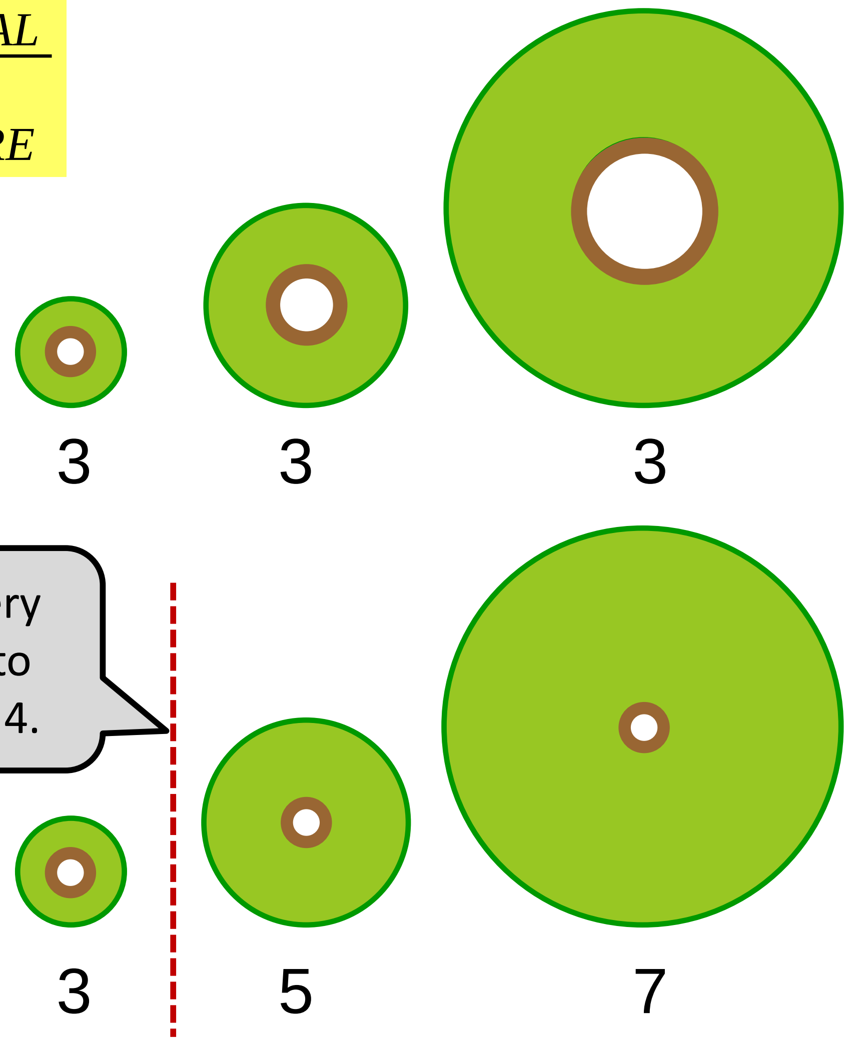
- Core Size
- Length
- Final Diameter

Cores

- Core Materials
- Core Compression
- Core Collapse

Roll Buildup Ratio

$$Buildup = \frac{r_{FINAL}}{r_{CORE}}$$



Buildup = 3 3 3

If core size increases proportional to diameter, buildup ratio doesn't change.

Low modulus or slippery products are difficult to wind if buildup ratio > 4.

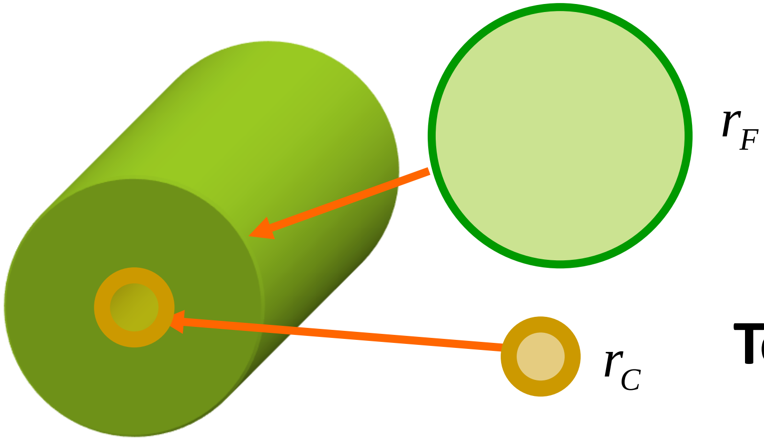
Buildup = 3 5 7

Winding longer lengths on the same core size increases buildup ratio.

Calculate Roll Size / Length

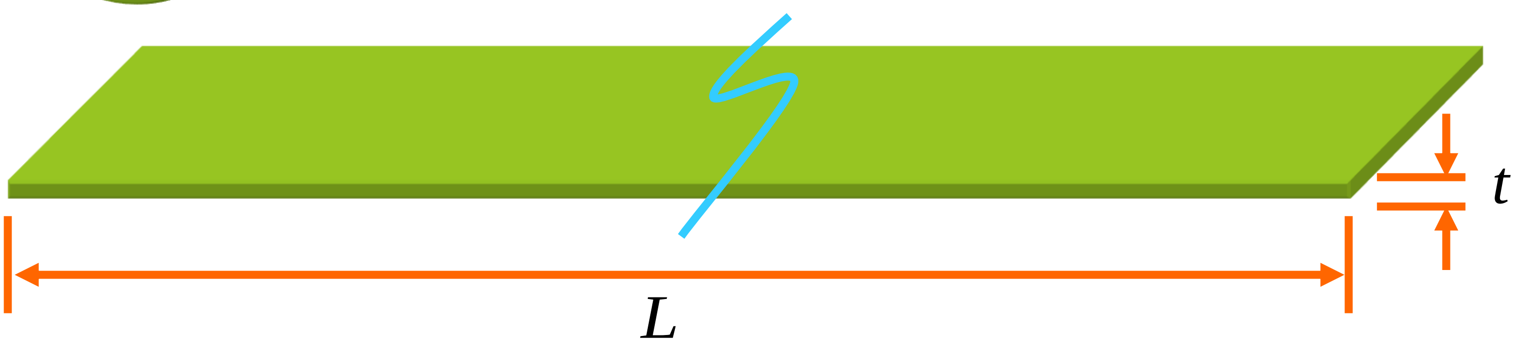
Set the side view cross-section areas equal to each other.

$$Lt = \pi\left(r_F^2 - r_C^2\right)$$



To find roll length: $$L = \frac{\pi\left(r_F^2 - r_C^2\right)}{t}$$

To find final roll radius*: $$r_F = \sqrt{\frac{Lt}{\pi} + r_C^2}$$



* This estimate may be too high or too low due to air entrainment or web thickness compression.

6.2 Material Properties (Web and Core)

Mechanical Properties

- Tensile-Elongation
- Stack Compressibility
- Poisson's Ratio
- Visco-Elasticity
- Temperature
- Moisture

Web Geometry

- Thickness
- Thickness Profile
- Width
- Laminates

Surface Properties

- Coefficient of Friction
- Roughness, Porosity

Roll Geometry

- Core Size
- Length
- Final Diameter

Cores

- Core Materials
- Core Compression
- Core Collapse

Cores

Core Variables

- Size
  - Inner diameter
  - Outer diameter
  - (Wall thickness)
- Width
  - = or > web width
- Materials
  - Paper spiral
  - Plastic
  - Metal

Core Requirements

- Support for winding pressure without buckling and minimal compression.
- Transmit center torque to core and web.
- Support roll weight and nip loads (with shaft or chucks).

Core Stiffness

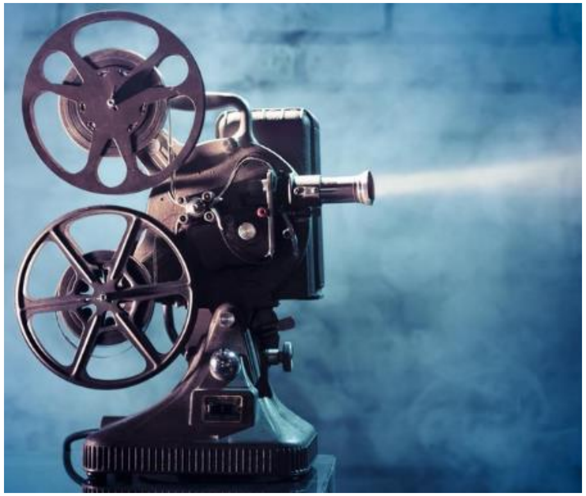
Core material and size determine radial and longitudinal stiffness.

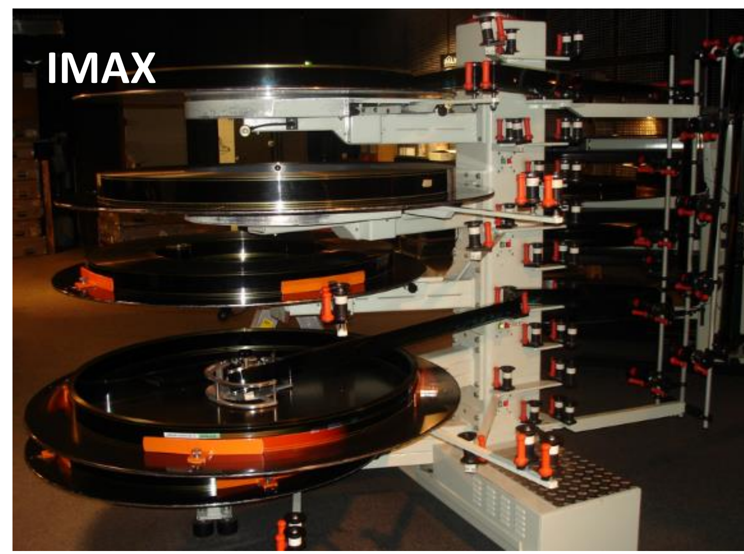


$$E_{CORE} = \left(\frac{r_o^2 - r_i^2}{r_o^2 + r_i^2}\right) E_M$$

E_{CORE} = Effective core modulus
E_M = Core material modulus
r_o = Core outer radius
r_i = Core inner radius


Metal and Plastic Reels (Flanged Cores)





Rubber Covered Cores

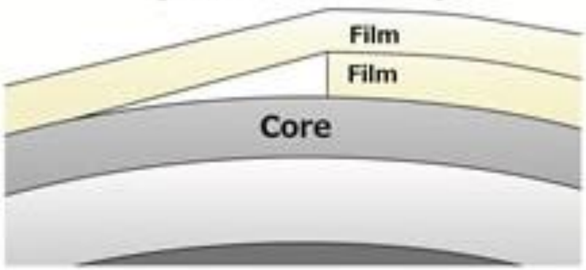
TJWALKER+ ASSOCIATES INC

Rubberized winding roller "e-CORE"

Film 38μm, Double-sided tape 35μm

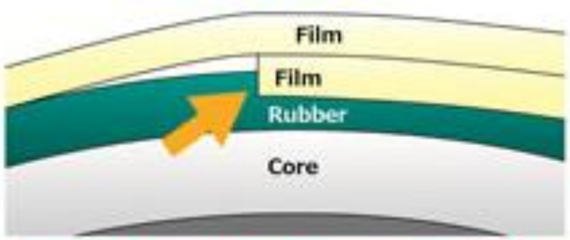
Winding roller from other company	200m waste
Rubberized winding roller from other company	70m waste
Rubberized winding roller from Katsura	35m waste

Conventional Winding Core (Hard surface)



Film overlapping mark transfers to the upper layers of the film.

e-CORE

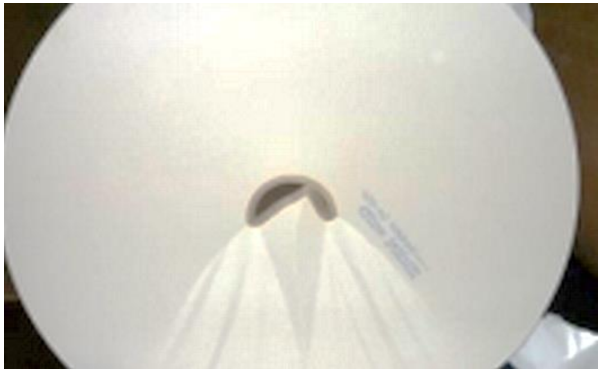


Rubber absorbs the overlapping film gaps and reduces film reject rate.

KATSURA **KATSURA ROLLER MFG. CO., LTD**

Core Mechanics (compression vs collapse)

A core will *collapse* in buckling failure if winding exceeds the critical buckling pressure.



$$P_{crit} = \frac{E'}{4}\left(\frac{t}{r}\right)^3$$

$$E' = \frac{E}{(1-\upsilon^2)}$$

P_{crit} = Critical buckling pressure
E' = Effective elastic modulus
E_M = Core material modulus
r = Core inner radius
t = Core wall thickness
ν = Poisson's ratio radial to MD direction

In-Process Cores

Often large diameter, reusable, steel or aluminum cores are used for rolls wound and unwound within the same plant or company.

Bigger Cores!

TJWALKER + ASSOCIATES INC

Big Core Advantages

- Smaller roll buildup = lower in-roll pressures.
- Smaller stress and strain variations from thickness variations = less bagginess.
- High torque capacity in near core layers = less cinching and telescoping.
- Lower saddle pressure and sag when core supported.
- Less core set curl.
- Lower deflections under equal load, less need for through-core shaft.

Big Core Disadvantages

- Increase cost per core, more expensive core chucks and shafts.
- Heavier, more difficult to handle cores and core shafts.
- Less length per final roll diameter.
- More core waste per revolution unused.
- The customer request small cores.

Coreless Winding

Some winding process have no core.

The product winds on a collapsible mandrel that is removed after winding is completed.

Cores

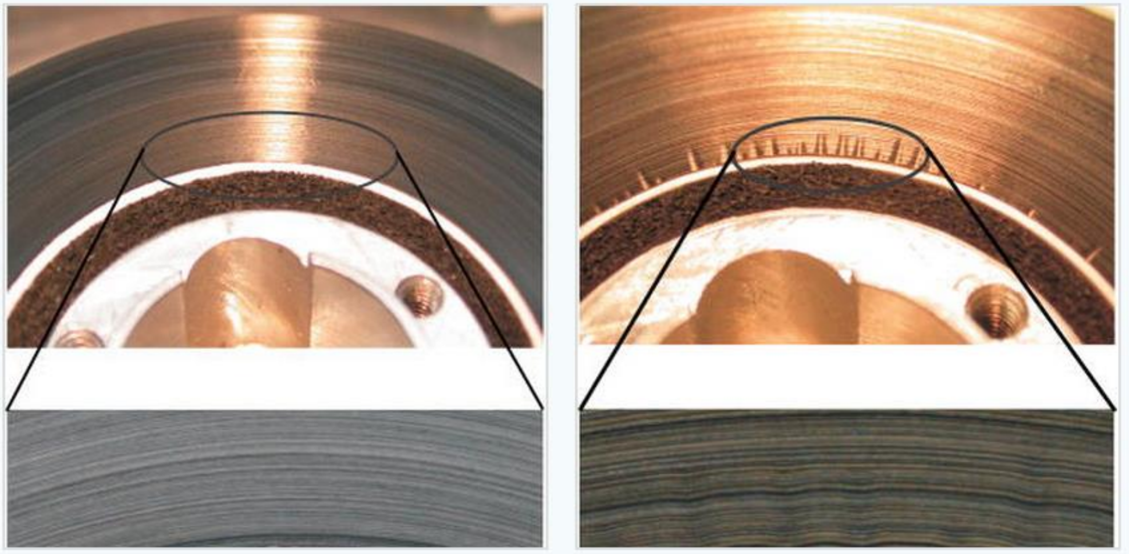
TJWALKER + ASSOCIATES INC

Core compression can create near-core buckling defects.

Near-core buckling occurs when the radial change is greater than the wound-on strain.

$$\frac{\Delta r}{r_i} > \epsilon_T$$

The exact buckling criteria of the near-core layers is not known, but any compressive strain may trigger this defect.



Corrugation and Buckling Defects in Wound Rolls
P. M. Lin and J. A. Wickert
J. Manuf. Sci. Eng. 128(1), 56-64 (Jul 13, 2005)

r_i = Initial core outer radius
Δr = Change in core radius from winding pressure.
ε_T = Wound-on strain of winding tension

Cores

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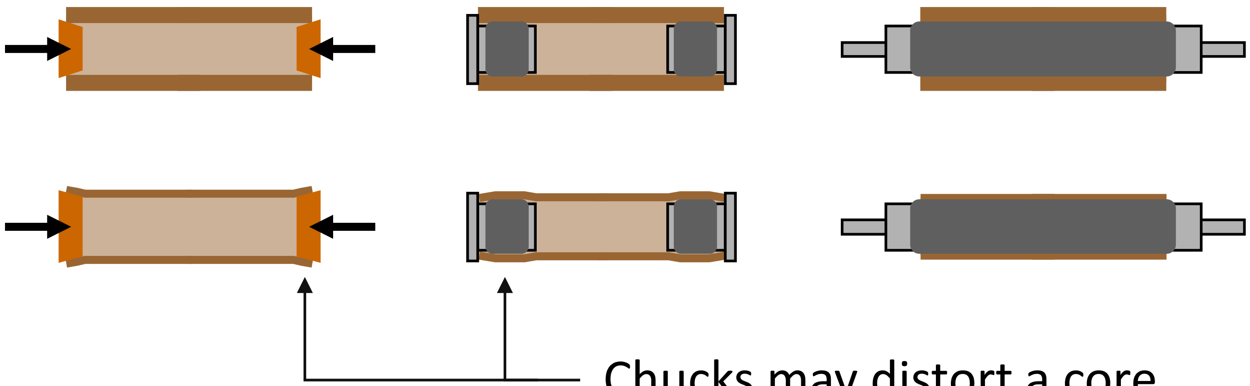
Core Support

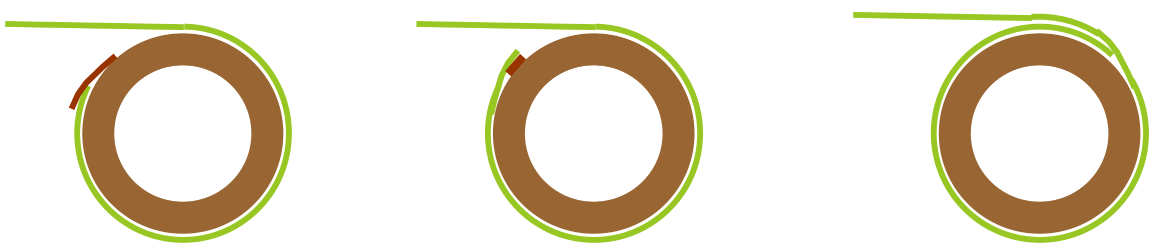
- End disks
- Chucks
- Shaft



Chucks may distort a core.

Web / Core Bond

- Tape
- Adhesive
- Tuck

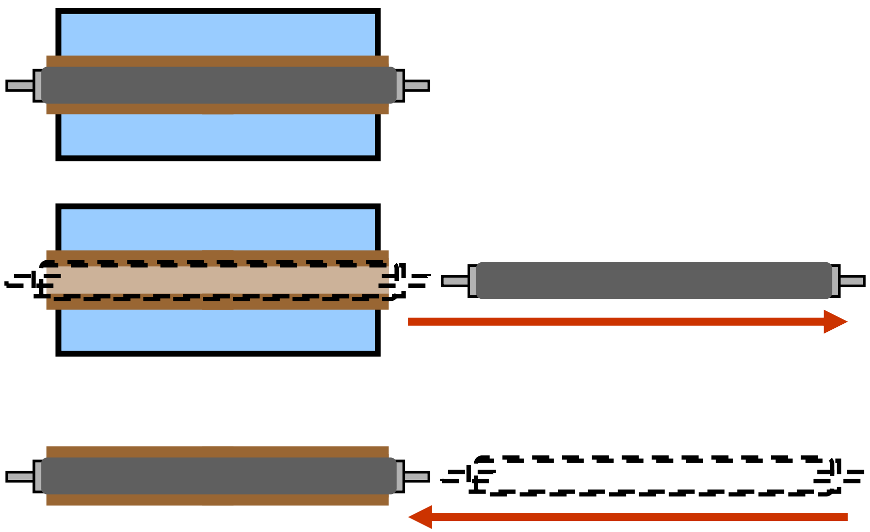


The unavoidable "step" at a core start may emboss through many layers.

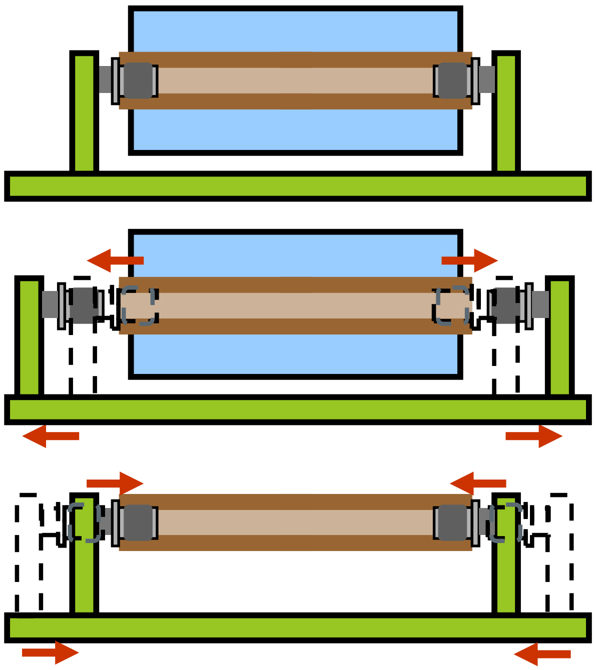
Shafted vs Shaftless

Shafts require more space than chucks to switch from finished roll to new core.

For most systems, the shaft must be removed completed from the roll and inserted into a new core.



Shaftless winders eliminate the need to handle the shaft between finished and new core, fitting in less space and improving productivity, safety, and ergonomics.



Advantages

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Shafted

- Shafts add stiffness, allowing smaller or thinner cores.
- Shafts can extend beyond web width, allowing fixed end supports, lowering equipment costs.
- Shaft have more contact area, allowing greater torque transmission without slippage.
- Extended shaft are used to unload rolls without contacting the roll's surface.

Shaftless

- Less floor space.
- No ergonomic problems of shaft handling.
- Reduced roll change cycle time.
- No added mass of roll handlng or acceleration inertia.

6.0 Winding

3. Winding a Roll

- Starting a Roll
- Managing Initial Contact
- Tension and Torque
- Winder Tensioning
- Co-Axial Winding

Key to Winding

<u>Start a roll</u> under tension, in-position, and wrinkle-free.

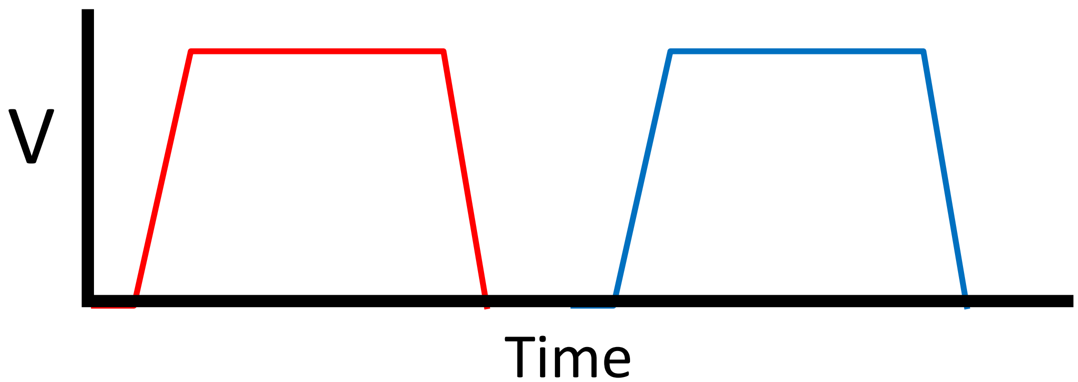
- Every roll needs to have a well-tensioned start
- Manage web lateral position before and into the winding roll
- Prevent wrinkled web within the winding roll from:
  1) Web handling before winding
  2) Roll transfer process
  3) Entry span into the winding roll

The Winding Process

Start Winding — Winding — Finish Winding

Option 1: Zero-Speed Transfer

Attached Web to Core at Zero Speed → Accelerate Web+Core to Wind Speed → Build a Roll → Decelerate Finished Roll → Cut Web at Zero Speed, Unload Roll



Slitter-Rewinders

Most slitter-rewinders make roll transfers at zero speed, decelerating the entire machine from unwind to rewind at the end of each roll.

Courtesy of Jagenburg

Courtesy of Catbridge

Fully Automatic Slitter Rewinder

- Full automatic slitter rewinders make roll transfers at zero speed.
- With turreted rewinds, automated core loading and roll unloading, and fast winding speeds, they can run up to five cycles per minute.

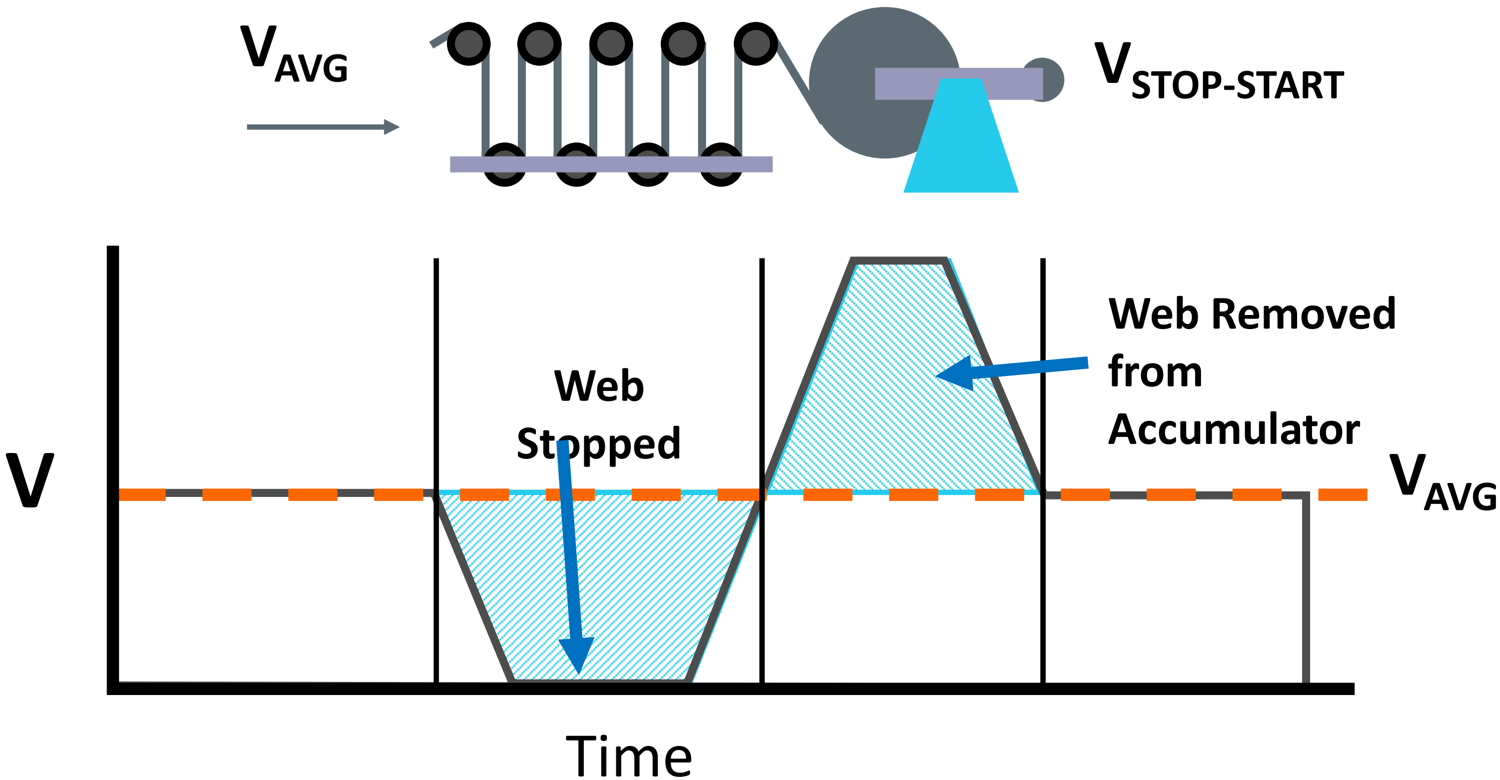
Courtesy of Ghezzo Annoni

Zero-Speed Roll Transfers

- Most slitter-rewinderss stop entire machine for roll transfers, sometime for several minutes.
- Fully automatic slitter-rewinders complete roll transfers in a few seconds.
- Zero-speed roll transfers should be well-aligned and wrinkle-free.
- Zero-speed if often the only choice for difficult to cut or splice materials (e.g. steel, textile, tear-resistant film laminates).

Accumulation-Dispense Balance

Many constant speed processes (e.g. extrusion, coating) use accumulators to make zero-speed transfers without changing main process speed.



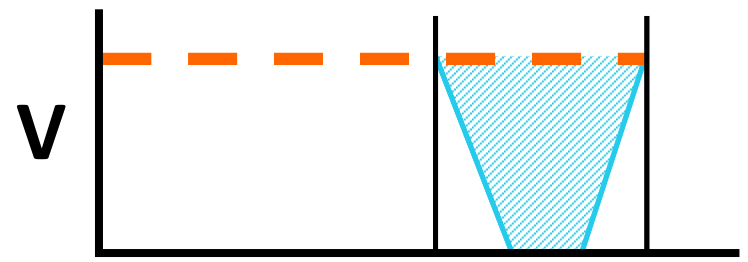
Accumulators

Courtesy of Menzel

Courtesy of Andritz

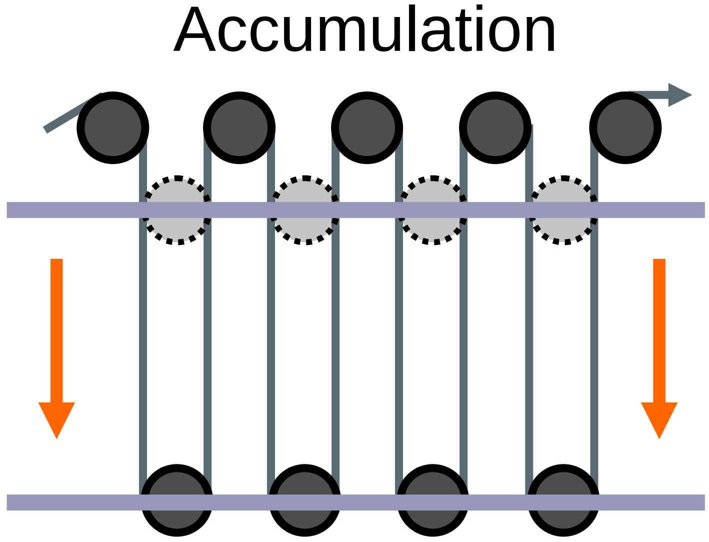
Accumulation Length

$$L_{ACCUM} = \left(t_{ACCEL} + t_{STOP}\right)V_{PROCESS}$$



$$\Delta L_{ACCUM} = n_{SPAN}\Delta L_{SPAN}$$

$$V_{ACCUM} = \frac{V_{IN} - V_{OUT}}{n_{SPAN}}$$



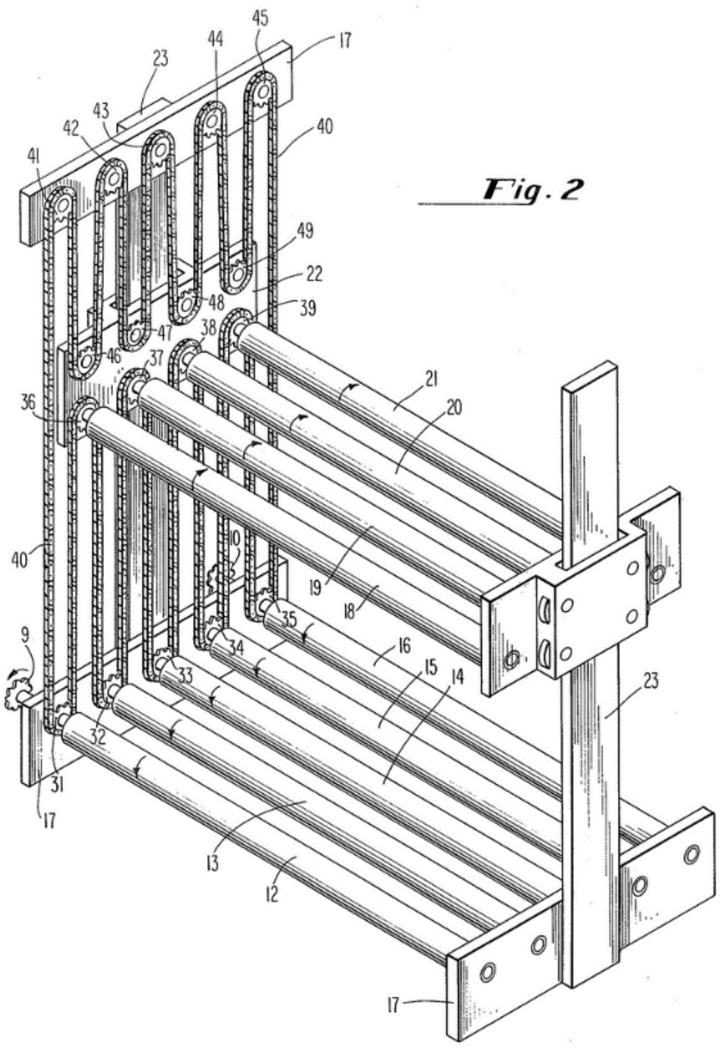
Driven Accumulator

Accumulators can have significant tension losses from roller drag (due to the large number of rollers), but more significantly from the inertial torque of accelerations.

Driving the accumulator rollers seems impossible, as each roller has a different acceleration rate (with rollers on the process side seeing little speed change and rollers on the winder sides seeing the full winder/unwind speed change, usually to a full stop up to 110% of line speed).

However, a chain system attached to each roller with thread path that has a dispensing geometry to offset the accumulating geometry can drive all the rollers at the appropriate speed, shifting the torque of roller drag and inertia away from the web and into the driven chain or belt.

U.S. Patent Mar. 1, 1977 Sheet 2 of 4 4,009,814

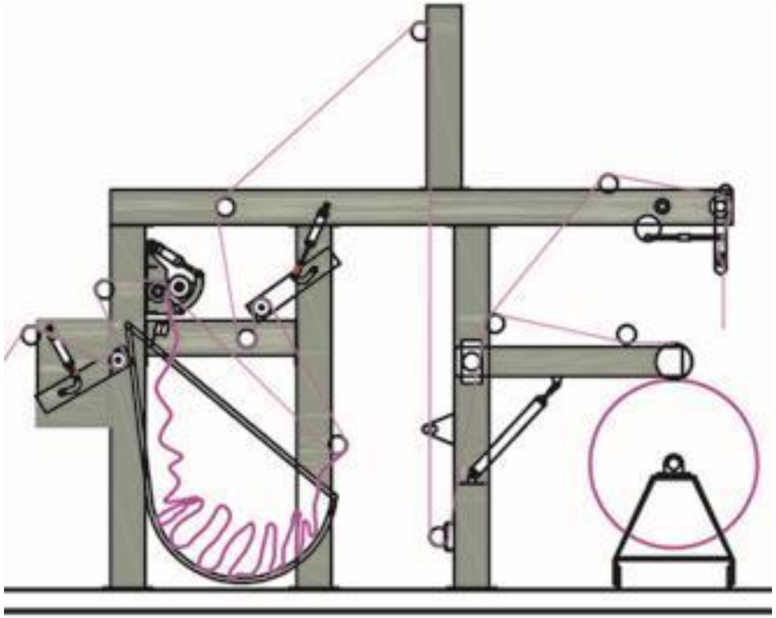


J-Box Accumulators for Fabric Webs

In wrinkle-resistant products, a J-box is a simple method to accumulate and dispense web between zero-speed splicing (or other variable speed processes) and constant speed processes.

J-Boxes violate the general rule of web handling to 'keep the web tensioned' and 'avoid zero tension handling.'

Courtesy of Monforts Fong's Textile Machinery Co., Ltd.

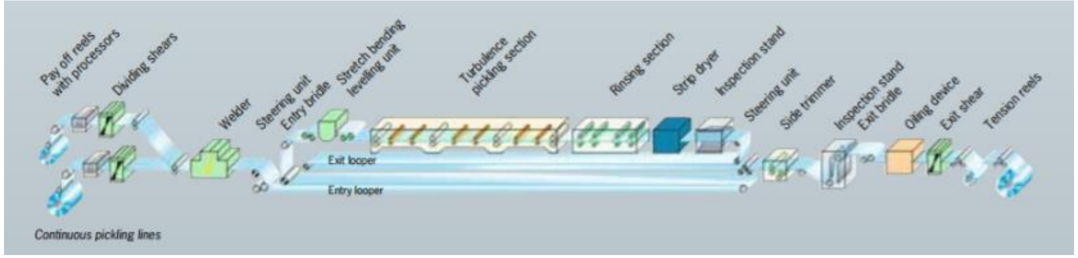


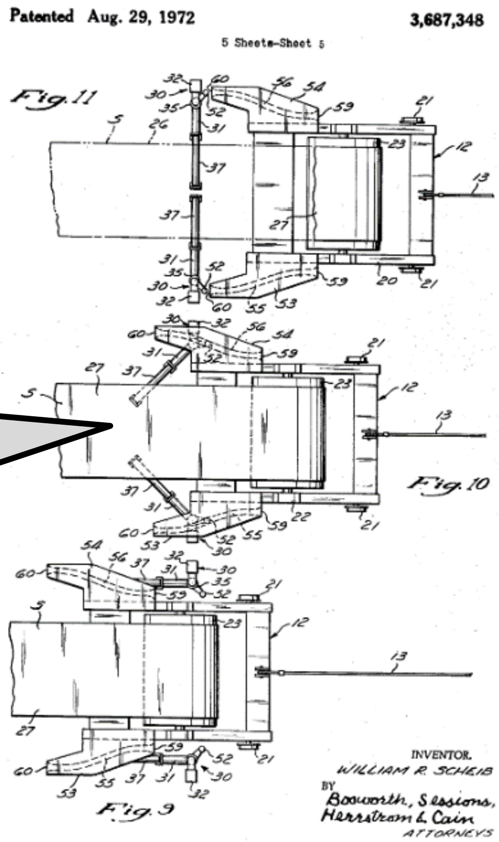
Courtesy of Yamuna Machinery

Steel Pickling Line Accumulator (Looper)

Courtesy of SMS Group

Courtesy of Siemens





Swing roller support horizontal strip and promote center tracking.

Zero-Speed Roll Transfers with Accumulators

- The accumulator length equal speed x time (the sum of acceleration time and transfer time).
- At some point, accumulator size is unreasonable large, requiring the main process to slow down or make at-speed roll transfers.
- Accumulators often create tension upsets from roller inertia and transfer of tension control.
- Accumulators often create significant lateral shifting, scratching, and wrinkling defects.

The Winding Process

TJWALKER+ ASSOCIATES INC

Start Winding

Winding

Finish Winding

Option 2: At-Speed Transfer

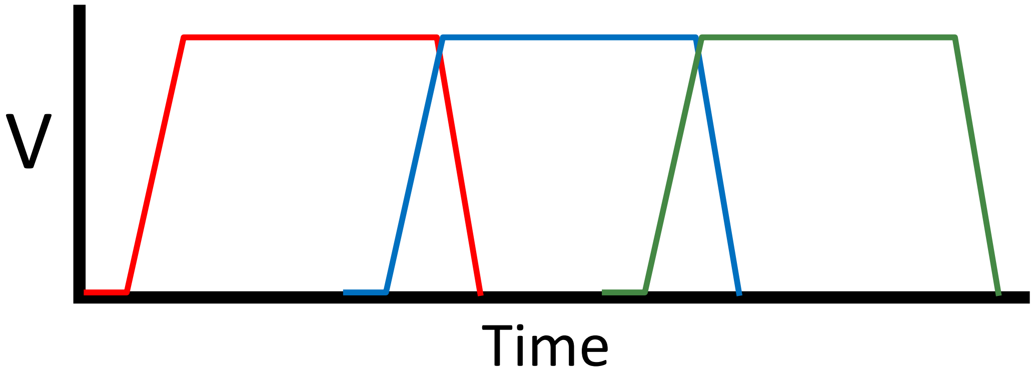
Accelerate Core to Speed

Attached Web to Core at Wind Speed

Build a Roll

Cut Web from Winding Roll at Wind Speed

Decelerate and Unload Roll



At-Speed Roll Transfers

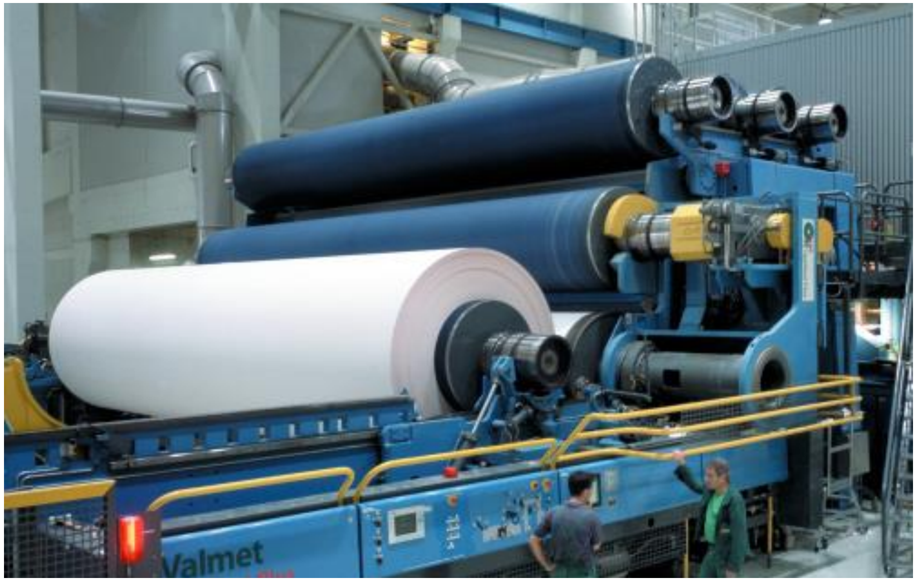
- At-speed roll transfers are common in paper and film making processes.
- At-speed roll transfers are used where speeds make accumulator size unreasonable (usually when accumulation length is over 100m).
- At-speed roll transfers are prone to waste, both at-core (e.g. wrinkling) and end of roll (indexing cycle lateral shifting, wrinkling, or post-splice material).
- At-speed roll transfers are best for easy to cut materials (thin, stiff) and low stack modulus web (which will hide poor roll starts after adding fewer layers).

At-Speed Roll Transfers

Film makers most often have turreted center winders with at-speed roll transfers.

Paper makers most often have pope-reel style surface winders with at-speed roll transfers.

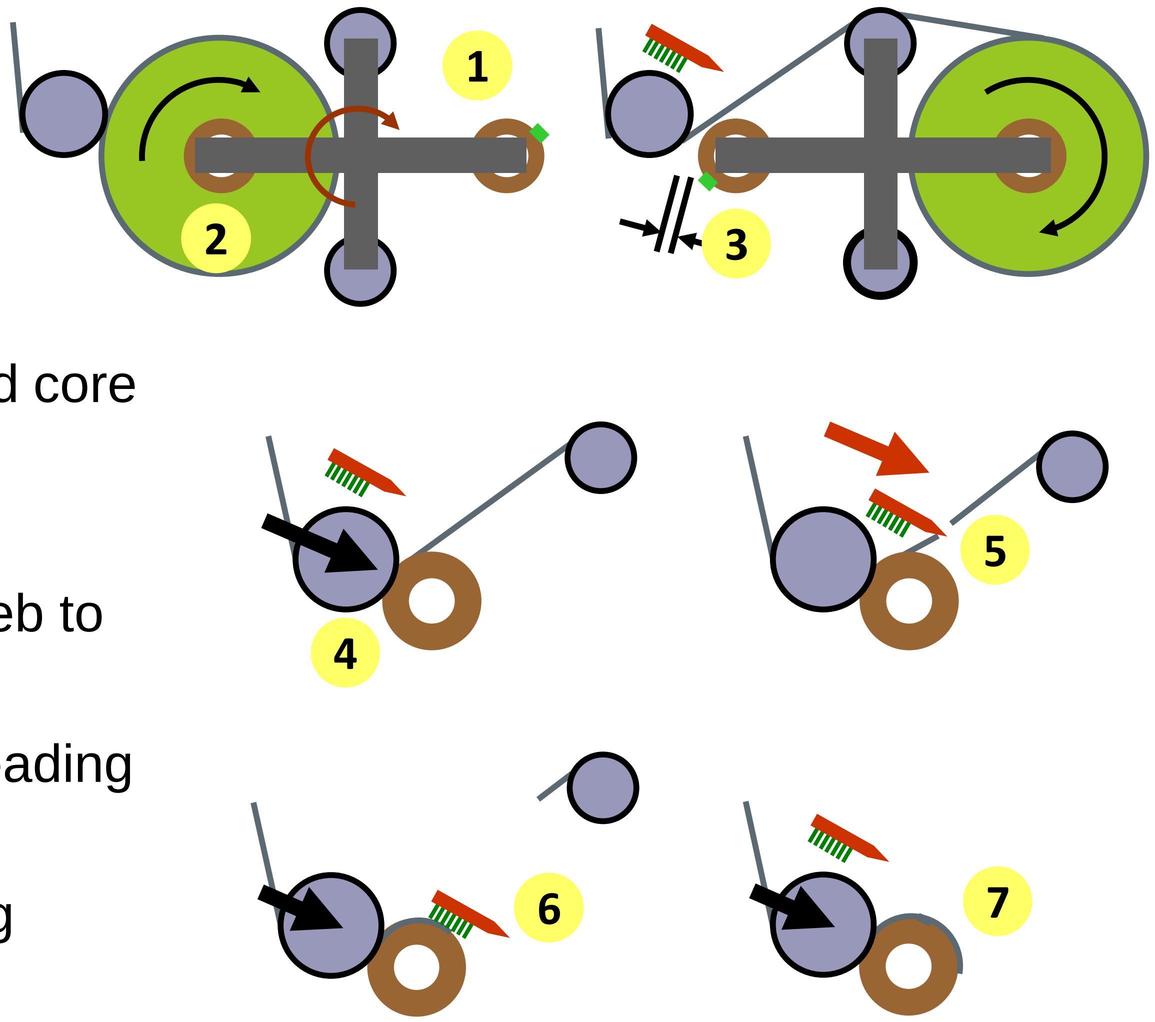
Courtesy of Bruckner



Courtesy of Valmet

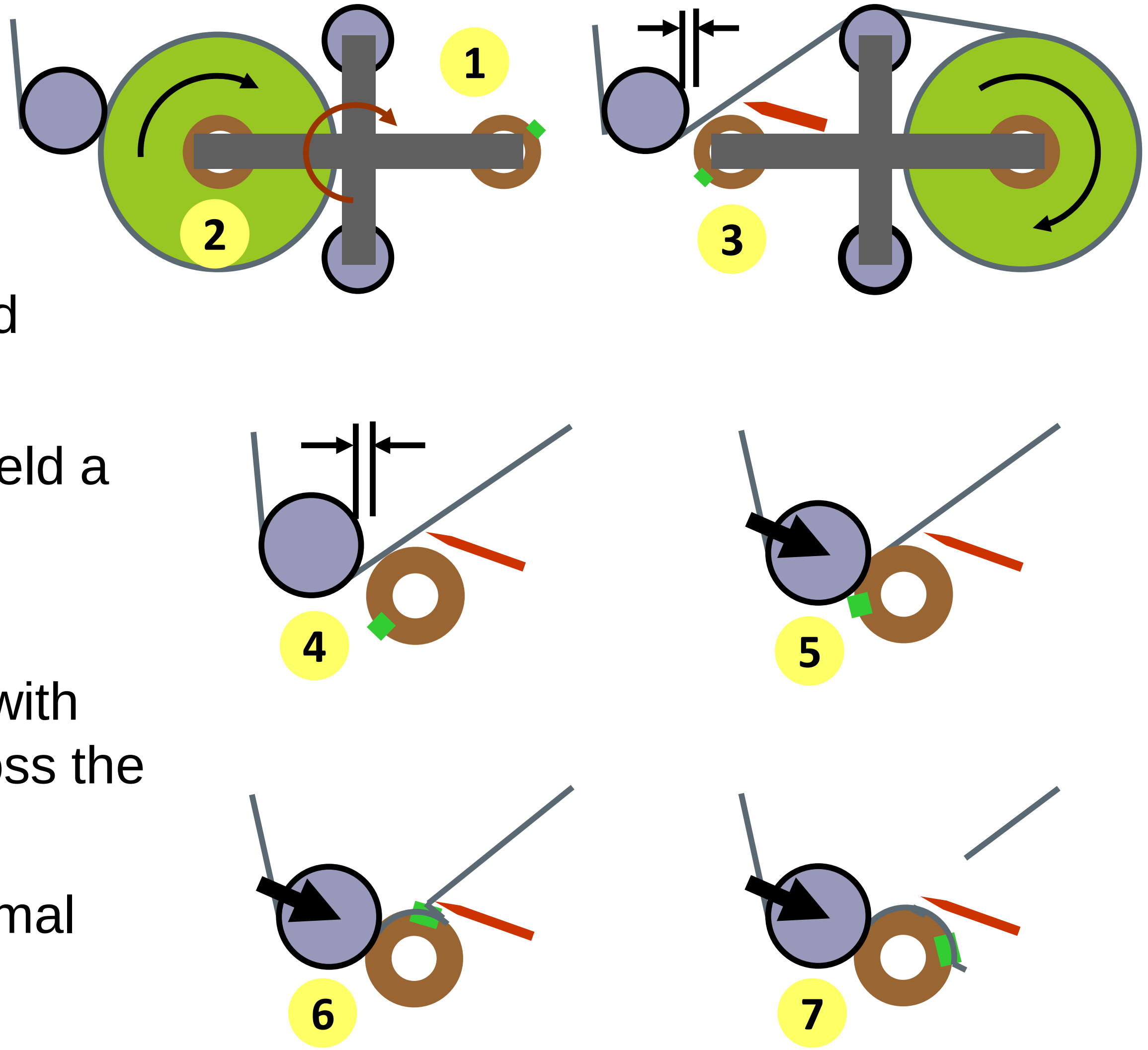
Flying Knife Auto-Transfer

1. Core may be prepared with adhesive to assist web-core bonding.
2. Target yardage triggers transfer cycle.
3. Turret indexes roll out and core in to wind position.
4. Bump roll nips the core.
5. Flying knife severs the web to the finished roll.
6. The brush smooths the leading edge to the new core.
7. Knife retracts and winding begins.



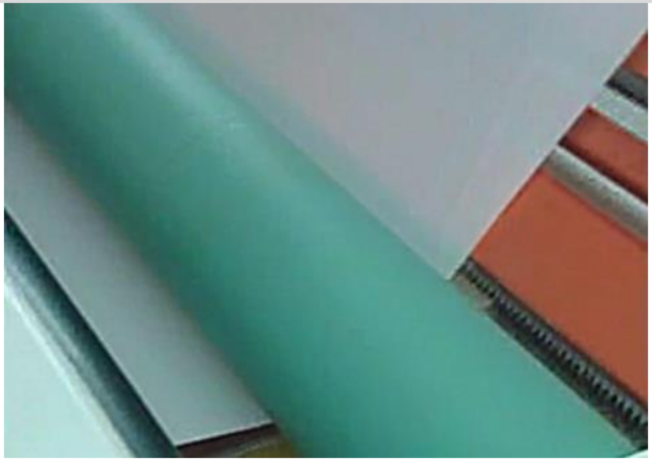
Stationary Knife Auto-Transfer

1. Core is prepared with adhesive strip.
2. Target yardage triggers transfer cycle.
3. Turret indexes roll out and core in to wind position.
4. Bump roll and knife are held a short gap from core.
5. Bump roll fires.
6. Core rotation grabs web with adhesive and pulls it across the knife.
7. New roll begins with minimal foldover.

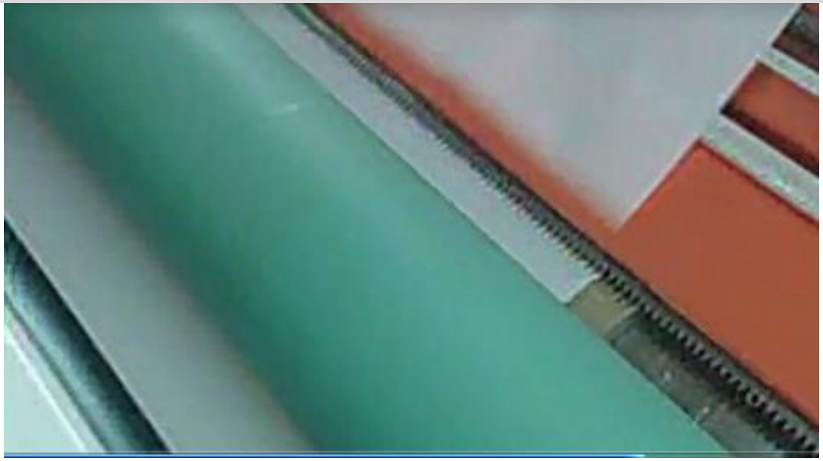


Black Clawson, Davis-Standard Stationary Knife

Just prior to tape grabbing web.

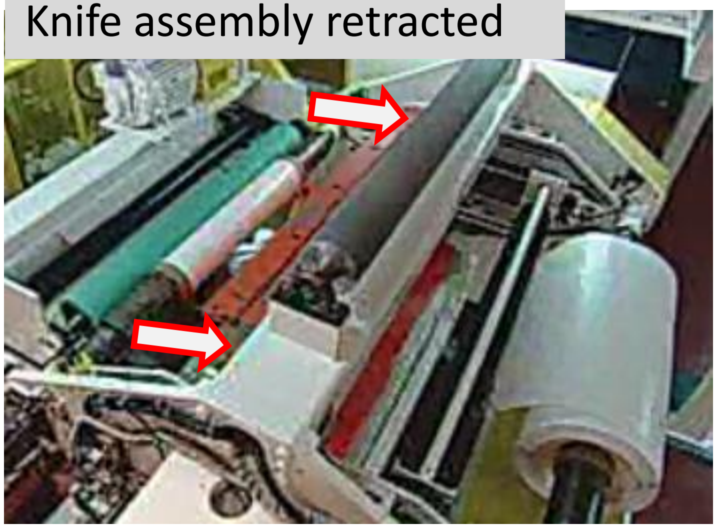


After tape pulls web into knife to cutoff.



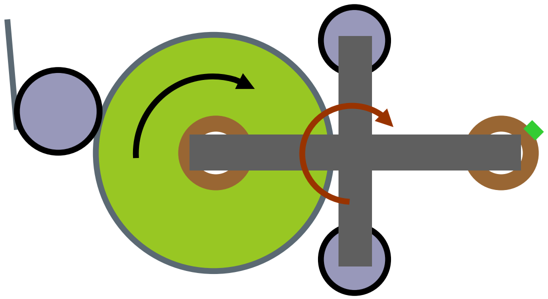
Knife assembly in cut position

Knife assembly retracted

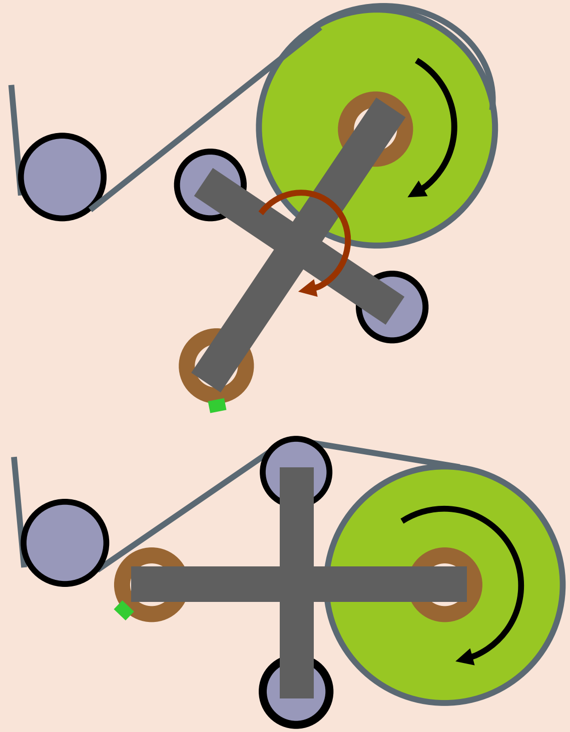


Roll Transfer Problems

Wrinkling and shifting should not be a problem in the primary winding position.



Out of the primary winding position, roll shape, long spans, misalignment, change in tensioning, and air lubrication may all lead to more wrinkles and shifting.



Unwinder Splicing

Flying Splice Factor	Key to Reliability
1. Splice Preparation	Ensuring roll is free of damage, generally cylindrical. Apply 2-sided bonding tape, cut leading tail flush to tape edge, tape down leading tail with tabbing tape. Minimize buckles in prepared roll to avoid raised features that will have resistance to air (be aerodynamic, no “sails”). The splice is aligned to the reference position (shown by the laser line), allowing the winder encoder to identify the rotational position of the splice.
2. Detect Unwind Roll Diameter	The ultrasonic sensor detects the new roll diameter. This diameter is used to 1) position the turret rotation to create the proper space between bump roller and new roll, and 2) determine the initial rpms to speed match the new roll to the web speed. NOTE: Errors in diameter measurement can create errors in both of these critical functions.
3. Turret and Bump Roll Positioning	The turret positions the prepared new roll close to, but not touching the web path of the expiring roll. The bump roll moves close to, but not touching, the new roll with the prepared splice. NOTE: The turret rotation requires the unwind roll to decelerate, then accelerate to let out the web length of the turret motion.

Unwinder Splicing

Flying Splice Factor	Key to Reliability
4. Accelerating to Line Speed	The new prepared roll must accelerate to line speed. Faster speeds will have more air drag forces exerted on the tabbing tape. Layers of the roll must not cinch (MD slip) to keep splice tape aligned to reference position.
5. Bump Roller Fires and Creates Full Width Bond to Expiring Web	The bump roller must press against the speed-matched roll, making contact ahead of the splice tape rotating into the contact point and maintaining contact as the splice tape point rotates through the contact zone. This requires accurate reference of the splice position by the encoder and timing of the bump roller start and length of engagement. The bump roller must also make sufficient contract pressure across the entire width of the new roll. If the new roll has significant cross-roll diameter variations, the bump roller may only contact on the large diameter side and fail to bond to the small diameter side. (We believe we have seen this problem, measuring a 0.25-in radius variations that correlated to one-sided (the small radius side) failure of the bump splice bond and splice tearout.
6. Splice Tape Bond > Tabbing Tape Strength	After the splice tape engages with the web, the bond of the splice tape must be great enough to break the tabbing tape. High spicing tape bond is a function of tape adhesive, bump roller pressure, contact time (which decreases with speed increases), and surface of the web (which if release coated, is more difficult to bond with).

Unwinder Splicing

Flying Splice Factor	Key to Reliability
7. Cutoff Knife Severs Old Roll	The cutoff knife must fire into the web from the expiring roll after the splice tape run through the bump roller contact zone. The web cutoff force is created by the web tension from the expiring roll interfering with the serrated blade position. The blade must have the proper position into the web path to enable the cut. Note: If the new roll speed is too high, slack may form after the bump and fall into the cutoff knife.
8. New Roll Takes Over Tension Control	Immediately after the bump splice, the new roll's motor must shift from speed match to tension control with dancer roller feedback.
9. Splice Tape Bond > Tension (and Other Applied Forces)	The newly formed bond of the splicing tape to the web must withstand tensioning, roller contact, nipped roller or process stresses, elevated temperatures, and reach the winder without failure.

3. Winding a Roll

- Starting a Roll
- Managing Initial Contact
- Tension and Torque
- Winder Tensioning
- Co-Axial Winding

The Winding Process

Start Winding | Winding | Finish Winding

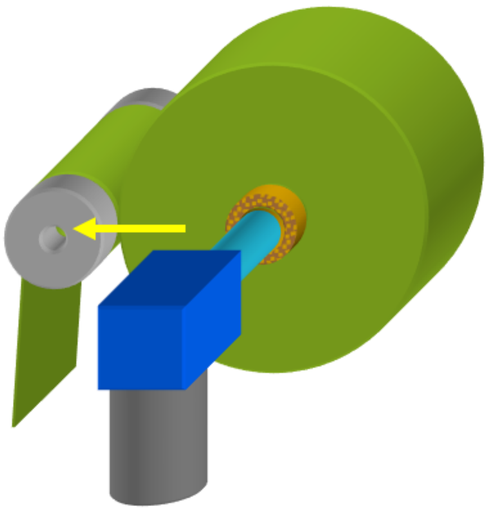
Attached Web to Core → Build a Roll → Cut Web and Unload Roll

Control Pre-Winding Tension

Align Web to Core

Hold Core Position

Transmit Center Torque without Internal Roll Slippage



Control Air Lubrication and Entrainment

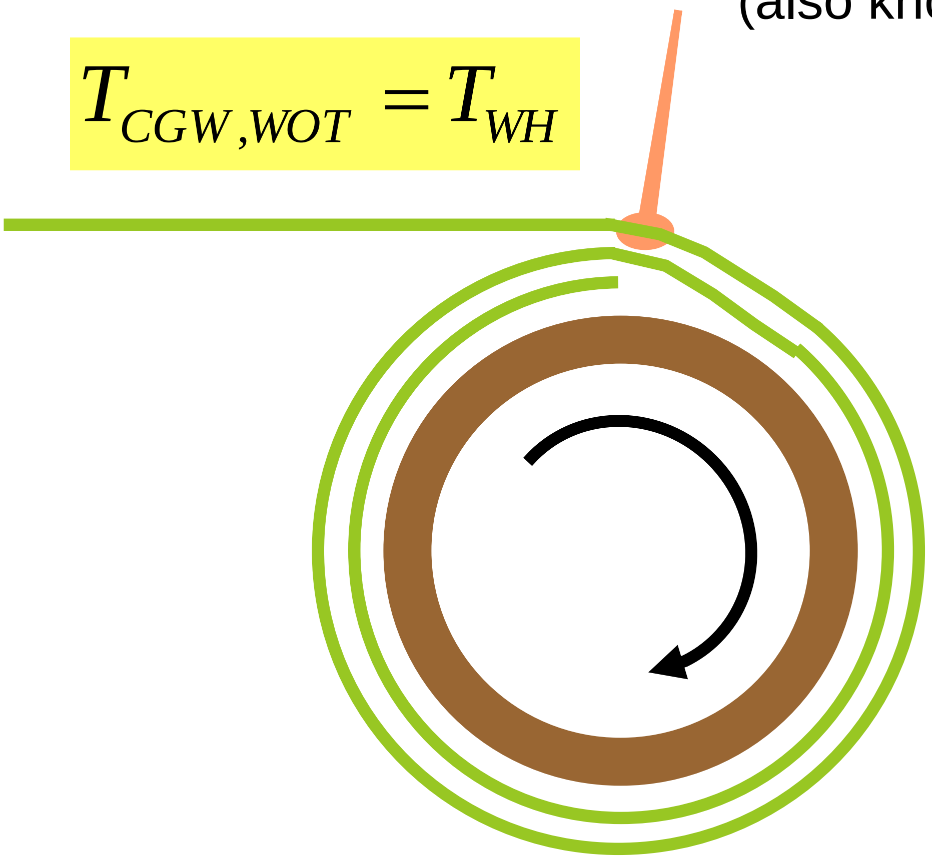
No Wrinkles

Wound-On-Tension

What is the web's tension as it enters the winding roll?

(also known as Wound-On-Tension or WOT)

$$T_{CGW,WOT} = T_{WH}$$



For center <u>gapped</u> winding…WOT equals the web handling tension.

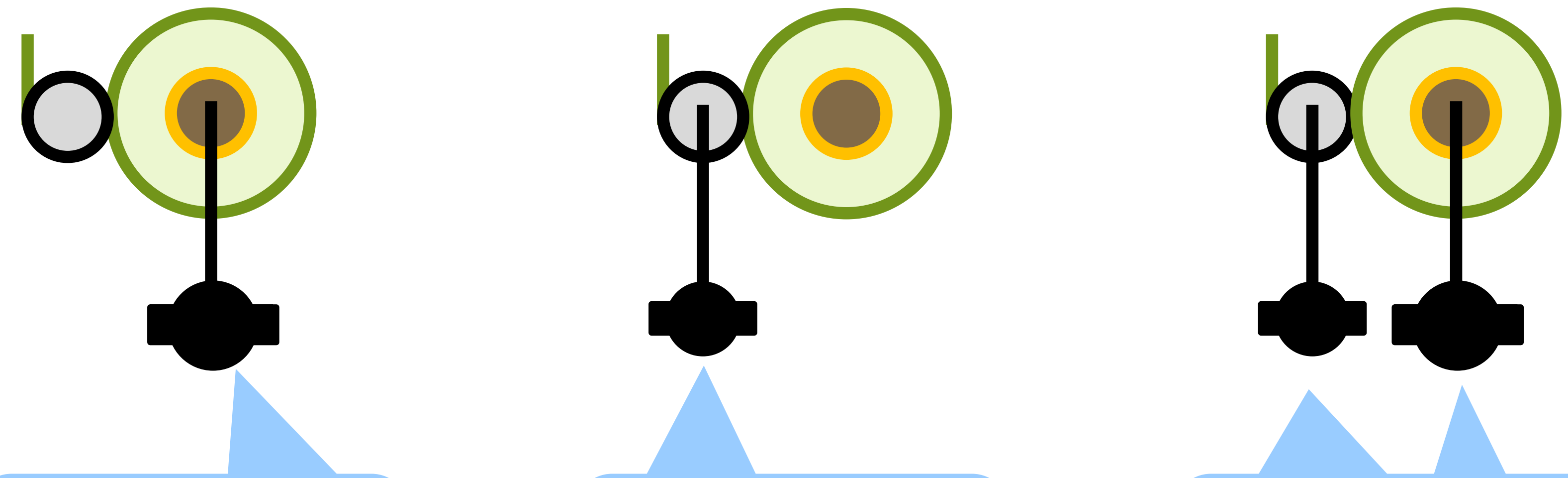
The WOT does NOT equal web handling tension for nipped winding, whether center or surface driven.

Winding Options: Center Driven

Center Driven Winder

Surface Driven Winder

Center+Surface Driven Winder

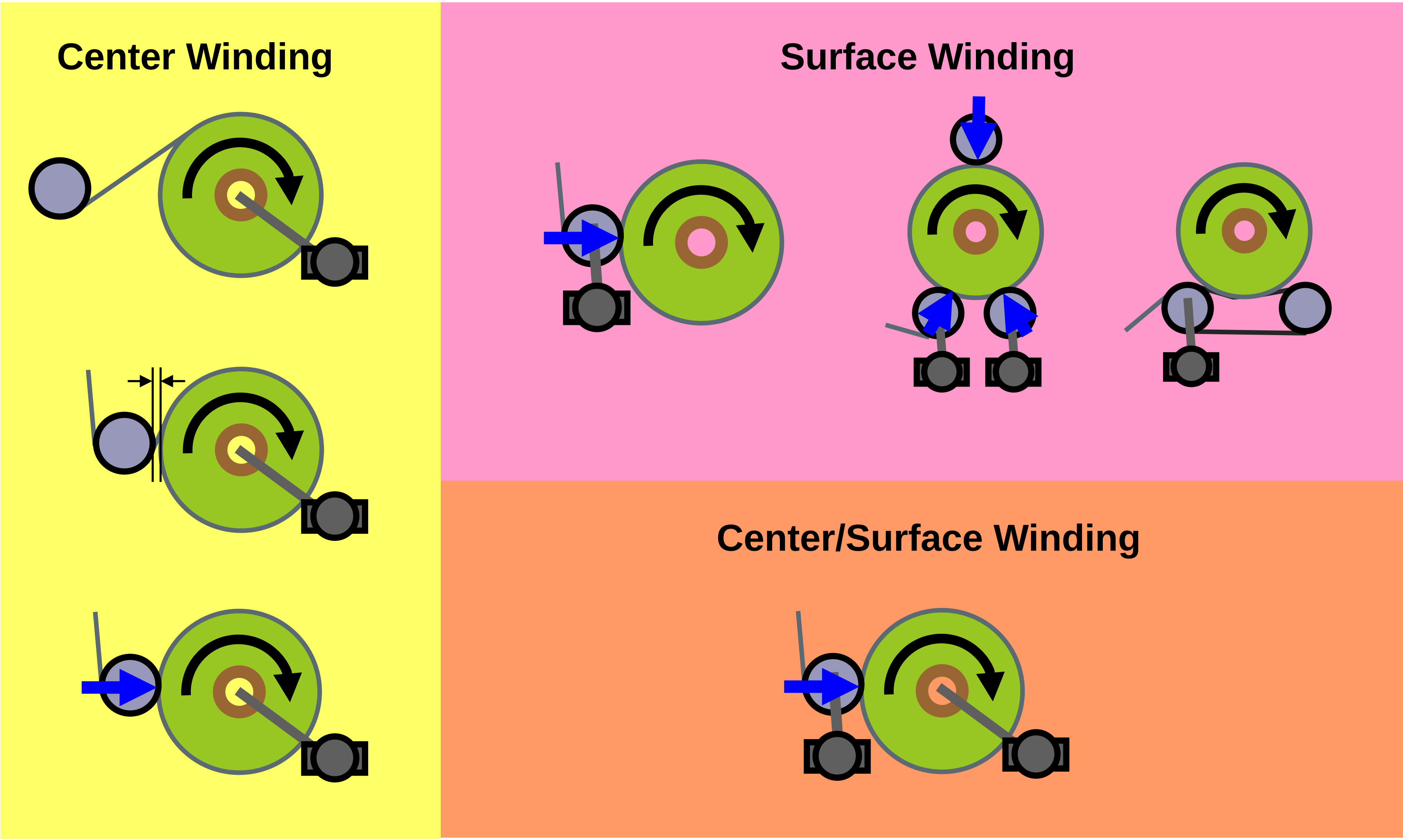


Winding roll is driven from its center.

Winding roll is driven from its surface.

Winding roll is driven from its center & surface.

Center/Surface Winding Definitions



Nip/Gap Winding Decision

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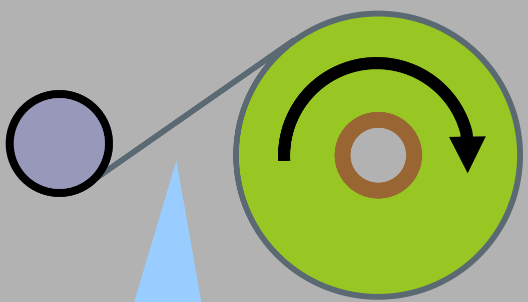
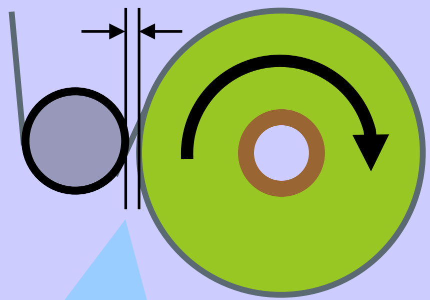
Free Span Winding



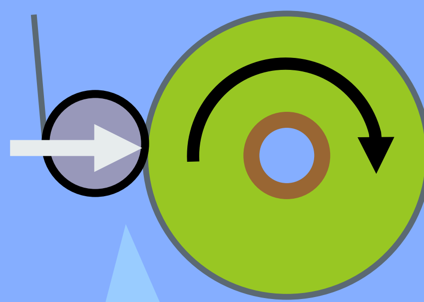
The winding roll has long entry span.

Controlled Gap Winding



A controlled gap and extremely short span is maintained between winding roll and last roller.

Nip Loaded Winding

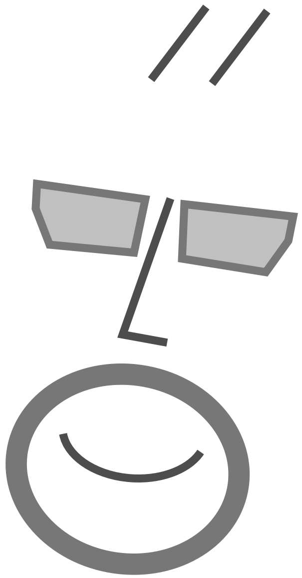


A roller is loaded against the winding roll with a controlled force.

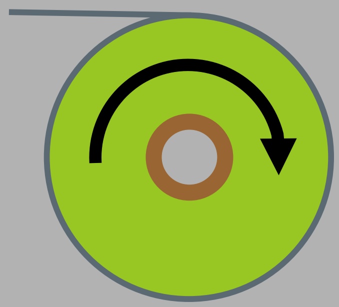
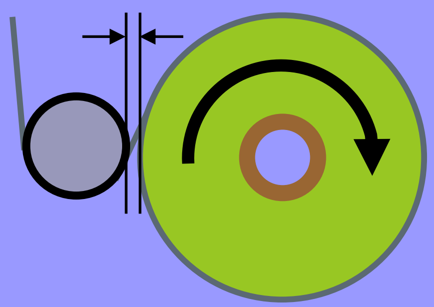
Winding Equipment Qs

What are the advantages of control gap winding?

When is control gap winding used?



Free Span vs. Control Gapped Winding

Free Span Winding	Controlled Gap Winding
	
Benefits:	2 Benefits:
• Simple design	• Improves alignment
• No translating parts	• Reduces wrinkles
• Low equipment cost	• <u>Doesn't compress winding roll</u>
• <u>Doesn't compress winding roll</u>	

Benefit #1: Prevent Wrinkling

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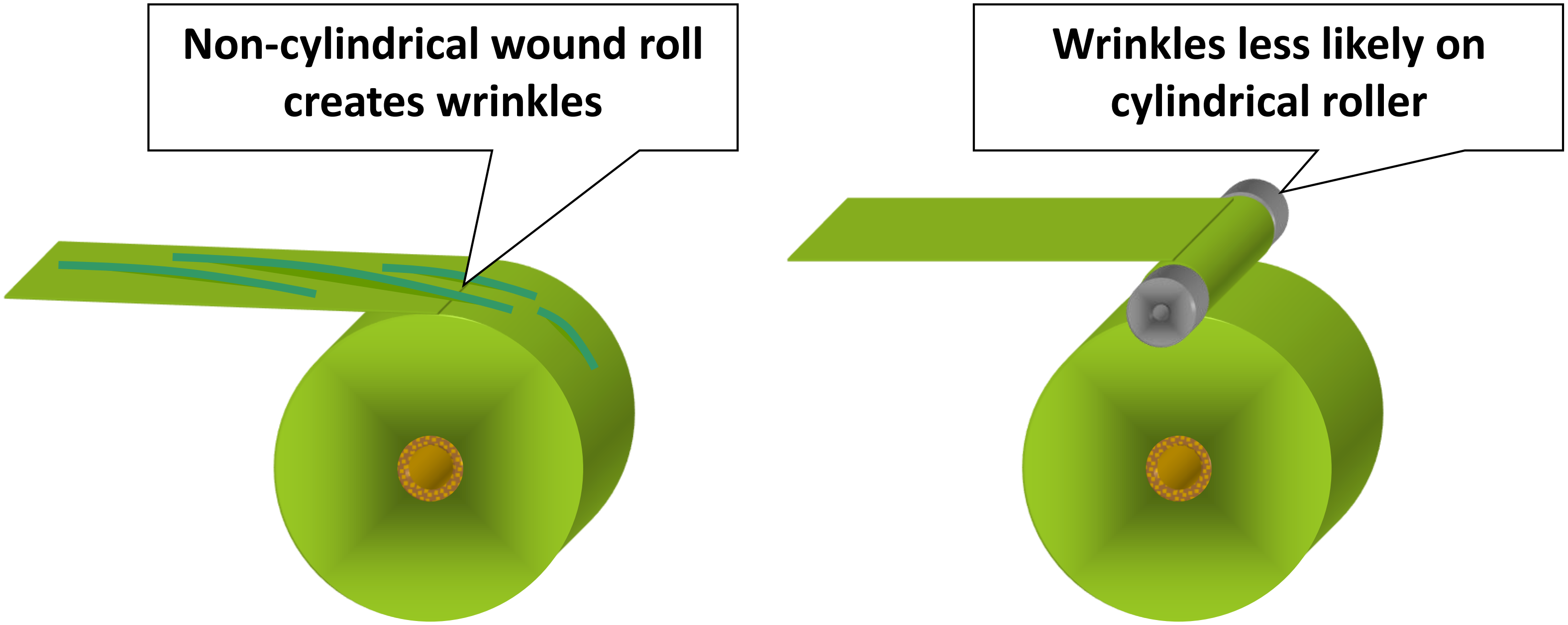
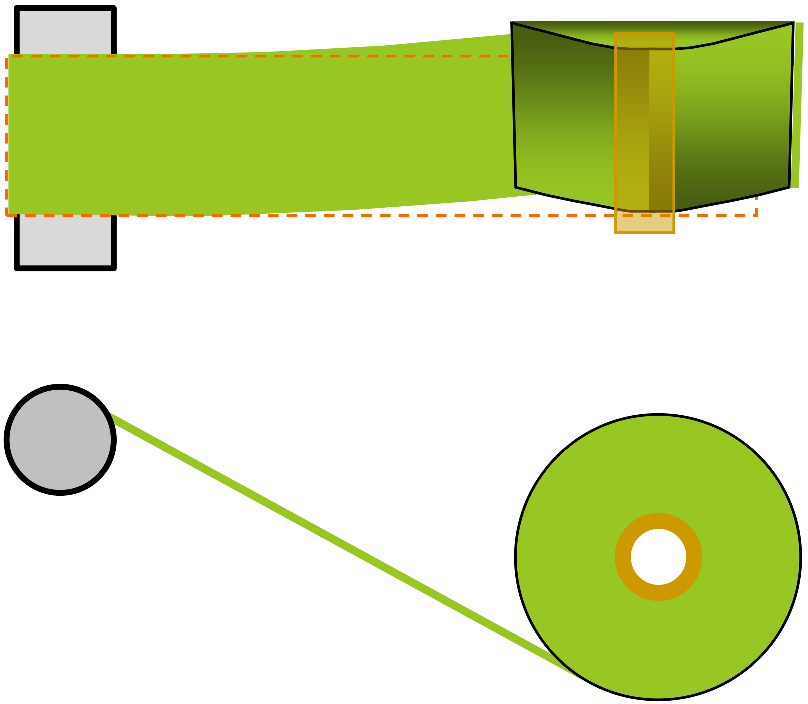
Nip or gap rollers can improve tracking and reduce wrinkling at winding.

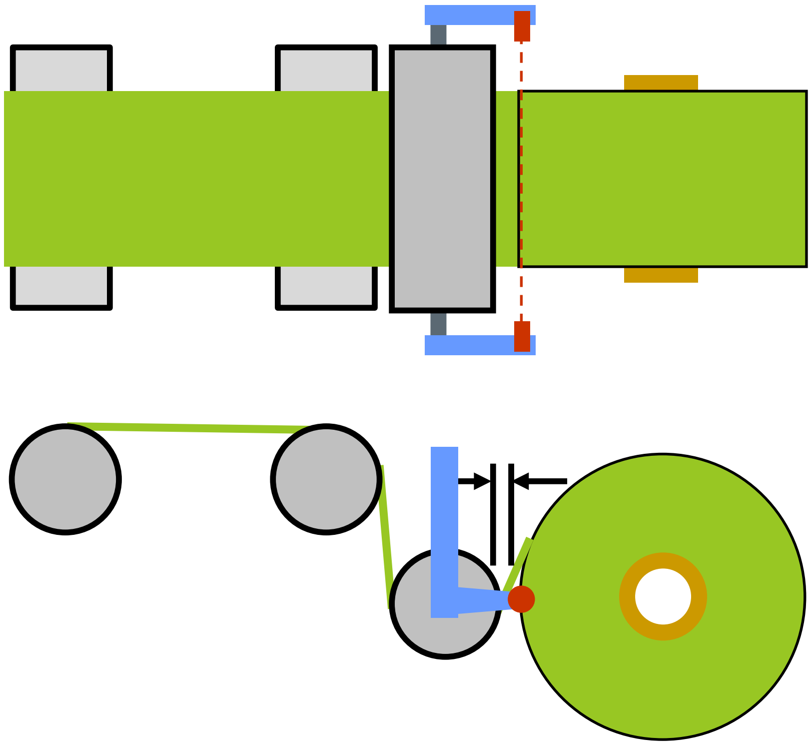


Benefit #2: Improve Alignment

Long spans are easily shifted by the winding roll misalignment or diameter variations



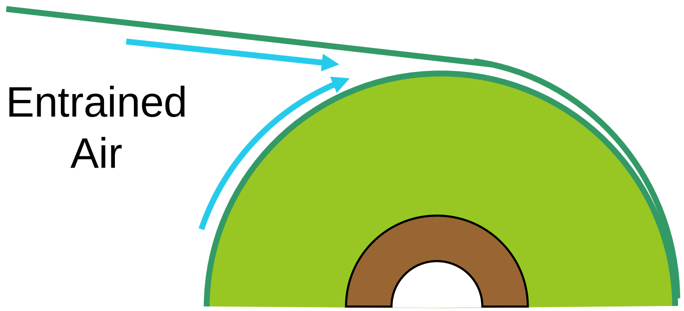
Gap or nip control makes it difficult for the winding roll errors to shift the web

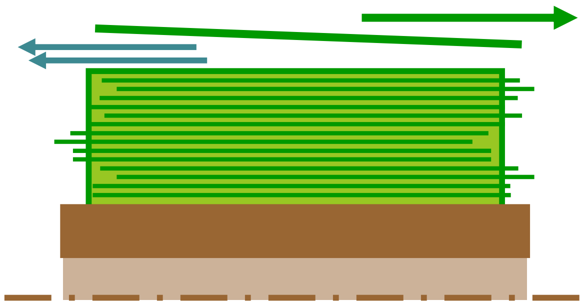


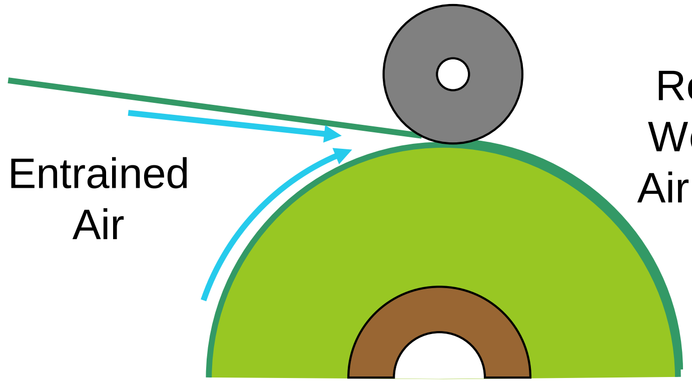
Benefit #2: Improve Alignment

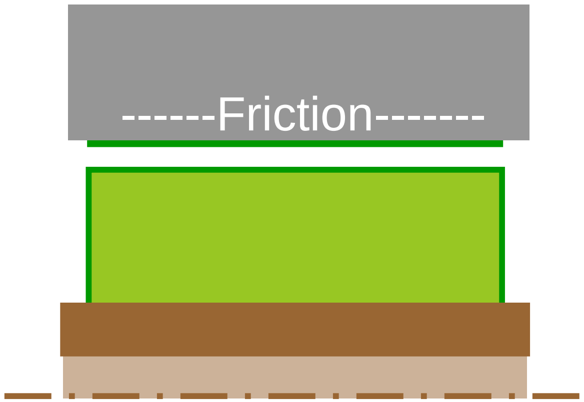
Nip rollers exclude entrained air (like a squeegee)…

…and oppose lateral sailing web motion.



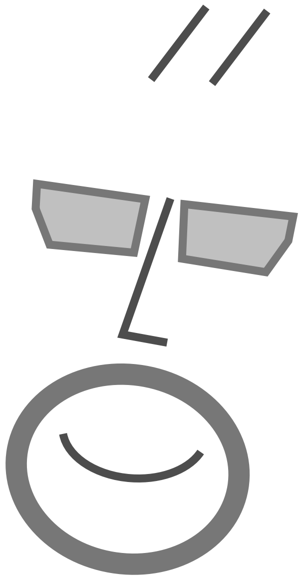






Winding Equipment Qs

What is the right gap distance and how is it controlled?



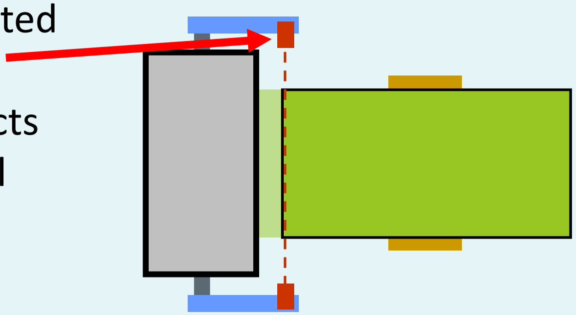
Gap distance is usually less than 25mm (1-inch).

Gap Roller Control Options

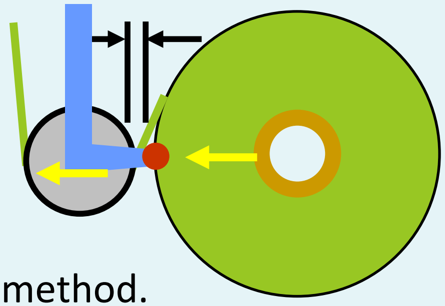
Gap winding requires a mechanism to position the gap roller relative to the roll's changing outer diameter, making gap control more complicated than force-loading air cylinders of nipped winding.

Method 1*: Gap Sensor

A sensor, mounted on gap roller assembly, detects the growing roll radius.



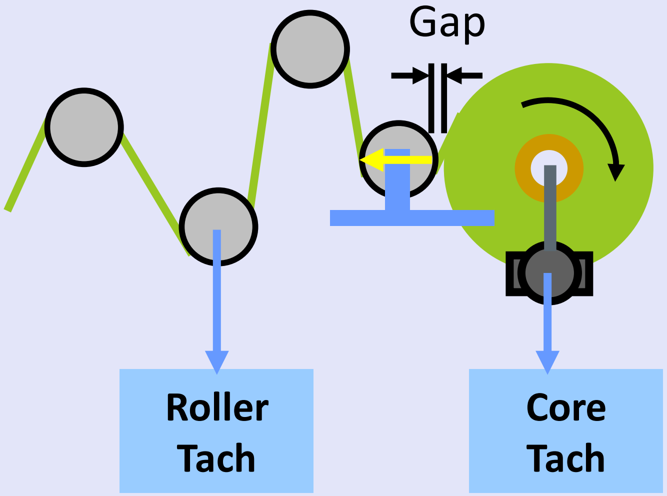
The gap roller assembly moves out as roll grows.



*This is the most used method.

Method 2: Gap Calculator

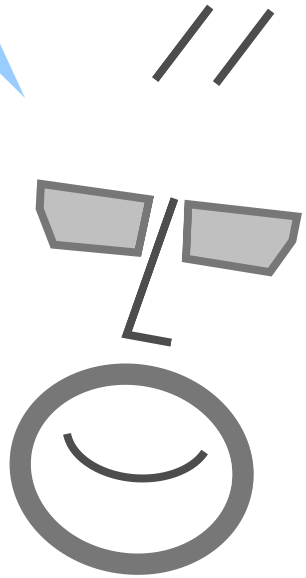
1. Roll diameter is calculated from tachometer info from core and non-slipping roller.
2. Gap roller is moved to calculated position.



Winding Equipment Qs

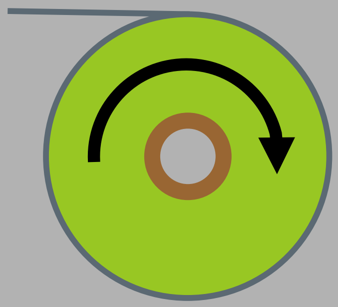
What are the advantages of nipped winding?

When is nipped winding used?



Nip/Gap Winding Decision

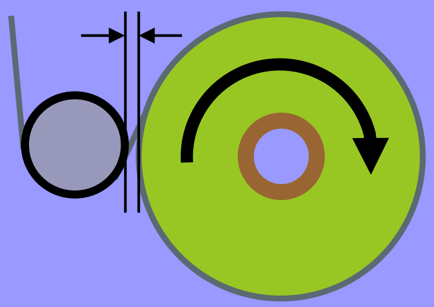
Free Span Winding



Benefits:

- Simple design
- No translating parts
- Low equipment cost
- <u>Doesn't compress winding roll</u>

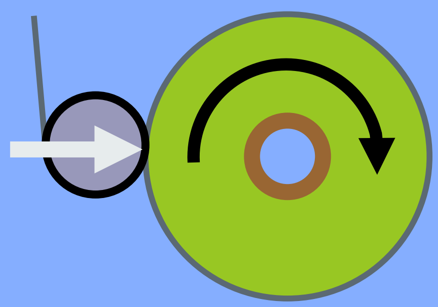
Controlled Gap Winding



2 Benefits:

- Improves alignment
- Reduces wrinkles
- Allows air lubrication
- <u>Doesn't compress winding roll</u>

Nip Loaded Winding

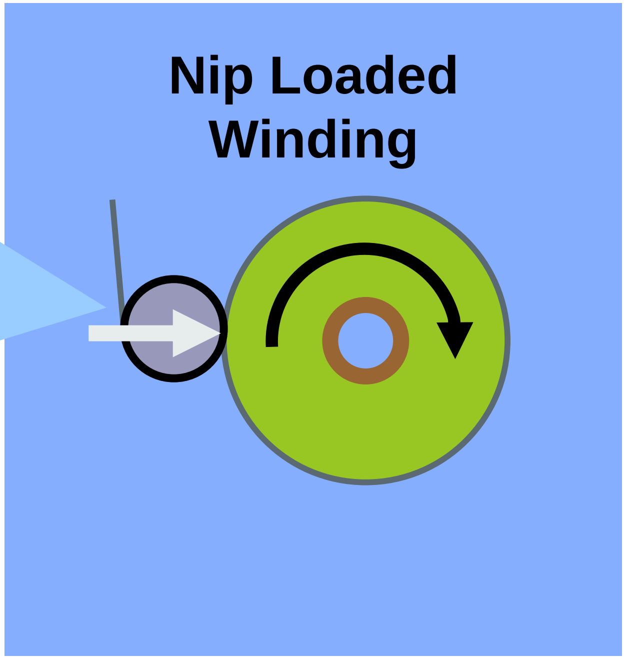


5 Benefits:

- Improves alignment
- Reduces wrinkles
- Squeegees air
- Increases roll tightness
- Promotes cylindricity

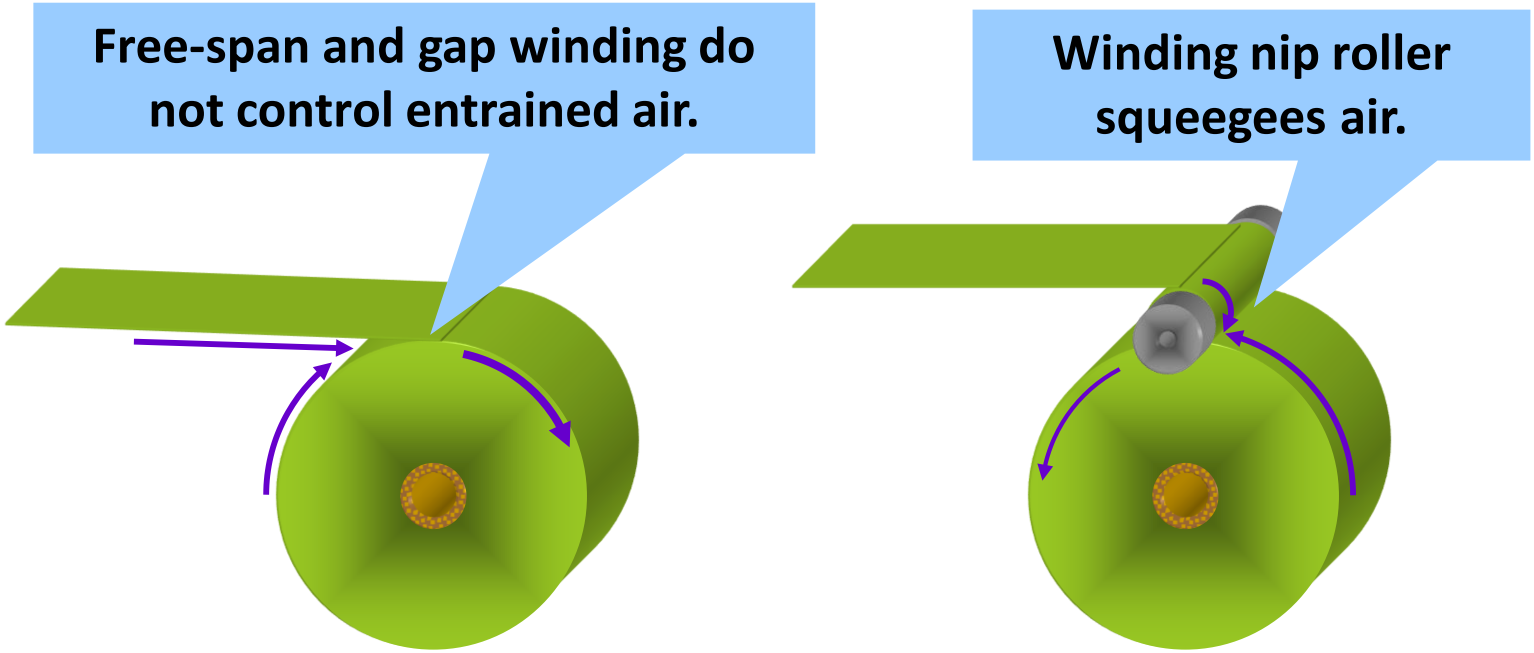
The Roller of Many Names

Pressure Roller
Contact Roller
Surface Roller / Drum
Ironing Roller
Lay-On Roller
Pack Roller
Squeeze Roller
Pinch Roller
Andruck (German)
Winding Nip Roller

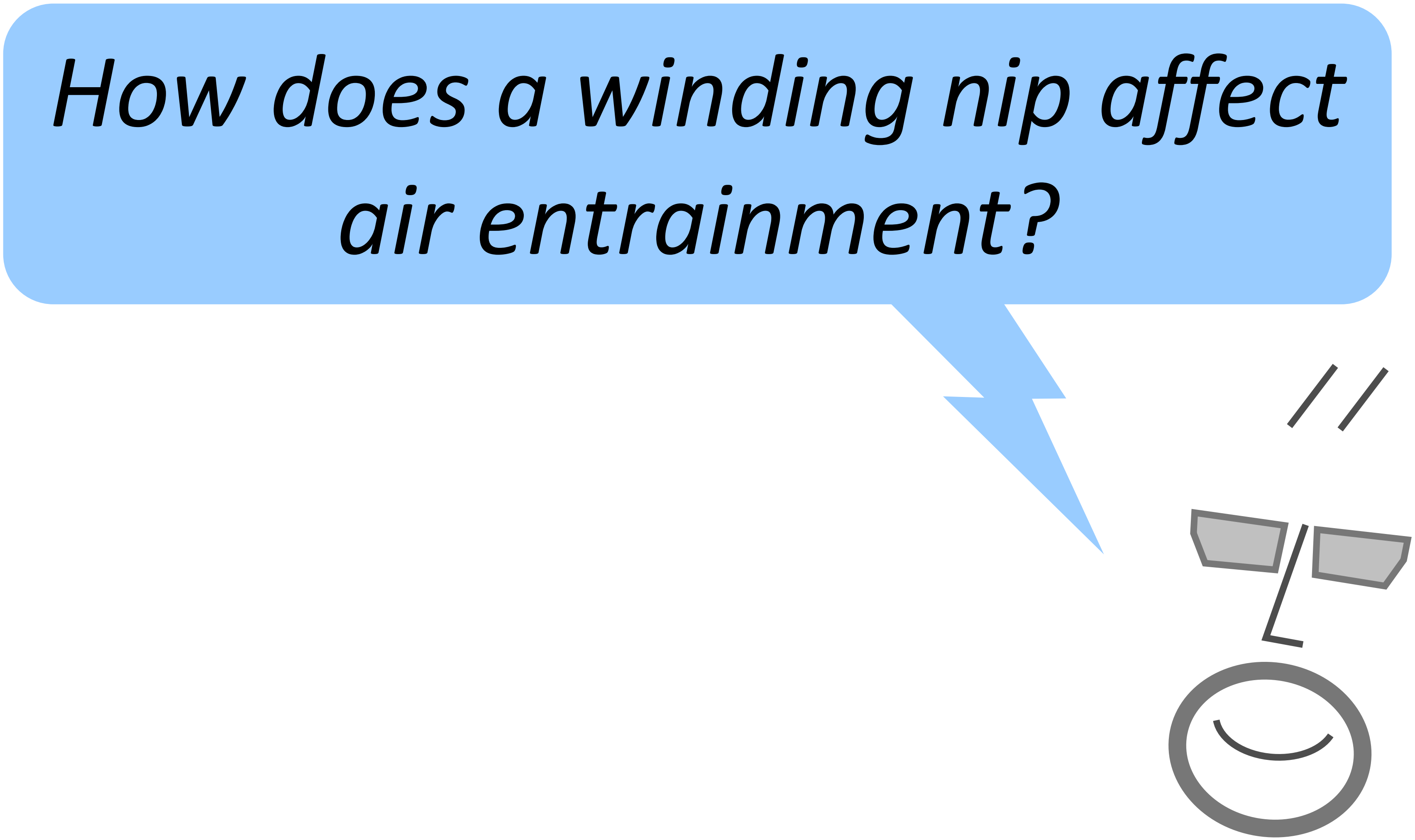


Benefit #3: Manage Entrained Air

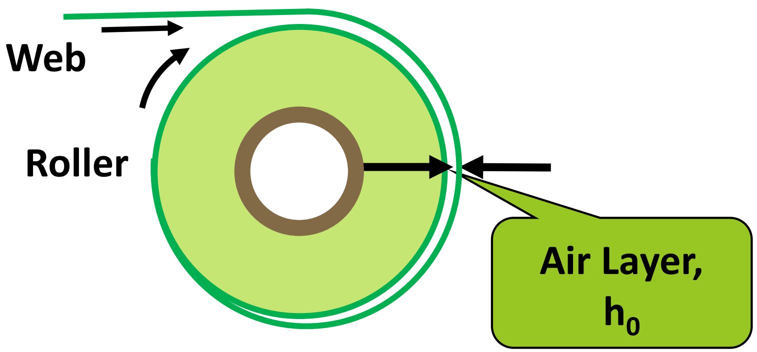
Nip rollers reduce air entrained in winding.



Winding Equipment Qs



Entrained Air in Gapped Winding



$$h_0 = 0.643r\left(\frac{6\eta(V_W + V_R)}{T}\right)^{2/3}$$

h_0	= Steady-state float height, in.
r	= Roller radius, in.
V_W	= Web speed, in/s
V_R	= Roller speed, in/s
T	= Tension, pli

No Winding Nip?
Air increases with:

- Larger radius
- Higher speed
- Lower tension

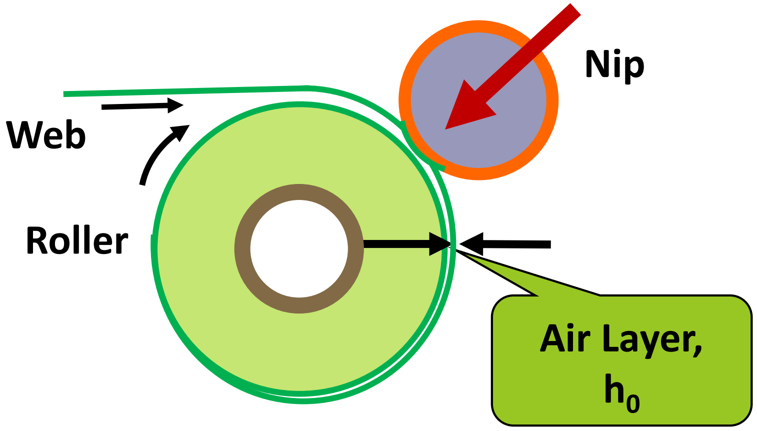
Entrained air increases proportional to radius.

2X radius = 2X entrained air.

Entrained air will be a function of speed-to-tension ratio (V/T):

50 fpm/0.25 pli = 400 fpm/2 pli

Entrained Air in Nipped Winding



$$h_o = 4\eta V \frac{r_{eq}}{N}$$

r_{eq}	= Equivalent roller radius, in.
η	= Air Viscosity,
V	= Process speed, in/s
N	= Nip load, pli

Winding Nip?

Air increases with:

- Larger 'effective' radius
- Higher speed
- Lower nip loads

Air increase with radius, but the nip roller radius may have a stronger effect that winding roll radius.

Entrained air will be a function of speed-to-nip ratio (V/N):

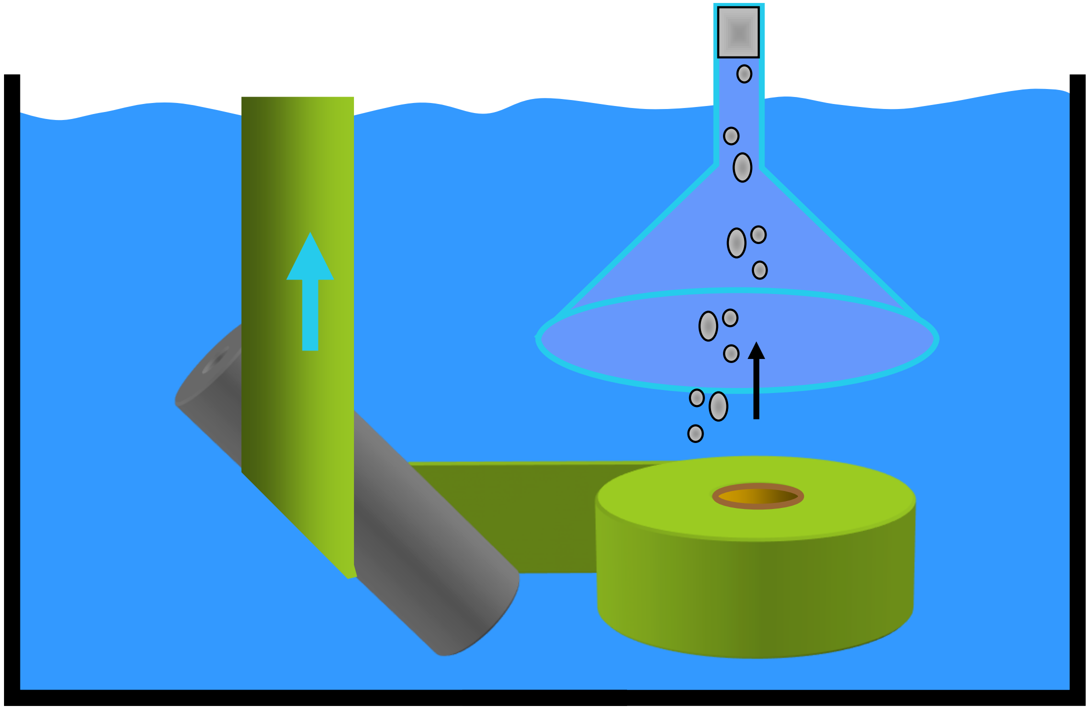
100 fpm/1 pli = 500 fpm/5 pli

Measuring Entrapped Air

The OSU WHRC measured entrapped air by unwinding wax-sealed rolls of film in an aquarium.

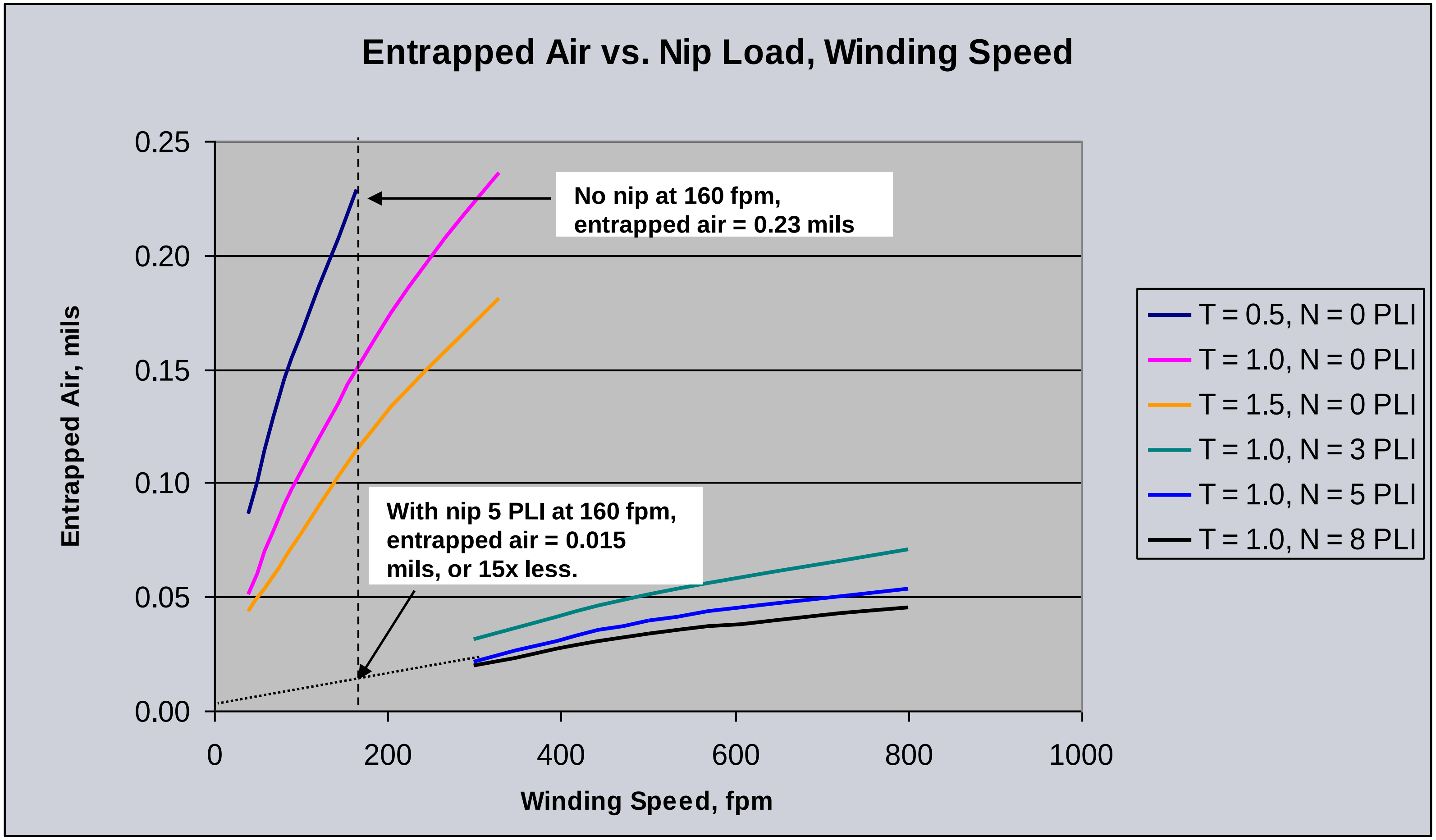
The "Bubblerimeter"

The released air bubbles were captured by a funnel and measured in a graduated cylinder.



J.K.Good, "Entrained Air Films in Center Wound Rolls – With and Without the Nip", Proceedings of the Fourth International Conference on Web Handling, 1997.

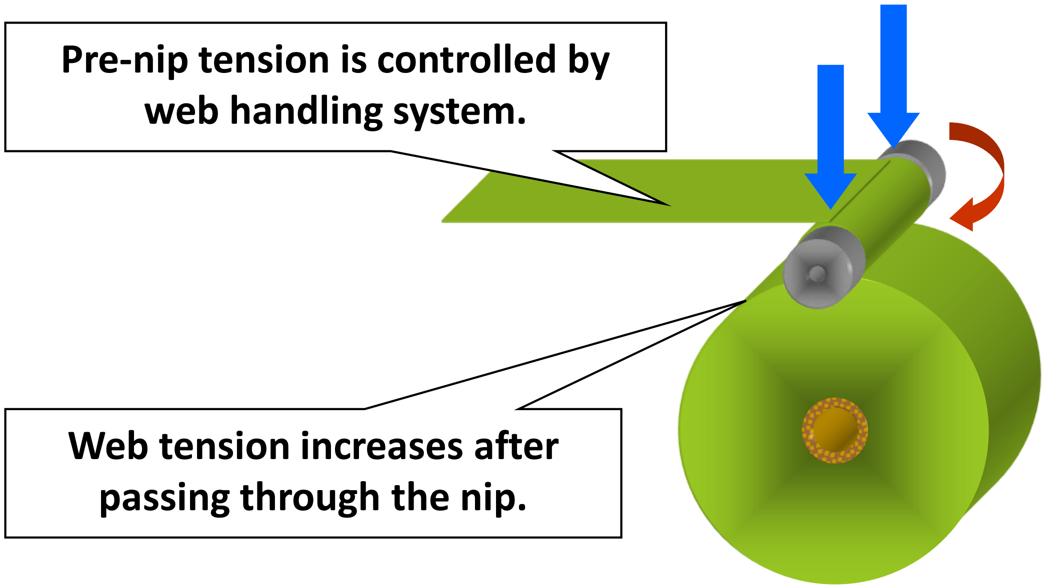
Measuring Entrapped Air



J.K.Good, "Entrained Air Films in Center Wound Rolls – With and Without the Nip", Proceedings of the Fourth International Conference on Web Handling, 1997.

Benefit #4: Increase Roll Tightness

Nip rollers increase wound roll tightness.



$$\sigma_{WIND} = \sigma_{WH} + \frac{\mu N}{t}$$

$$T_{WIND} = T_{WH} + \mu N$$

$$T_{WOT} \approx \frac{M_{CW}}{wr} + \mu_K N$$

$$\sigma_{WOT} \approx \frac{M_{CW}}{twr} + \frac{\mu_K N}{t}$$

(T and N in PLI)

J.K.Good, etal, "Stresses Within Rolls Wound in the Presence of a Nip", Proceedings of the First International Conference on Web Handling, 1993.

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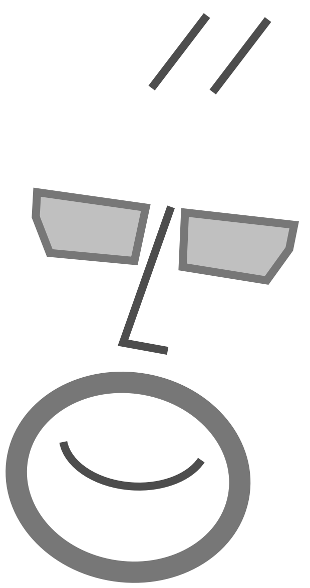
Winding Process Qs

What is nip-induced tension?

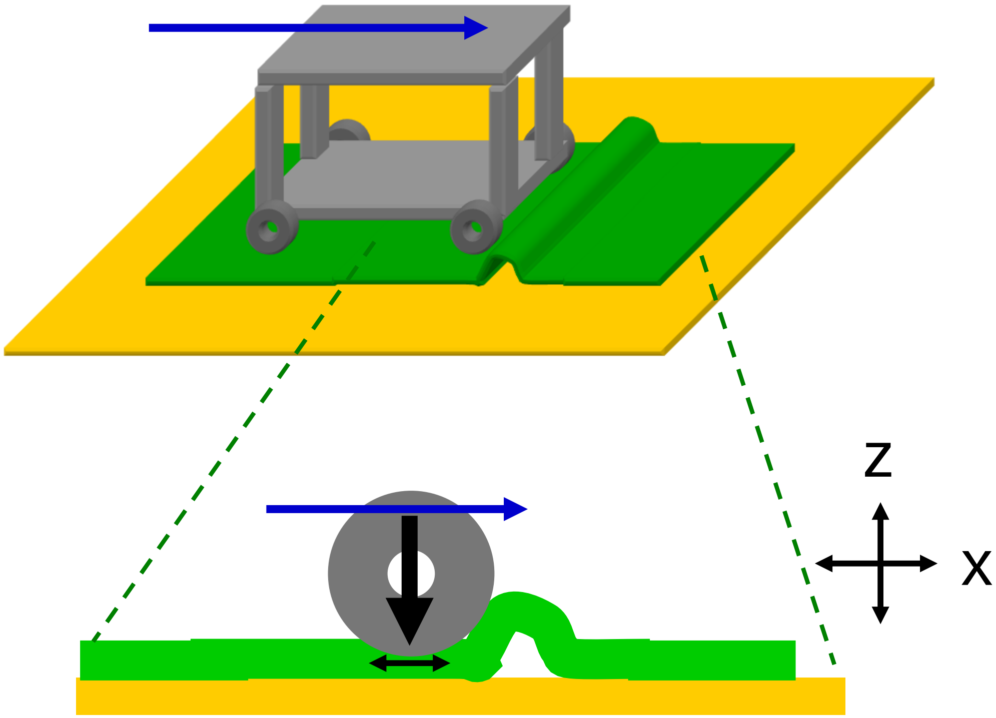


Nip Induced Tension

A cart rolled across a rug will form a wrinkle or buckle in front of it.

The nip force compresses the rug in the Z-direction and elongates it in the X-direction.

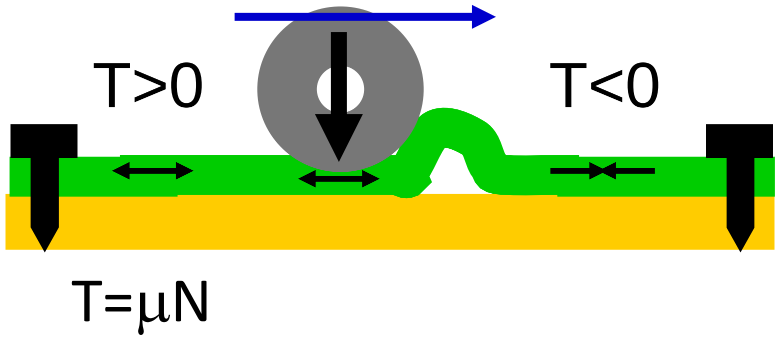
The wheel pins the elongated material to the ground, pushing the excess material forward..



Nip Induced Tension

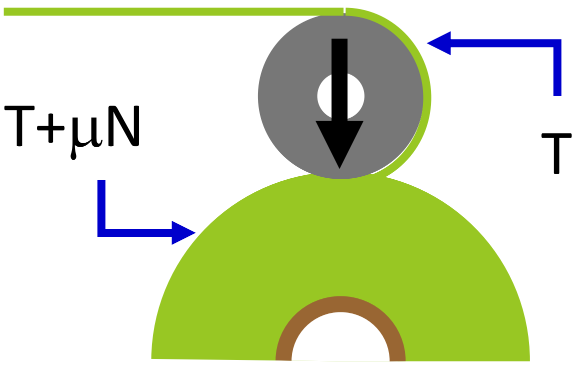
If the rug was fixed to the floor on both ends, its tension would increase after the nip rolled over it and decrease ahead of the nip.

The tires ability to tension the rug is limited by friction – the frictional force between the rug and floor (μN).



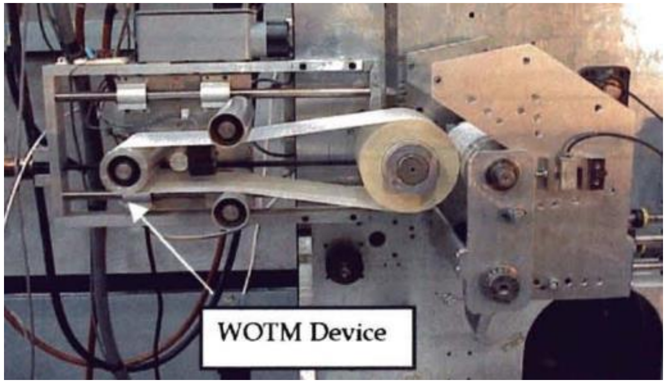
The winding nip has the same effect.

The web that passes under the nip, slips relative to the layer below it, increasing tension by approximately μN.



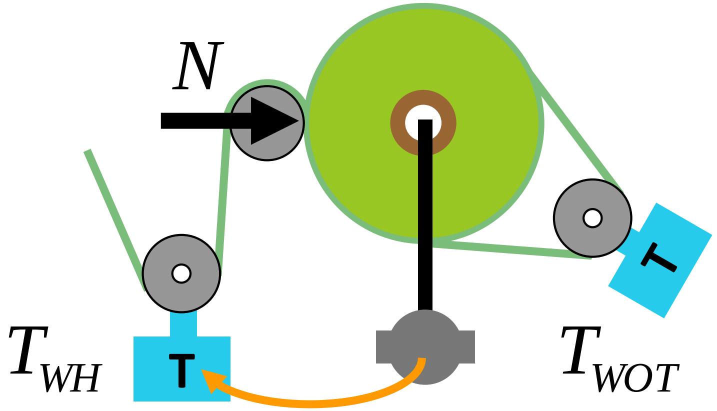
Measuring Wound-On Tension (WOT)

For nip winding, WOT is measured by diverting the top layer of the winding roll around a tension measuring roller.

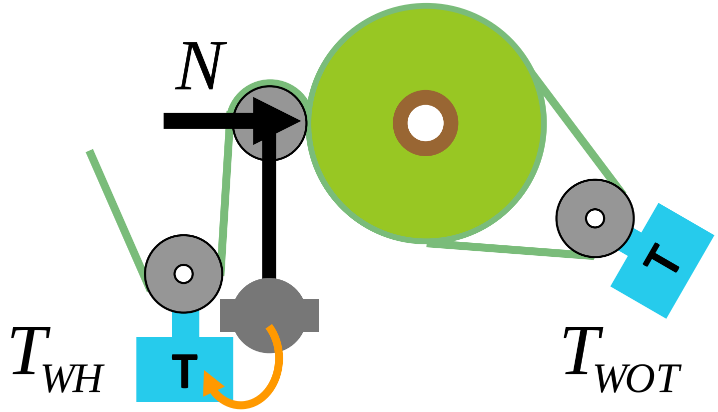


Nip-Induced Tension is the difference between the web handling tension and the wound-on tension.

$$\Delta T = T_{WOT} - T_{WH} = NIT$$

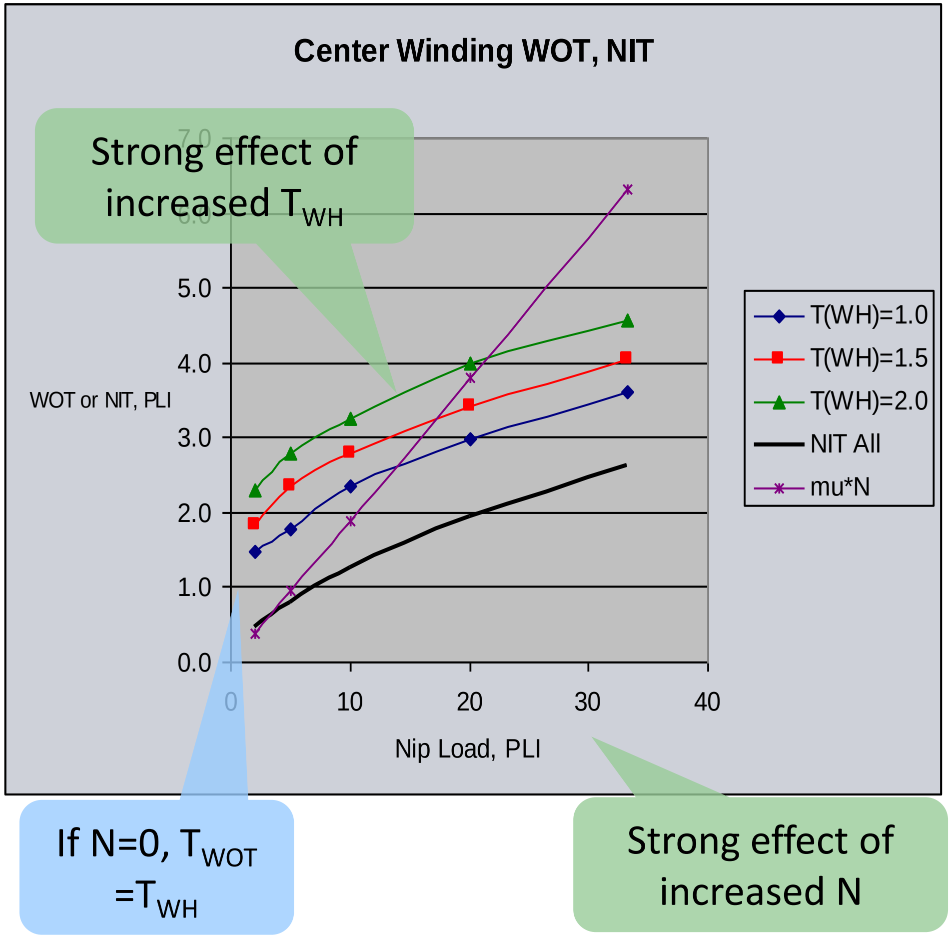


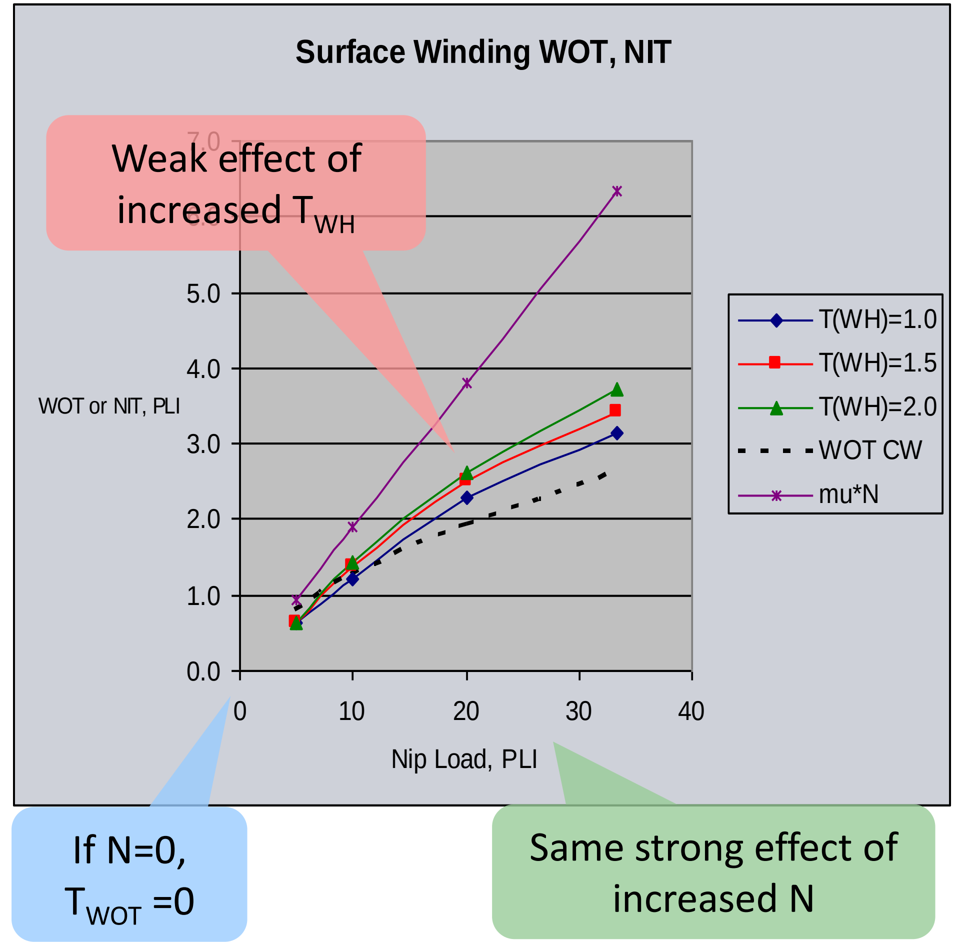
Measuring Center Winding WOT



Measuring Surface Winding WOT

NIT for Center & Surface Winding





J.K.Good, "Modeling Nip Induced Tension in Wound Rolls", Proceedings of the Fourth International Conference on Web Handling, 1997.

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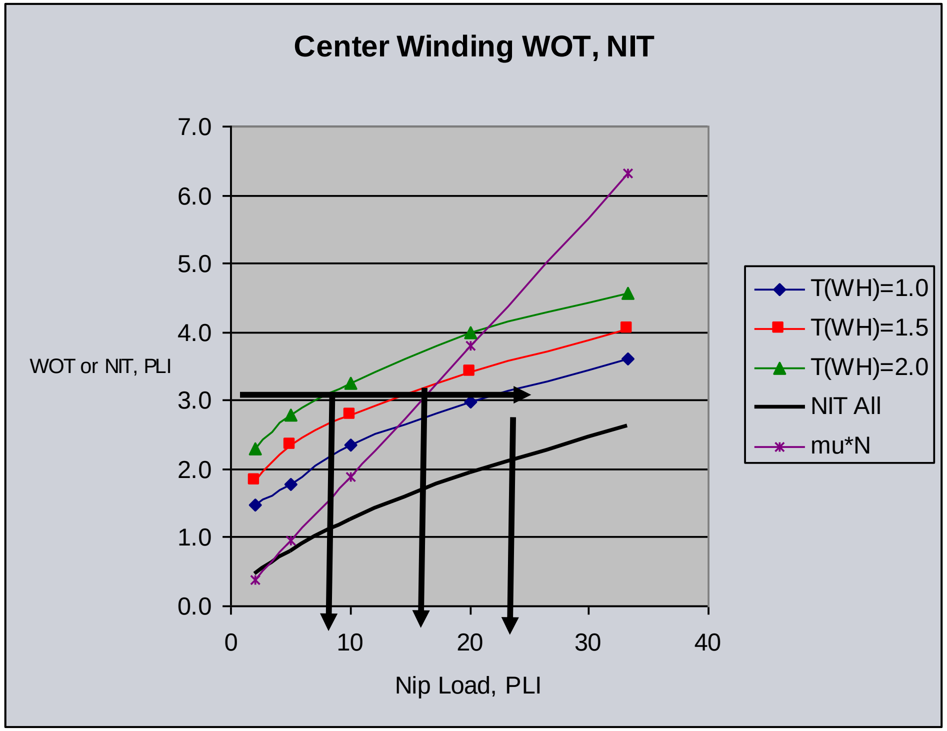
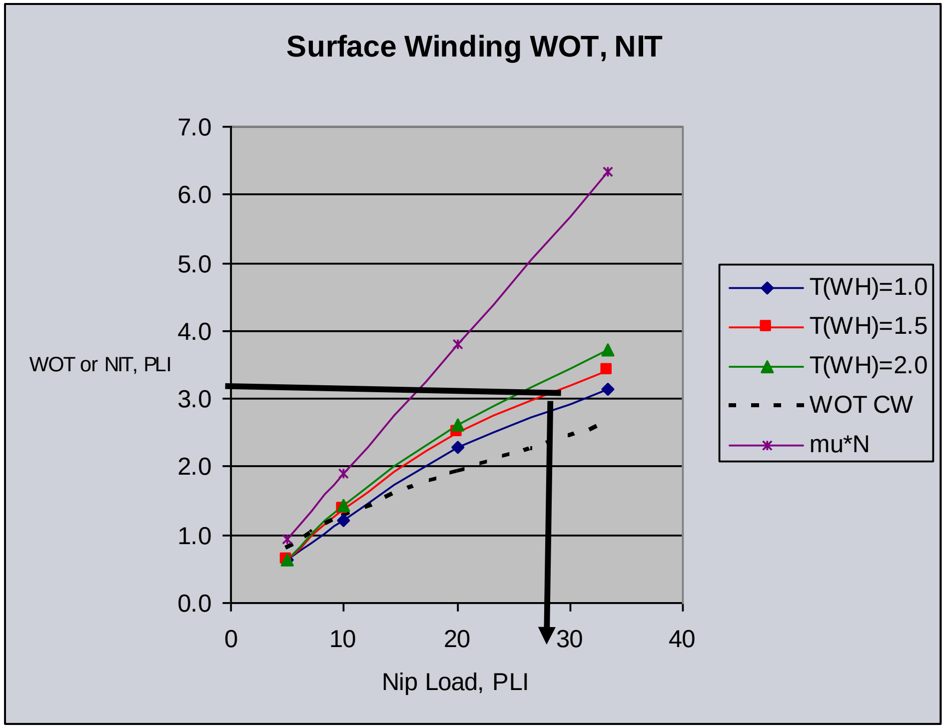
NIT for Center & Surface Winding





J.K.Good, “Modeling Nip Induced Tension in Wound Rolls”,
Proceedings of the Fourth International Conference on Web Handling, 1997.

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Nip Induced Tension and Web-Web COF

We expect nip-induced tension to increase with higher web-web coefficients of friction, and it does from 0.1 to 0.3 COF.

However, beyond 0.3, NIT decreases with higher COF, and become nearly insignificant for high friction webs.

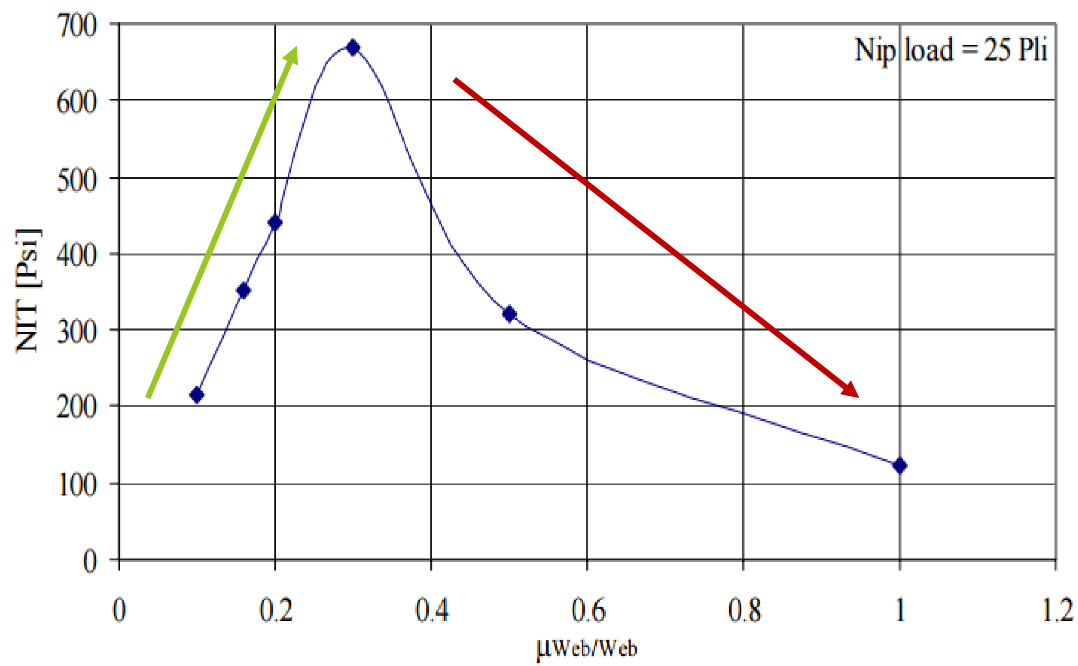
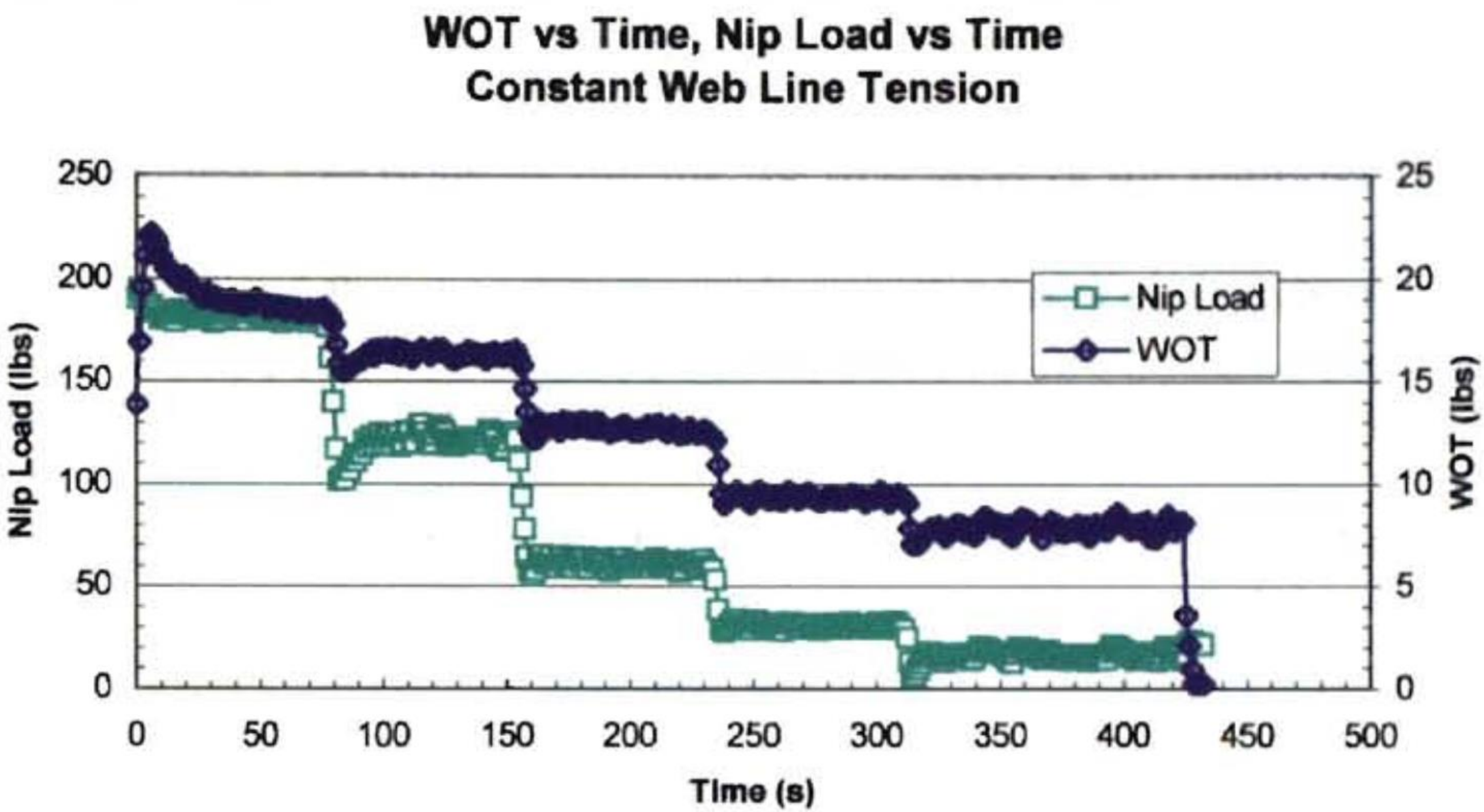


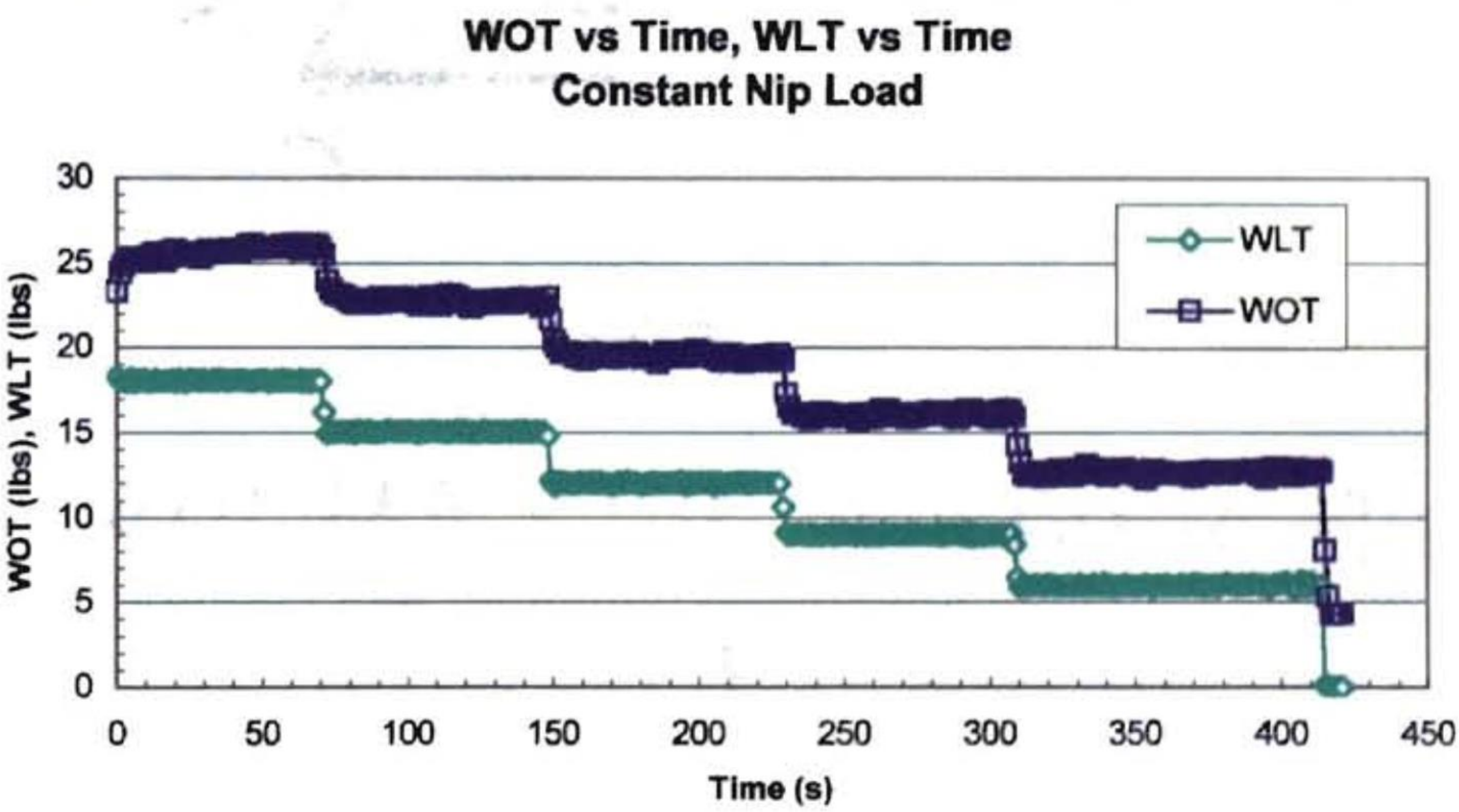
Figure 6.26: Effect of the coefficient of friction between web layers on the NIT.

"The Developmnet of Wound-On-Tension in Webs Wound into Rolls," B. K. Kandadai, 2006

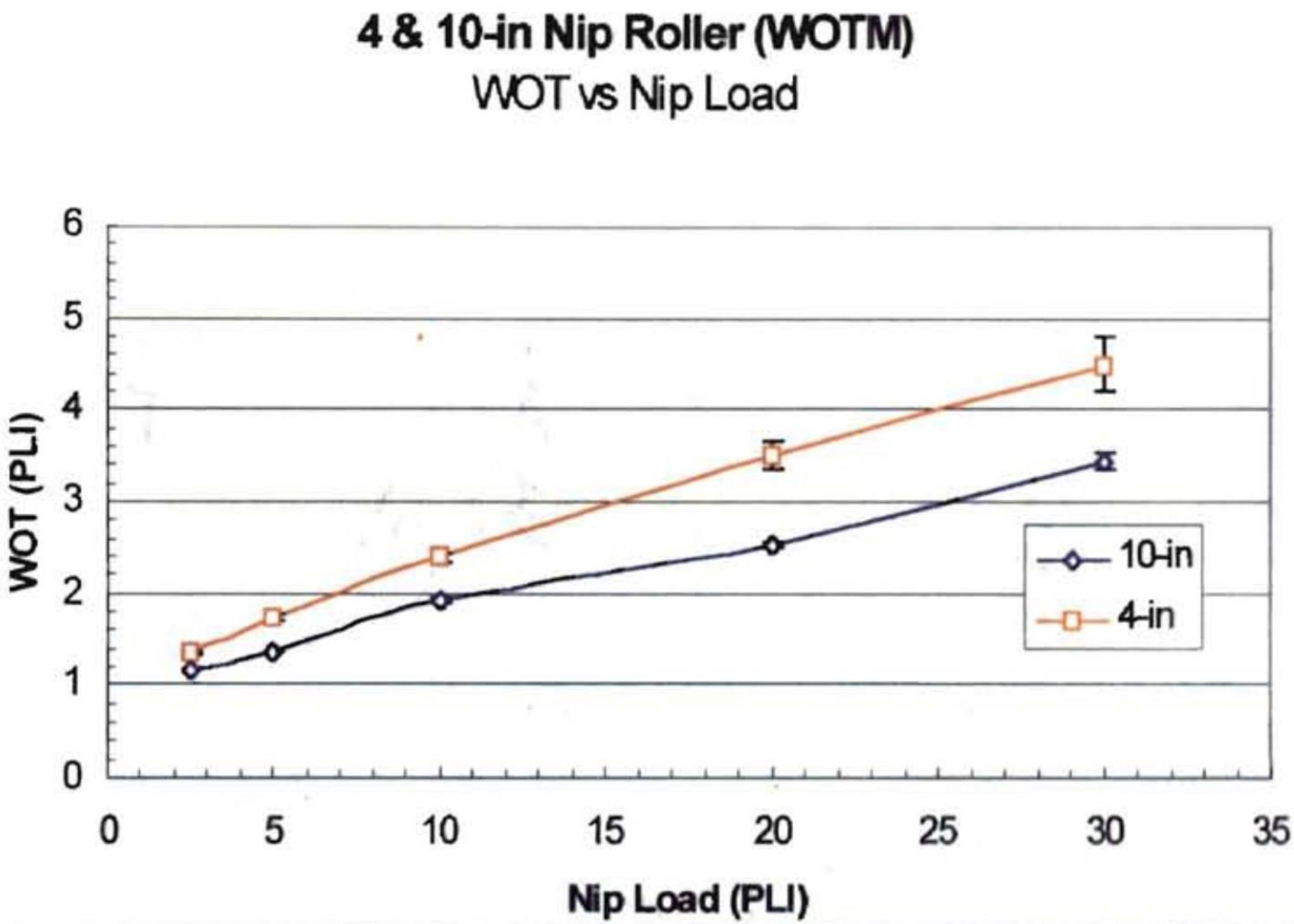
Wound-On Tension vs Nip Load



Wound-On Tension vs. Web Handling Tension



Wound-On Tension and Nip Diameter



Comparison of WOT vs Pre-WOT Wrap Angle

Reducing pre-WOT wrap angle increased measured WOT.

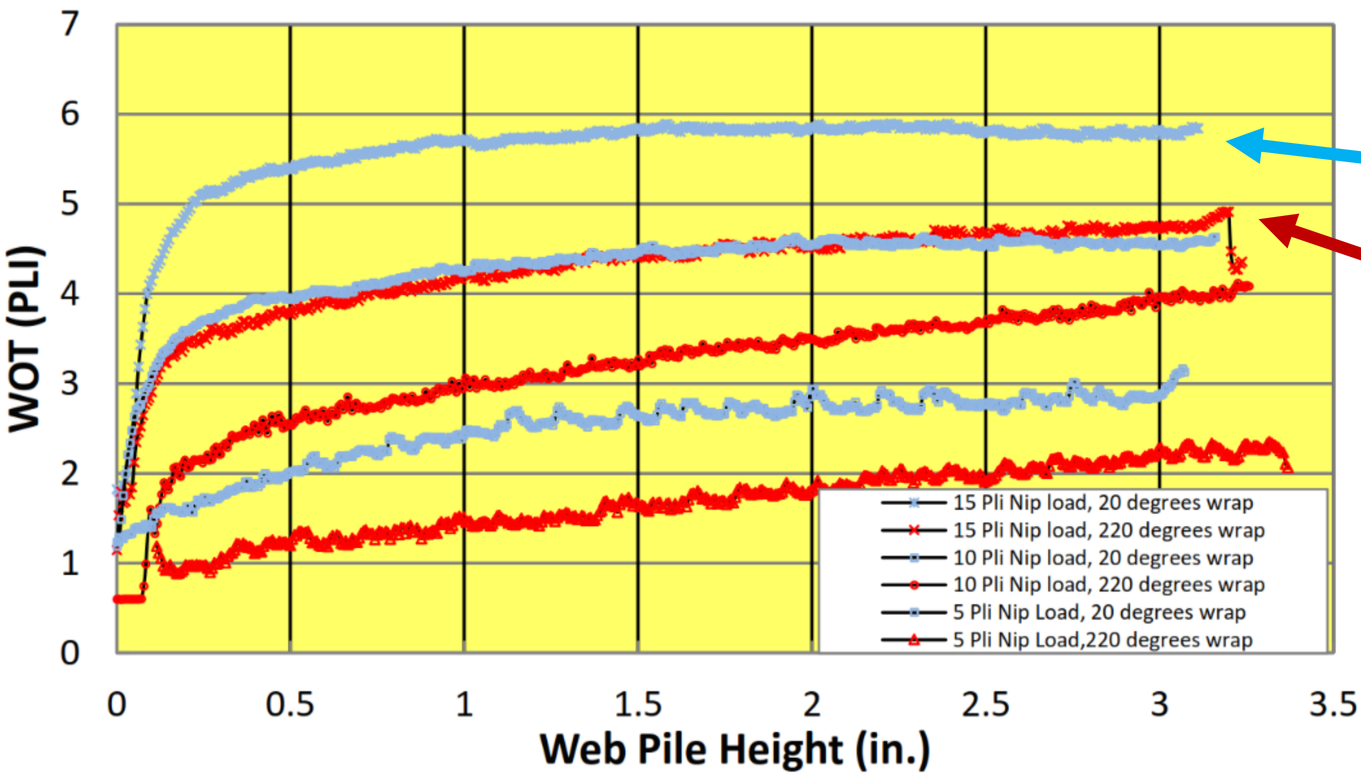
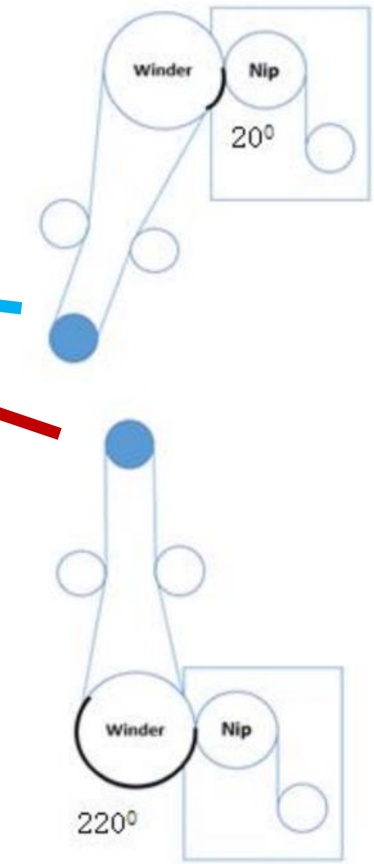


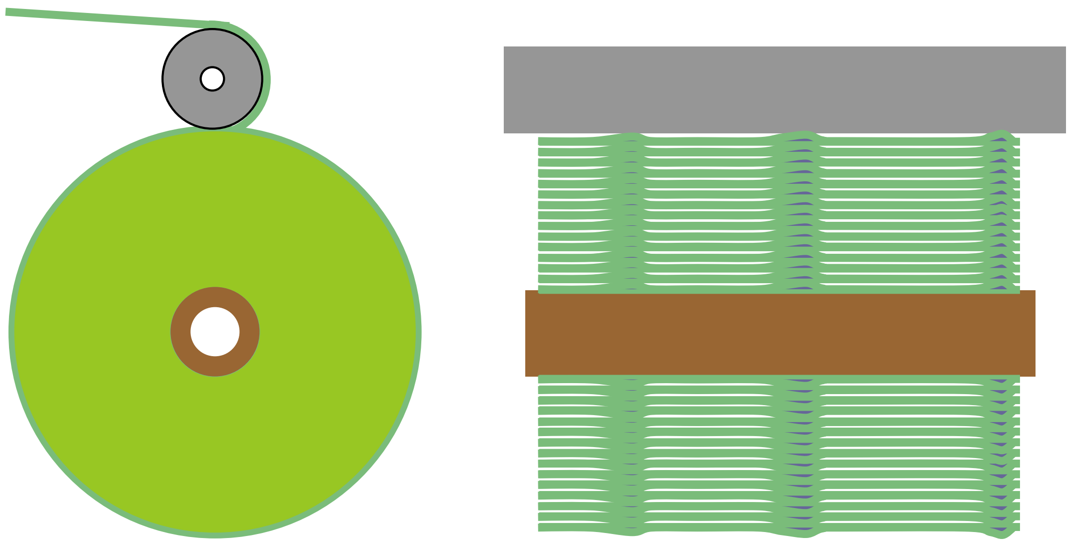
Figure 3-4 Comparison of WOT measurement for different wrap angles



DEVELOPMENT OF NON-INTERFERING WOUND-ON-TENSION MEASUREMENT METHOD, W. Jiang, OK State, 2012

Benefit #5: Promote Cylindricity

A rigidly parallel winding nip promotes cylindricity.
High nip loads on high thickness lanes will compress the stack.
Lower nip loads on low thickness lanes will allow more entrapped air.



Nipping Winding Rolls

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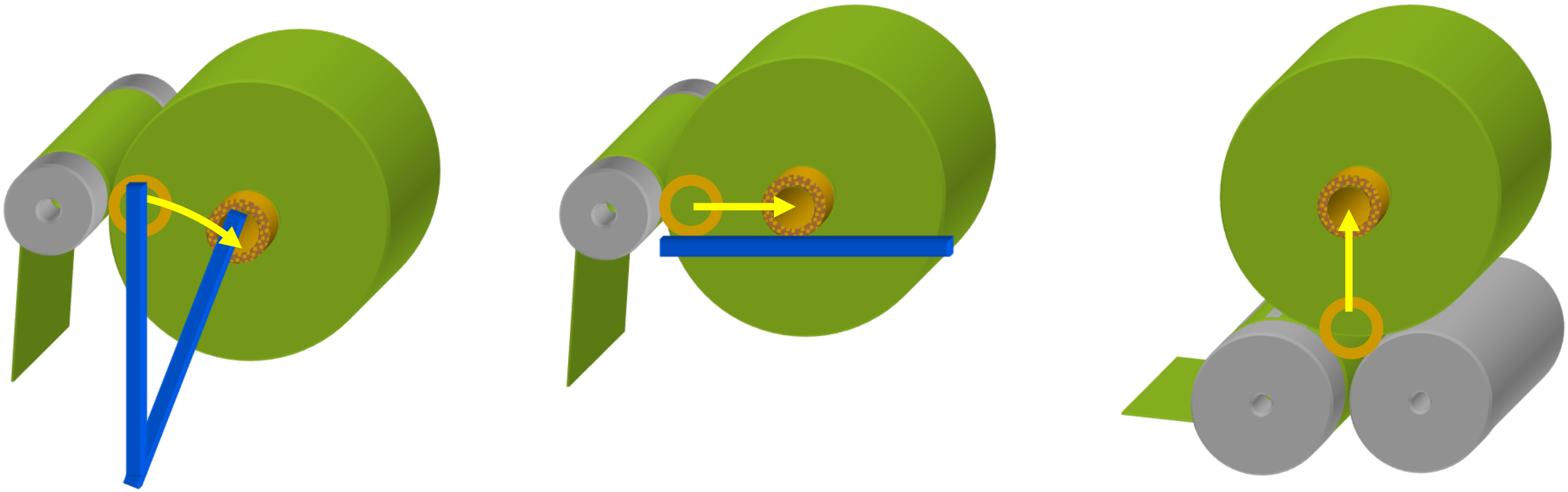
How to follow a winding roll's diameter?

Surface Winders:

Maintaining contact is a simple, natural trait, pushing the core out as the roll grows.

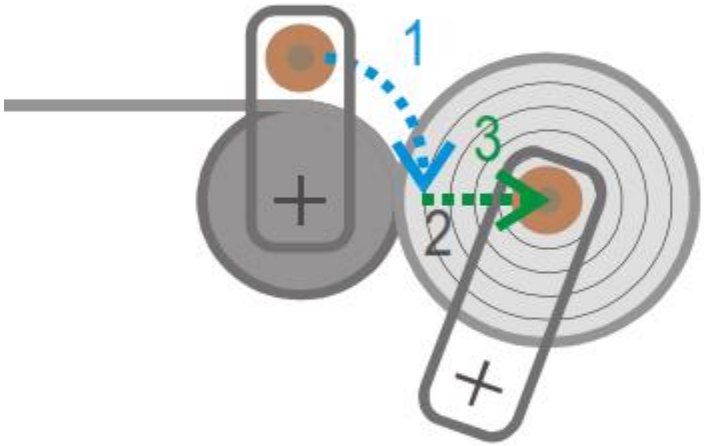


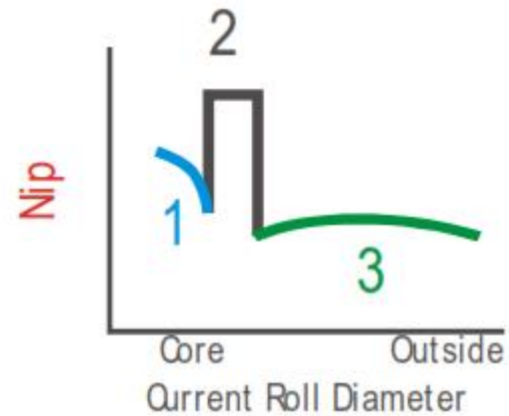
Pivot Arm Surface Winder

Linear Surface Winder

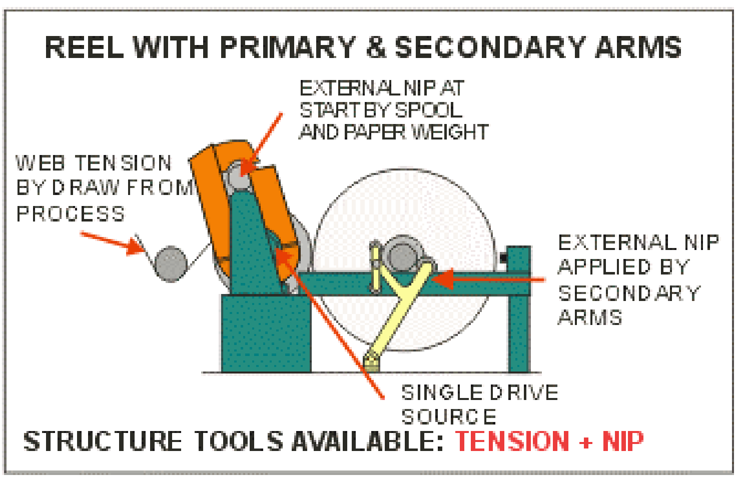
Two Drum Surface Winder

Surface Winder Nip Variations





Graphic courtesy of David Roisum



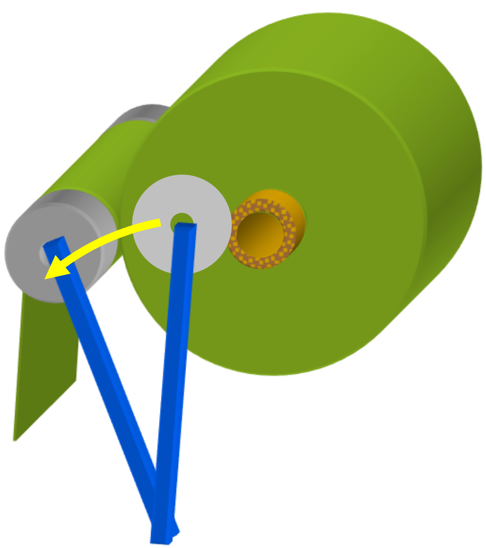
Graphic courtesy of Paper Industry Web

Nipping Winding Rolls

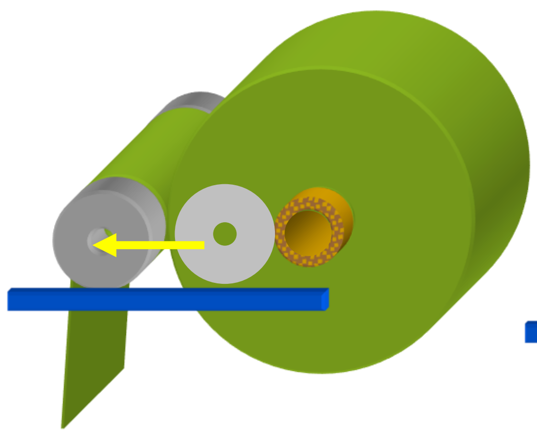
How to follow a winding roll's diameter?

Center Winders:

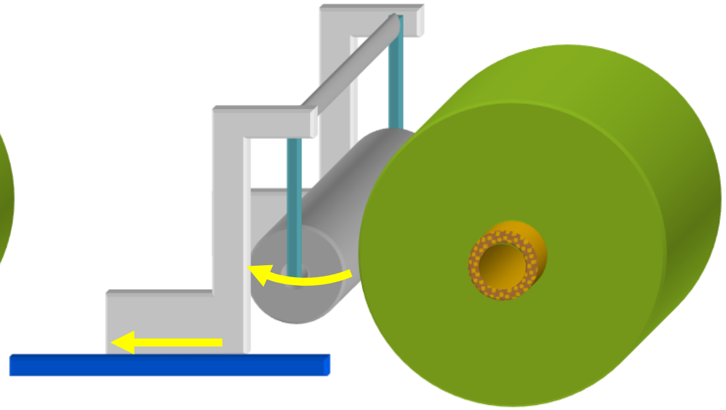
Pivoting arms are simple, but need geometry compensation to maintain constant force. Linear assemblies naturally maintain constant force.



Pivot Arm
Nip Roller



Linear
Nip Roller



Growing roll pushes back nip arm, triggering carriage motion

Linear Nip
Carriage

Hard or Soft Winding Nip Rollers

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A general rule would be:

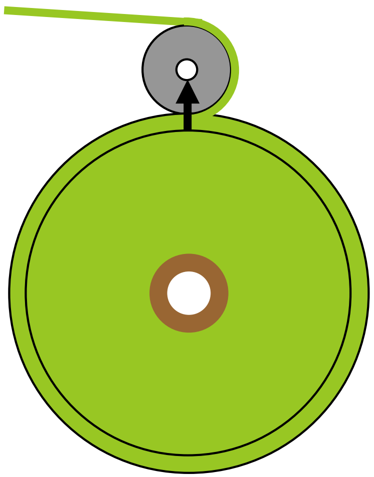
- Hard nipping rollers for soft rolls (nonwovens, soft films, uncoated papers)
- Soft (compliant) nipping rollers for hard rolls (foils, uncoated films, coated papers)

The goal is to have either the winding roll or roller to indent 20 mils (0.5mm) or more under normal nipping loads to reduce nip load variations from misalignment, diameter differences, and other variations.

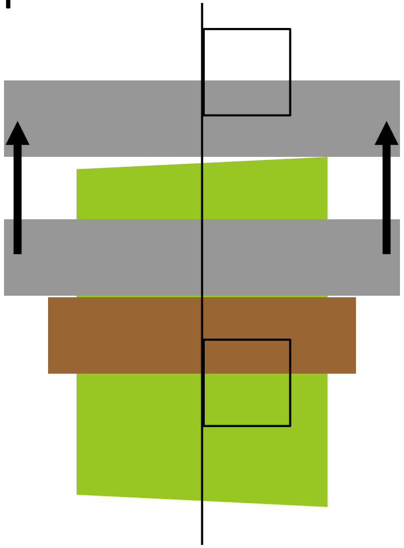
* If you wind both hard and soft rolls, use a soft nip roller.

Nipping Roller Design

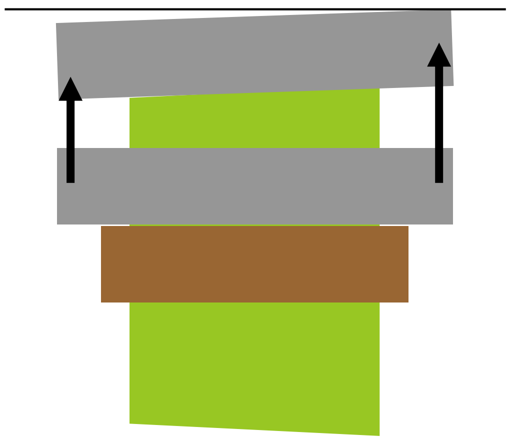
Should a winder nip be held parallel to the core or allowed to float to the roll's shape?



Nip roller held parallel to the core



Nip roller floating to the roll's shape

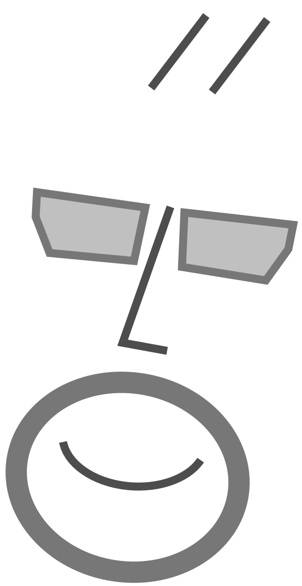


For compressible rolls, holding the nip roller parallel to the core will force the roll into cylindricity.

Avoid wrinkles from nip roller misalignment with near zero or 180 degree wrap.

Winding Equipment Qs

What is the best wrap angle for a winding nip roller?



Best Winding Nip Entry Plan?

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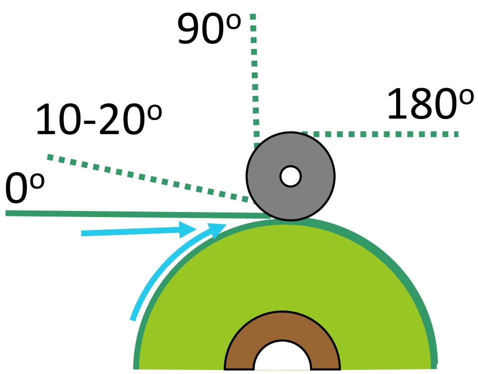
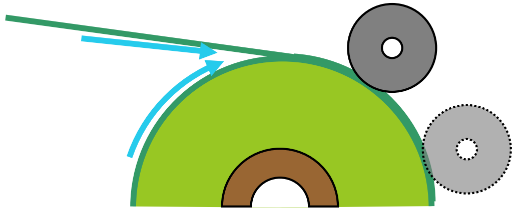
1. Considering Air:
   - Wrap the nip at least 10 degrees to assure the nip controls the contact line.
   - A 'late' or unwrapped nip will still have some air exclusion benefits, except with adhesive coated product that would trap air bubbles upon contact.



2. Considering Nip-Induced Tension

   Nipping 'late' may have reduced nip-induced tensioning.



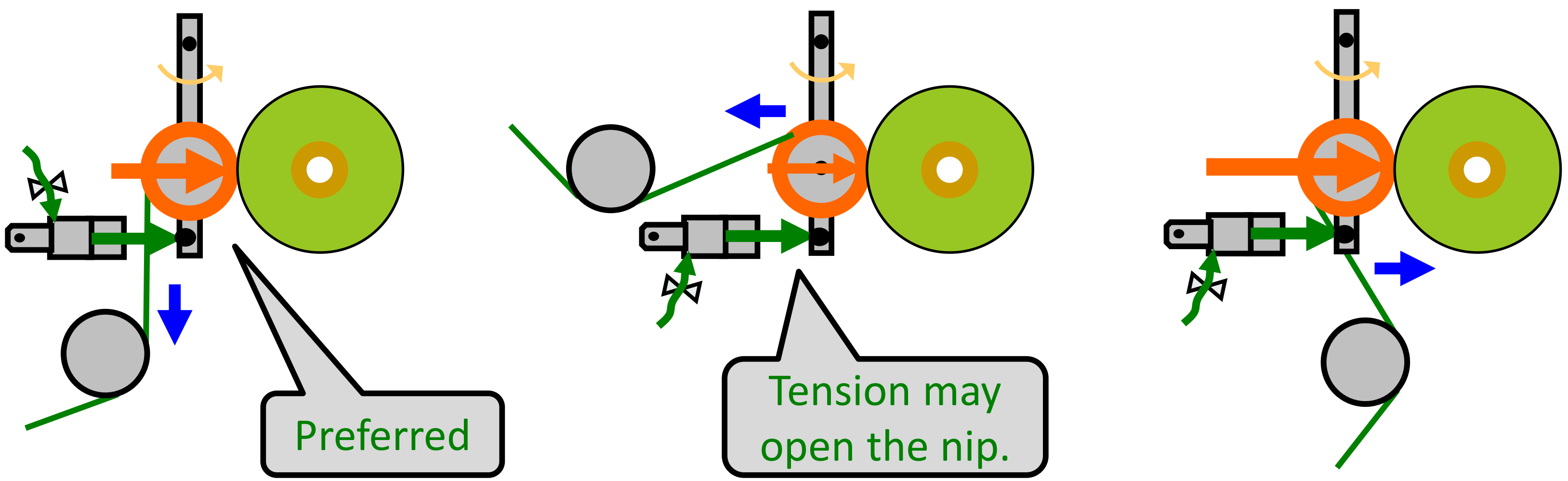
Nips and Tension: Neutral, Subtractive, or Additive

If the winding nip roller is wrapped with anything other than 180-degrees, the web tension will subtract from (or add to) the air cylinder and roll weight forces, reducing the effective nip load.

Tension Neutral:
Tension is perpendicular to nipping direction

Tension Subtracts from Nip:
High tension force may pull open a nip with low loads.

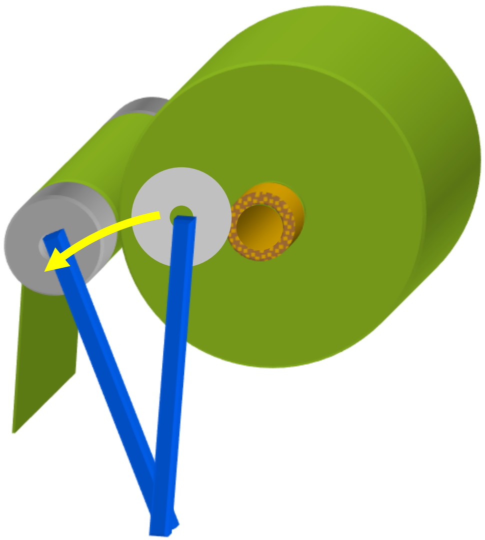
Tension Adds to Nip:
Tension adds to force closing the nip.



Best Winding Nip Entry Plan?

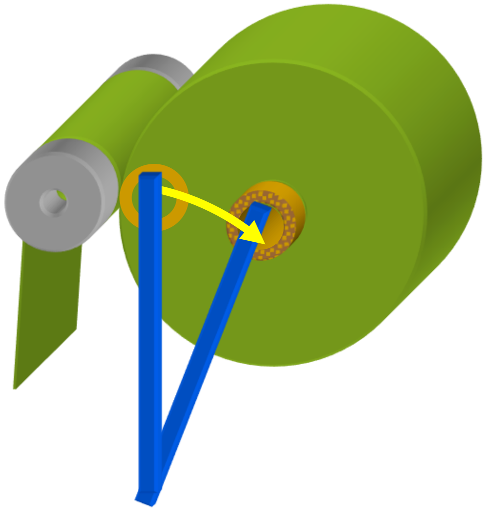
3. Considering Nip Load Independent of Tension

If the core is fixed and the nip roller moves to create load (common in most center winders):



Nipped rollers commonly have wrap angle near zero or 180-degrees to have tension perpendicular to nip load. This prevents high tension from reducing the load or lifting the nip roller.

If the core and winding roll is loaded to a fixed nip roller (common in most surface winders):



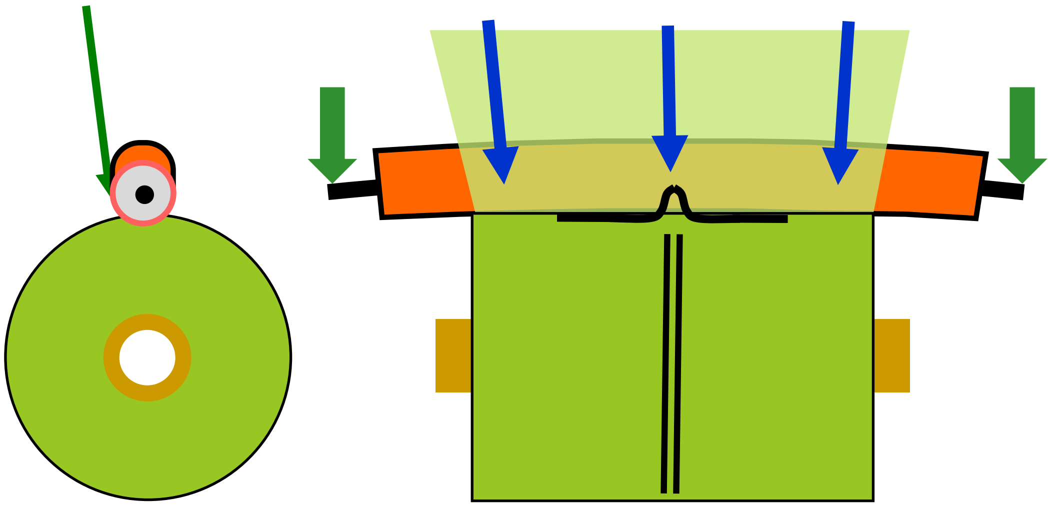
The nip load is independent of tension.

Best Winding Nip Entry Plan?

4. Considering Wrinkling and Alignment

Any pre-nip wrap on the nip roller creates tension stiffening and frictional benefits to prevent wrinkles and lateral shifting.

Avoid wrap angle between 45 and 135 degrees when nip roller deflection may be significant. A 90-degree wrap angle into a deflecting nip will gather the web like a bowed roller pointing upstream.



3. Winding a Roll

- Starting a Roll
- Managing Initial Contact
- Wound-On Tension
- Tension, Torque, and Taper
- Co-Axial Winding

Winding Equipment Qs

What controls in-roll stresses and pressures?

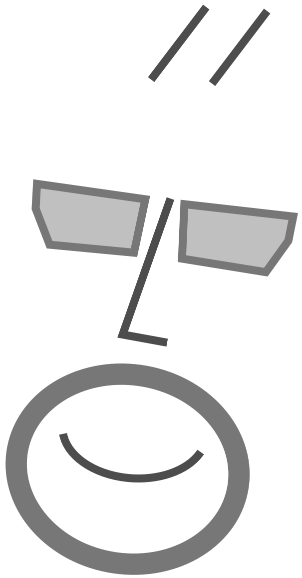
Center Gapped Winding (CGW)

CNW: Center Winding with Nip

S1W: Surface Winding (Single Nip)

S2W: Surface Winding (Two-Drum)

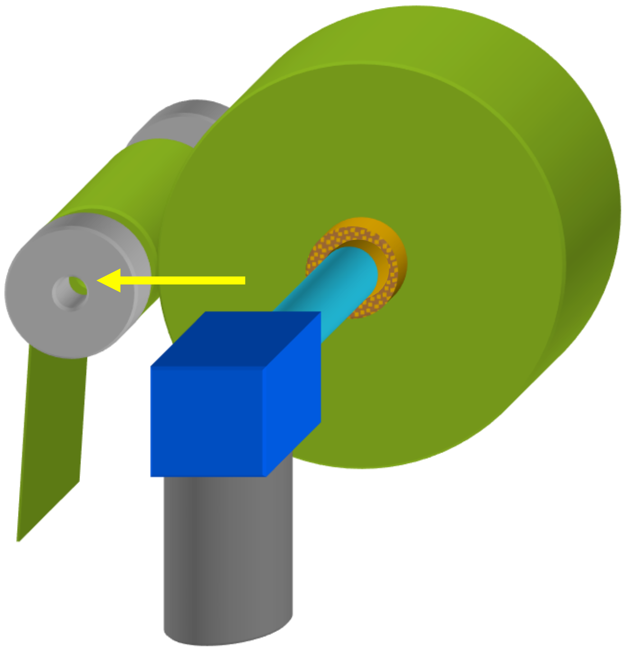
CSW: Center-Surface Winding



Wound-On Tension

Roll tightness is created by a combination of the applied center torque and the surface nip load.

$$T_{WOT} \approx \frac{M_{CW}}{rw} + \mu_K N$$



M_{CW} = Center Wind Torque (in-lbs or N-m)

r = Radius (in or m)

w = Width (in or m)

T_{WOT} = Wound On Tension (lb/in or N/m)

μ_K = Web A-B Kinetic Coefficient of Friction

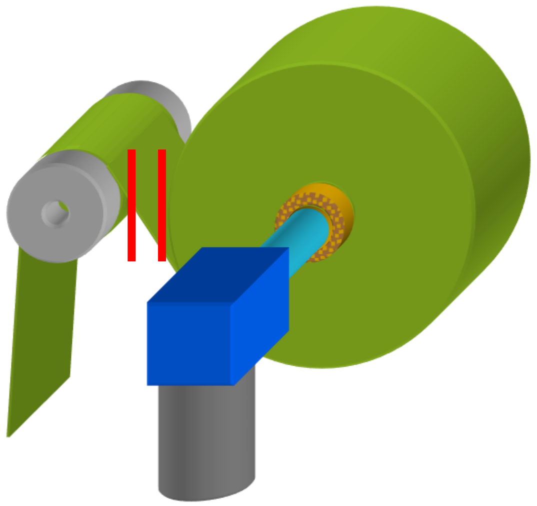
N = Winding Nip Load (lb/in or N/m)

Center Gapped Winding (CGW)

Free span and gapped center winders have no nip-induced tensioning. Roll tightness is created 100% by applied center torque.

.

$$T_{GCW,WOT} = \frac{M_{CW}}{rw}$$



Web handling tension and roll tightness are controlled by the same variable, limiting optimization of each.

$$M_{CW} = T_{WH} rw$$

M_{CW} = Center Wind Torque (in-lbs or N-m)

r = Radius (in or m)

W = Width (in or m)

T_{WOT} = Wound On Tension (lb/in or N/m)

Center with Winding Nip (CNW)

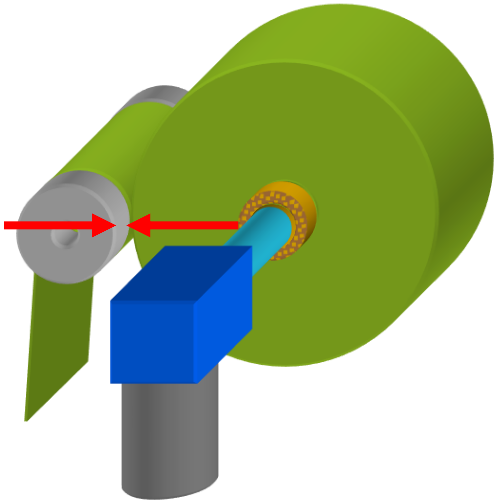
Nipped center winders create roll tightness from the combined effects of center torque and nip load.

$$T_{NCW,WOT} \approx \frac{M_{CW}}{rw} + \mu_K N$$

Rolls can be made tighter or looser without changing web handling tension.

$$M_{CW} = T_{WH} wr$$

Tension is commonly tapered vs increasing radius to limit change in applied center torque.



M_{CW} = Center Wind Torque (in-lbs or N-m)

r = Radius (in or m)

W = Width (in or m)

T_{WOT} = Wound On Tension (lb/in or N/m)

μ_K = Web Coefficient of Friction

N = Winding Nip Load (lb/in or N/m)

T_{WH} = Web Handling Tension (llb/in or N/m)

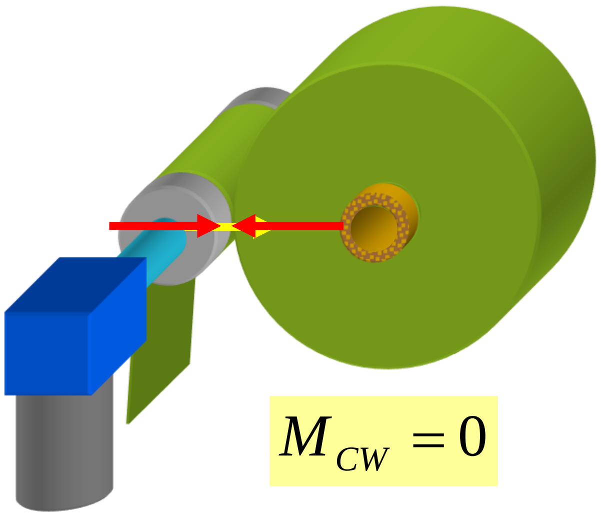
Surface Winding with One Nip (SW1)

Surface winders have no applied center torque, creating nearly all roll tightness from nip-induced tension.

$$T_{SW1,WOT} \approx \mu_K N$$

Upstream tension contributes to roll tightness if the surface roller bond to the web is created by forces other than tension (gravity, adhesion, mechanical grip).

Upstream tension proved insignificant in film and paper winding with N<10PLI (1750N/m)



$$M_{CW} = 0$$

M_{CW} = Center Wind Torque (in-lbs or N-m)

r = Radius (in or m)

W = Width (in or m)

T_{WOT} = Wound On Tension (lb/in or N/m)

μ_K = Web Coefficient of Friction

N = Winding Nip Load (lb/in or N/m)

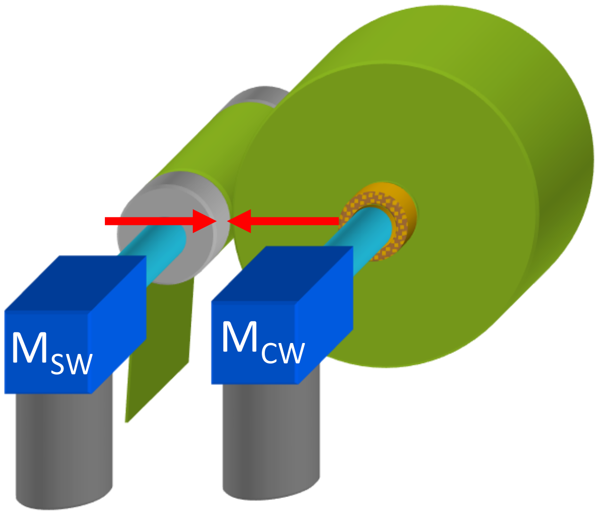
Center-Surface Winding (CSW)

Center-surface winding create rolls of equal tightness to center nipped winding.

$$T_{CSW,WOT} \approx \frac{M_{CW}}{rw} + \mu_K N$$

The surface torque of C-S winders allows web handling tension and applied center torque to be independently optimized with the limits of web to surface nip and wrap traction.

$$T_{WH} = \frac{M_{CW}}{wr_{roll}} \pm \frac{M_{SW}}{wr_{nip}}$$



M_{CW} = Center Drive Torque (in-lbs or N-m)

M_{SW} = Surface Drive Torque (in-lbs or N-m)

r = Radius (in or m)

W = Width (in or m)

T_{WOT} = Wound On Tension (lb/in or N/m)

μ_K = Web Coefficient of Friction

N = Winding Nip Load (lb/in or N/m)

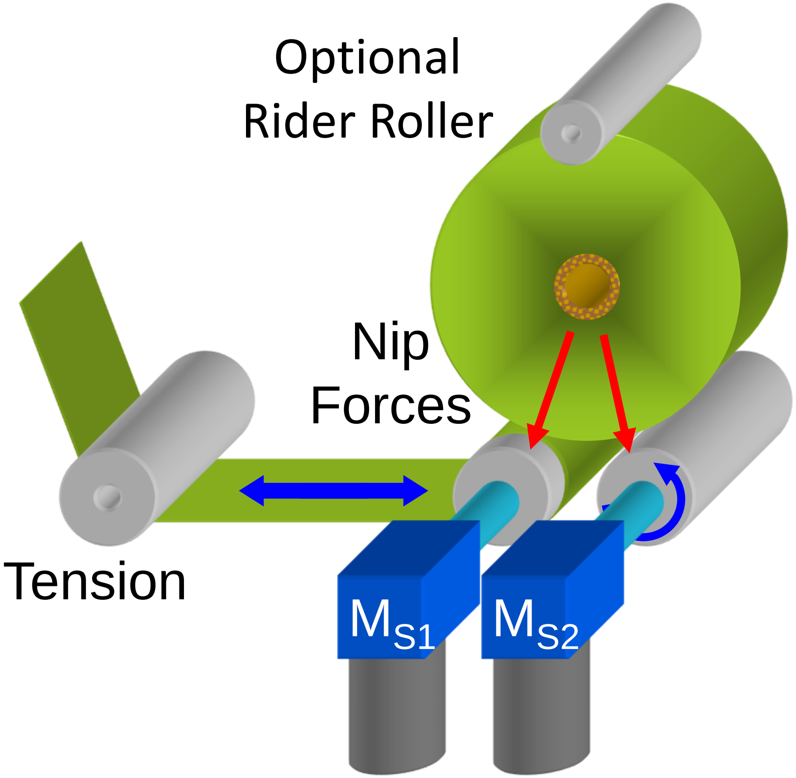
T_{WH} = Web Handling Tension (llb/in or N/m)

Two-Drum Surface Winding (SW2)

Two drum surface winders have many winding variables. Roll tightness is dependent on web handling tension, nip load, and second roller torque.

$$T_{2SW,WOT} = T_{WH,eff} + \frac{M_{S2}}{r_2 w} + \mu N_{max}$$

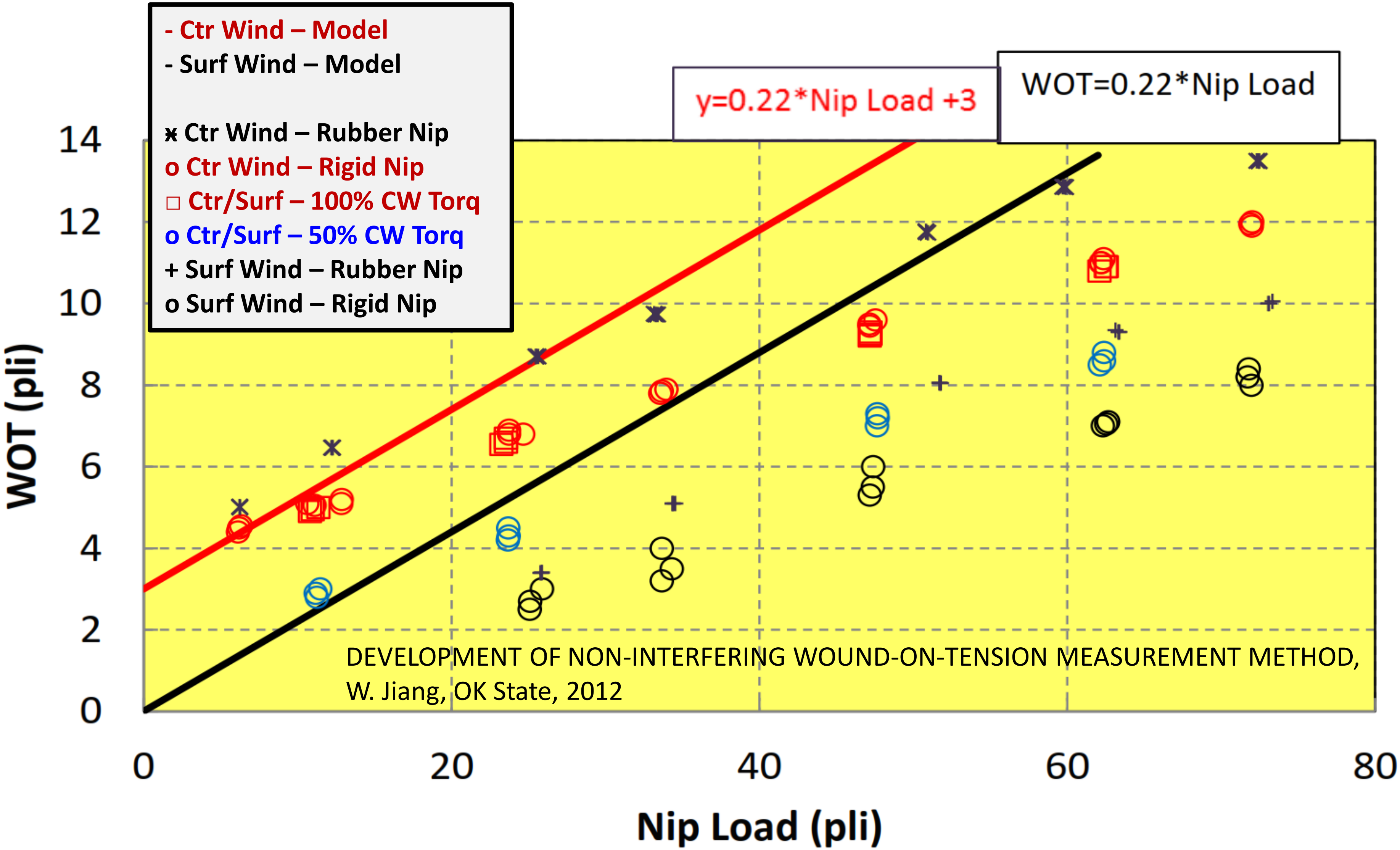
1) A portion of web handling tension dependent on slippage in the contact zone.
2) 100% of assisted torque.
3) Nip-induced tension from the greatest nip load from either rider roller or roll weight.



M_{S1} = First Drum Torque (in-lbs or N-m)

M_{S2} = Second Drum Torque (in-lbs or N-m)

Wound-On Tension vs. Winding Method



Wound-On Tension vs. Winding Method

In order from tightest to loosest winding method:

Rubber nip induces more tension than rigid nip (both center and surface winding).

ж Ctr Wind – Rubber Nip, 100% CW Torque

o Ctr Wind – Rigid Nip, 100% CW Torque

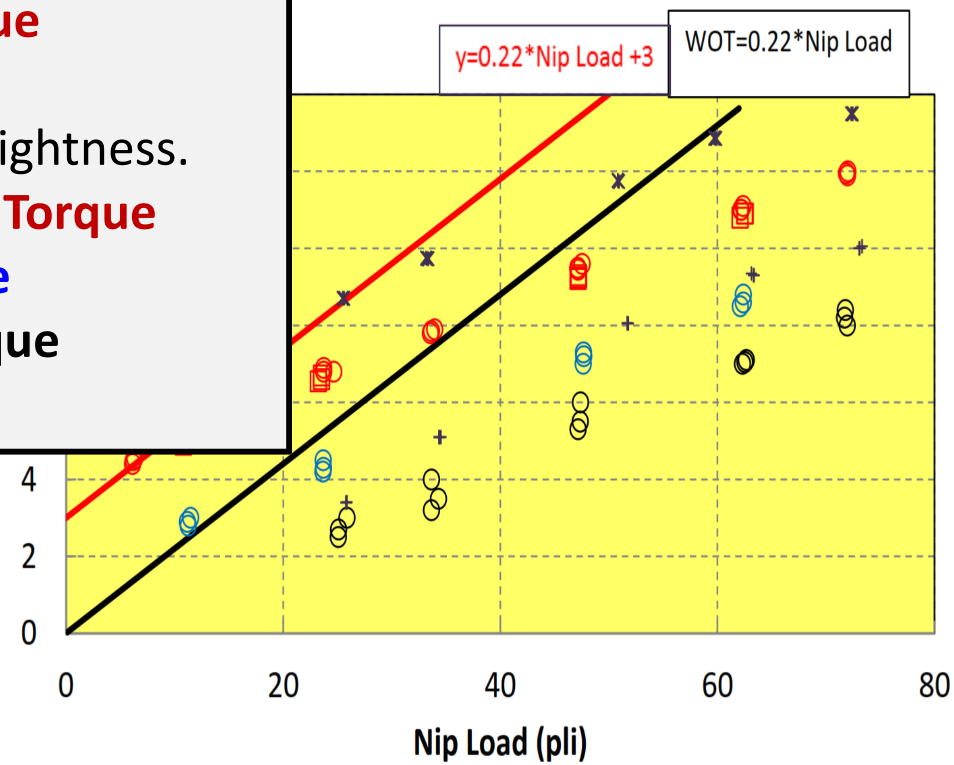
Decreasing center torque reduces roll tightness.

□ Ctr/Surf – 100% Ctr. / 0% Surf. Wind Torque

o Ctr/Surf – 50% Ctr. /50% Surf. Torque

+ Surf Wind – Rubber Nip, 0% CW Torque

o Surf Wind – Rigid Nip



3. Winding a Roll

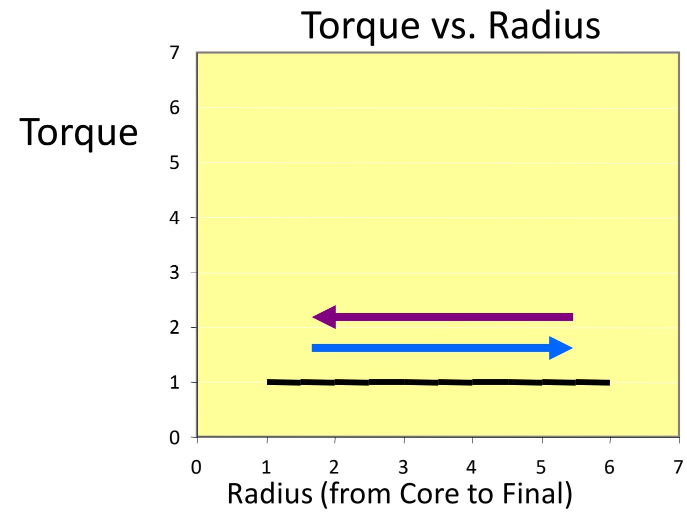
- Starting a Roll
- Managing Initial Contact
- Tension and Torque
- Winder Tensioning
- Co-Axial Winding

Torque and Tension vs Radius

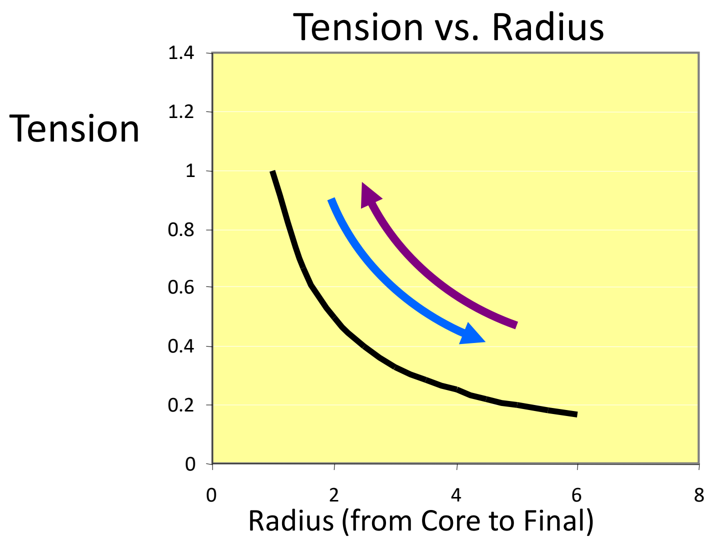
If center torque is held constant, tension will change inversely with roll radius.

Constant center torque creates decreasing tension at winding.

Constant center torque creates increasing tension at unwinding.

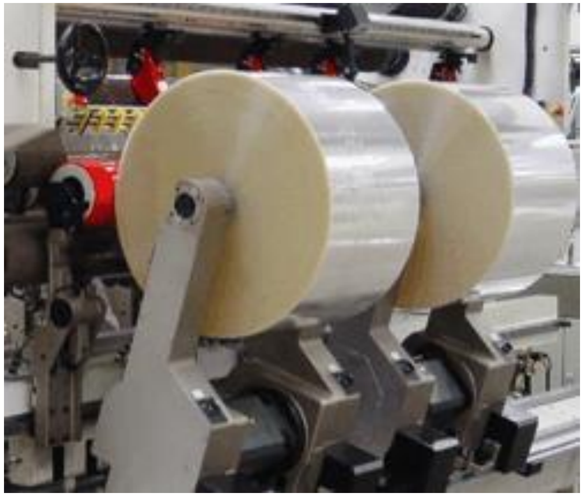


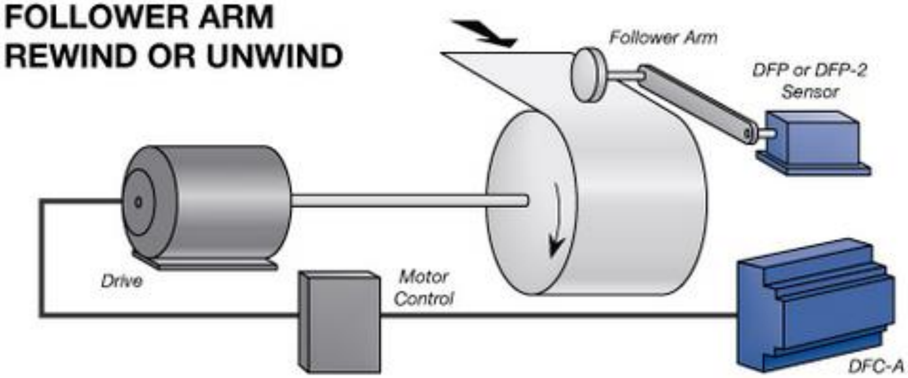
$$M = Twr$$

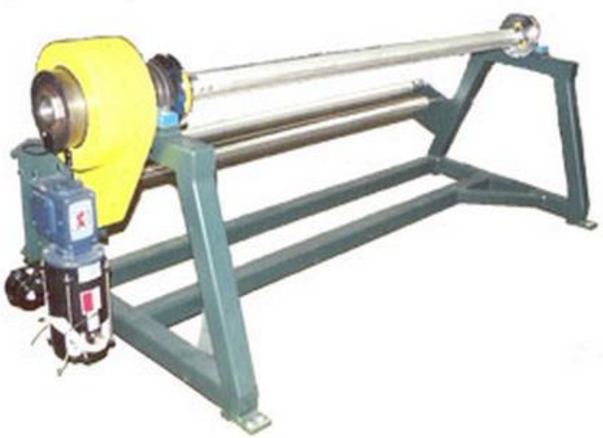


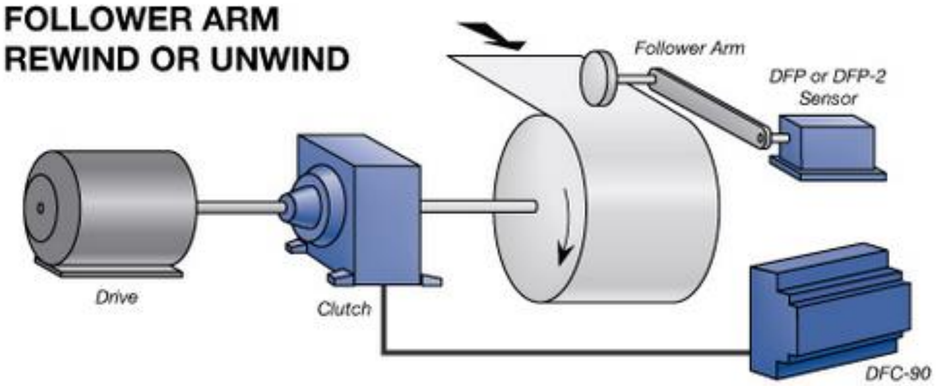
$$T = \frac{M}{rw}$$

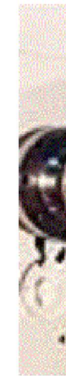
Open-Loop Torque Controlled Winders











Images courtesy of Magpowr, US Webcon, Atlas

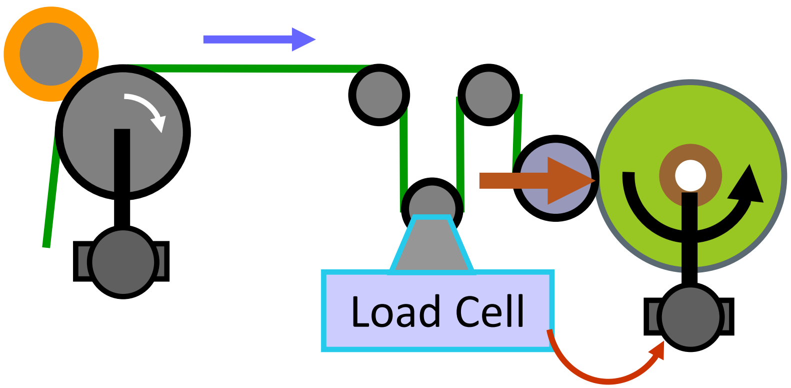
Torque Control Winding with Differential Shafts

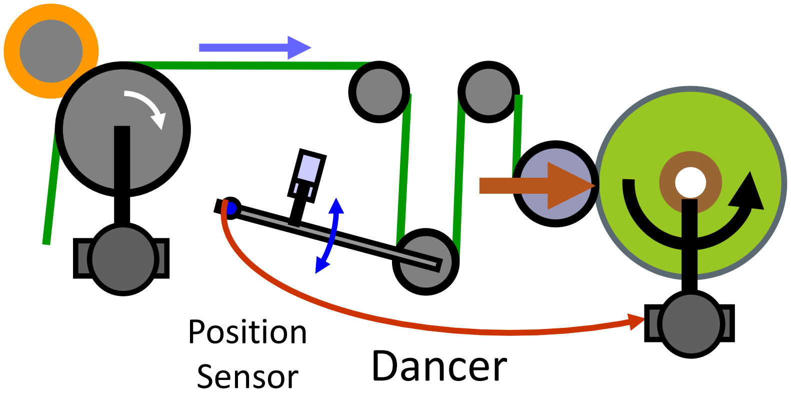
In differential winding, each roll has local applied torque to allow each roll to turn at speed required for roll-to-roll variations in length or diameter.

Images courtesy of HCI Converting Ltd.

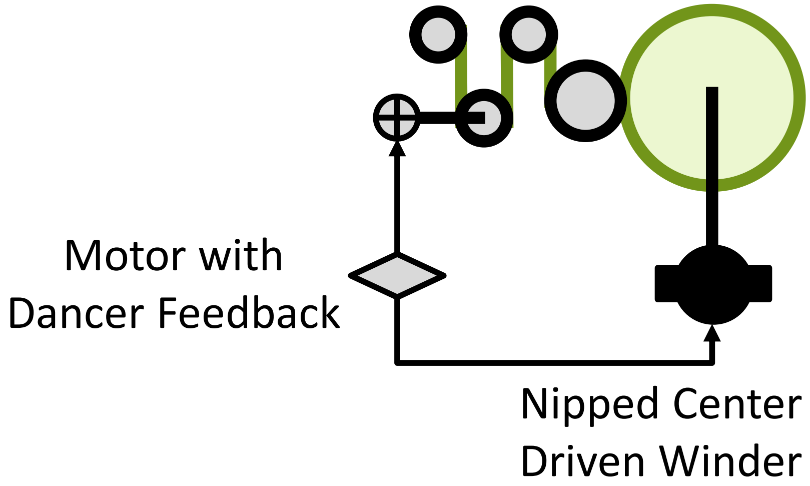
Closed-Loop Control of Winding Tension

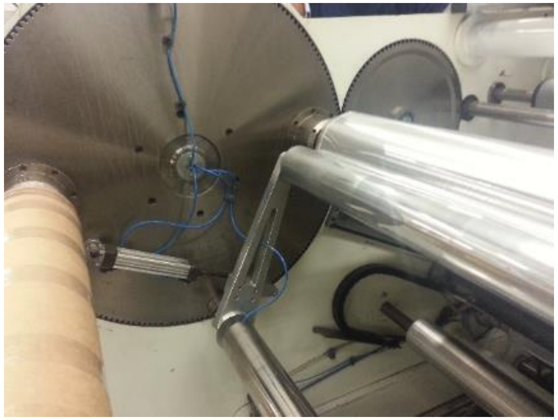
Closed-loop tension control with either load cell and dancer roller feedback is common center, surface, and center-surface winding.



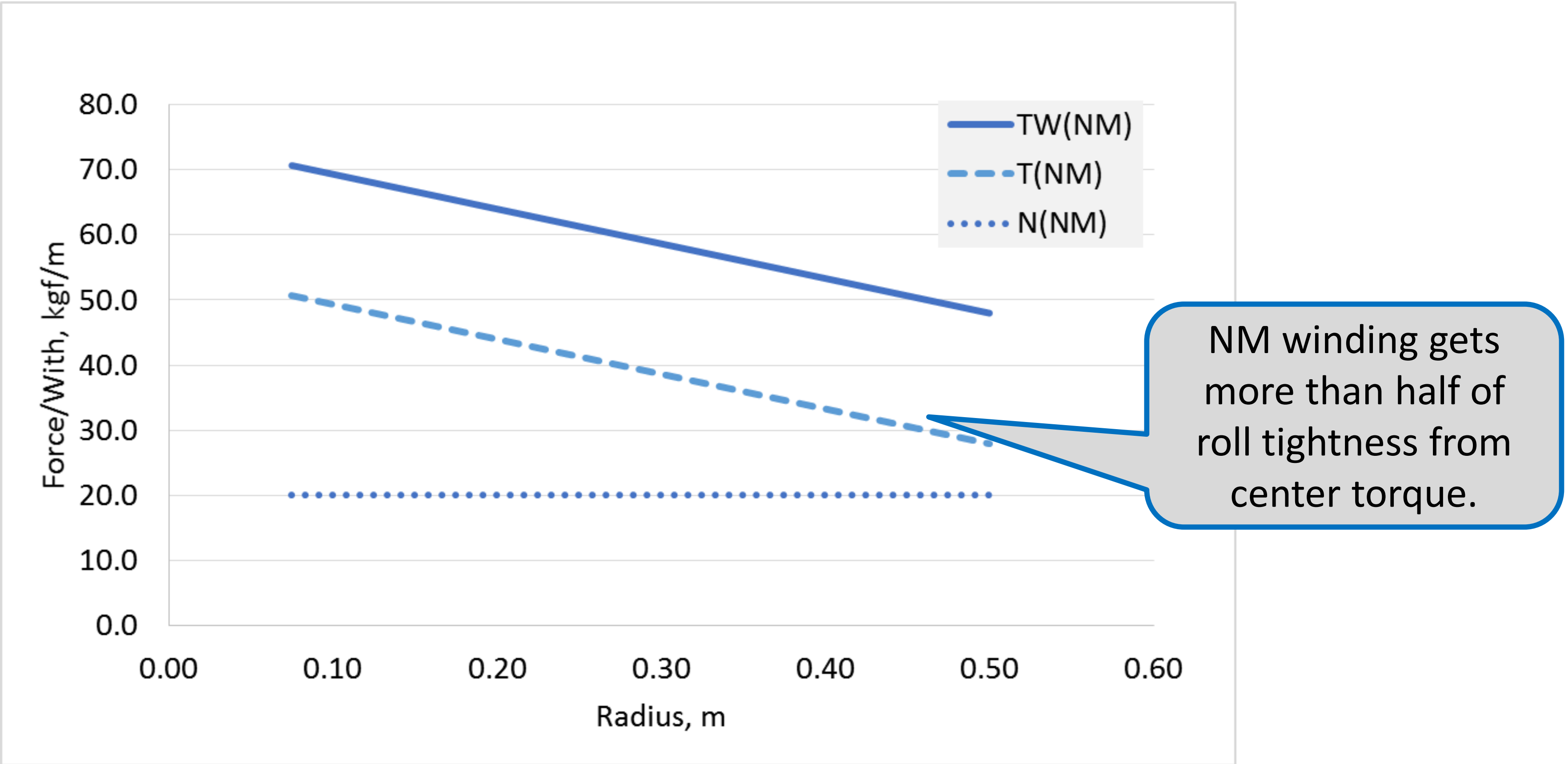


Normechanica Winding Control





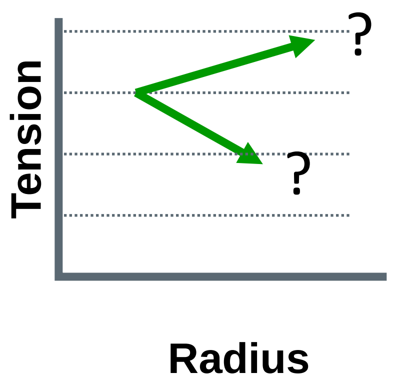
Winding Tension of Example (NM) Winder



Taper Tension

Taper tension is an automated system to vary winding tension as a function of roll radius.

The taper tension function is defined in a winder's controls programming.



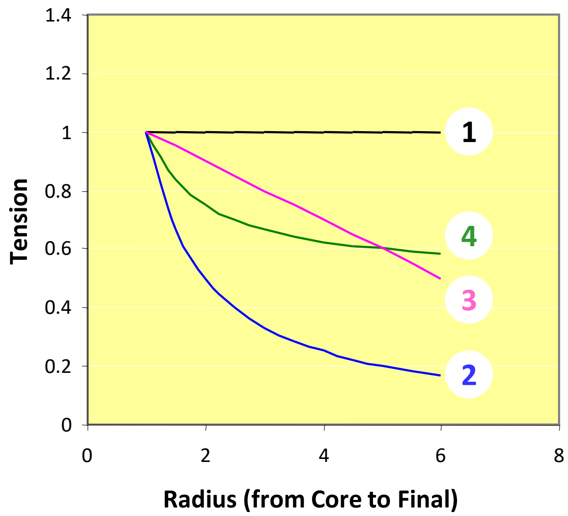
The two most common tension taper functions are *linear* and *hyperbolic*.

Torque-based taper is a linear change in **torque** vs. radius, commonly found on torque controlled winders.

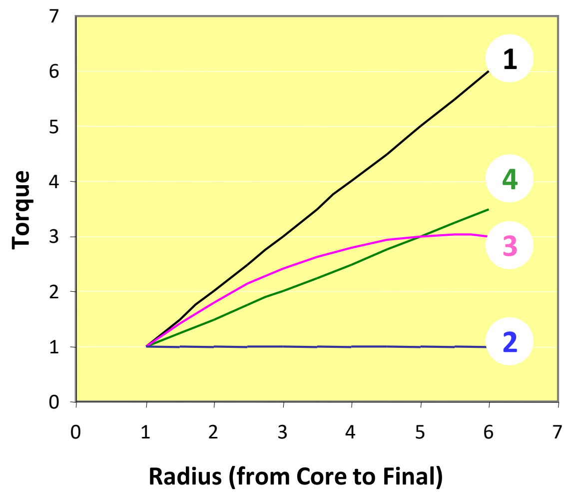
Tension-based taper is a linear change of **tension** vs. radius, commonly found on closed-loop tension control winders.

Tapering Tension

Tension vs. Radius



Torque vs. Radius



Tension Tapering is defined as varying tension vs. radius, usually along a linear tension or torque function.

<u>Curve Legend</u>

1 Constant Tension

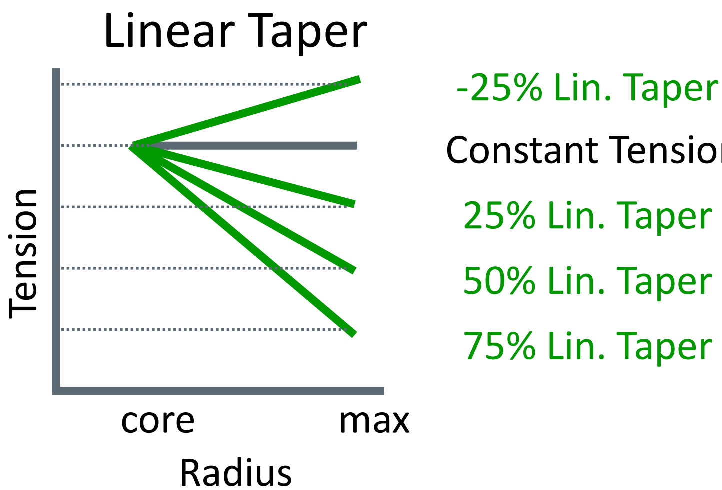
2 Constant Torque

3 50% Tension-based Taper

4 50% Torque-based Taper

Tension-Based Taper

Tension increases or decreases linearly with radius.



$$T_X = T_C - T_C\left(\frac{P_{LIN}}{100}\right)\left(\frac{x - r_C}{r_{MAX} - r_C}\right)$$

If $P_{LIN}=0$ $$T_X = T_C$$

If $P_{LIN}=100$ $$T_X = T_C - T_C\left(\frac{x - r_C}{r_{MAX} - r_C}\right)$$

T_X = Tension at radius x

T_C = Tension at the core

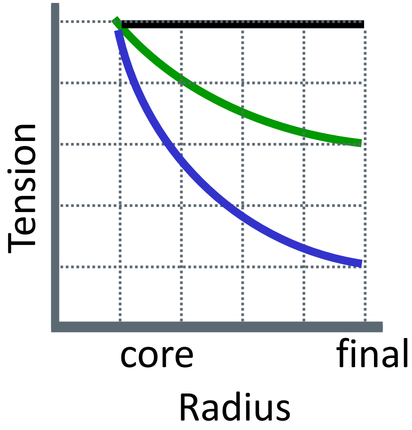
P_{LIN} = Percent taper (what percent of starting tension is lost during roll buildup)

X = Varying radius

R_C = Core radius

R_F = Final radius (Sometimes maximum radius is used here.)

Torque-Based Taper Tension



If P_{HYP}=100 $M_X = M_C$ Torque is constant.

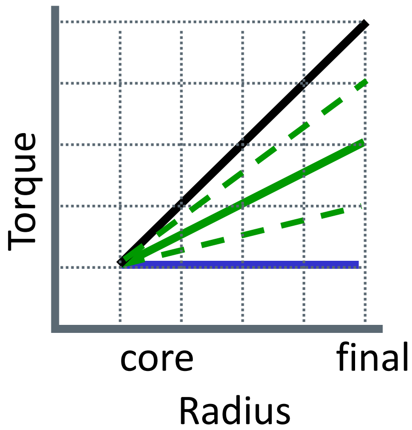
If P_{LIN}=0 $M_X = T_C w x$ Torque increasing directly with radius.

$$M_X = T_C w r_X + T_C w \left(\frac{P_{HYP}}{100}\right)(x - r_C)$$

Torque at r = x

Constant Torque

Torque Decrease as Roll Builds



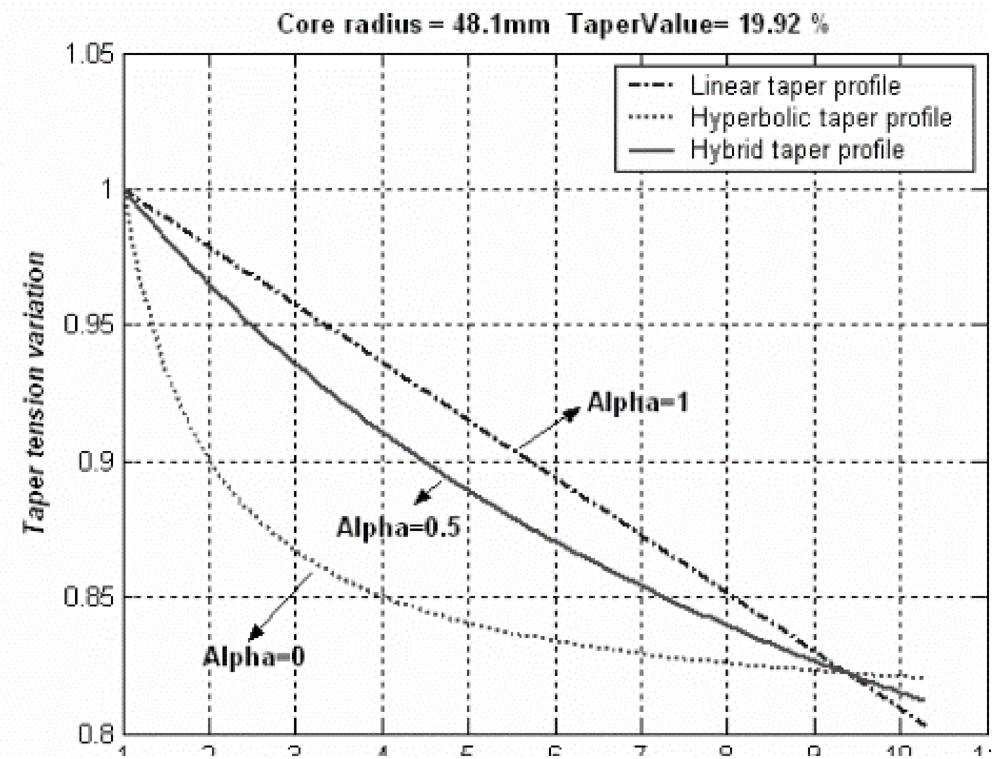
$$T_X = T_C + T_C \left(\frac{P_{HYP}}{100}\right)\left(1 - \frac{r_C}{x}\right)$$

Tension at r = x

Core Tension

Tension Decrease as Roll Builds

Alternate Taper Tension Profiles



$$\sigma_{linear}(r) = \sigma_0 \left[1 - \left(\frac{taper}{100} \right) \frac{(r-1)}{(R-1)} \right]$$

$$\sigma_{hyperbolic}(r) = \sigma_0 \left[1 - \left(\frac{taper}{100} \right) \left(\frac{r-1}{r} \right) \right]$$

$$\sigma_{hybrid}(r) = \sigma_0 \left[1 - \left(\frac{taper}{100} \right) \frac{(r-1)}{(r + \alpha \cdot (R - r - 1))} \right]$$

σ_0 =	Core tensile stress (psi or kPa)
$\sigma_x(r)$ =	Winding tensile stress vs roll radius (psi or kPa)
Taper =	Percent decrease in tension from core radius to final radius
r =	Core radius (in or m)
R =	Final roll radius (in or m)
α =	Hybrid taper shape factor (α = 1 is linear, α = 0 is hyperbolic)

Effect of taper tension profile on the telescoping in a winding process of high speed roll to roll printing systems †

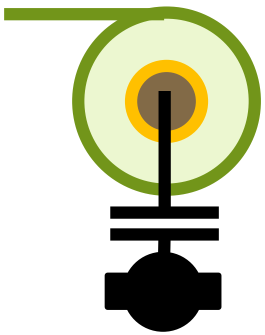
Changwoo Lee[1], Hyunkyoo Kang[2], Hojoon Kim[2] and Keehyun Shin[2,*]

[1]*Flexible Display Roll to Roll Research Center, Konkuk University, 1 Hwayang-Dong, Gwangjin-Gu, Seoul 143-701, Korea*
[2]*Department of Mechanical and Aerospace Engineering, Konkuk University, 1 Hwayang-Dong, Gwangjin-Gu, Seoul 143-701, Korea*

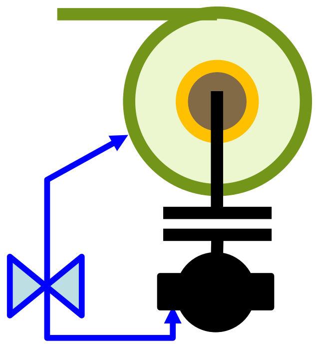
Benefits of Taper Tension

- Reduce winder torque range.
- Build tight initial layers to reinforce the core as a foundation for continued winding.
- Increase pressure and torque capacity near the core.
- Reduces maximum core pressure in low modulus film roll, reducing high pressure defects (starring, core crushing, impressions, blocking).

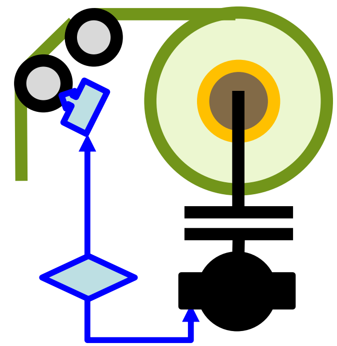
Winding Options: Center Driven



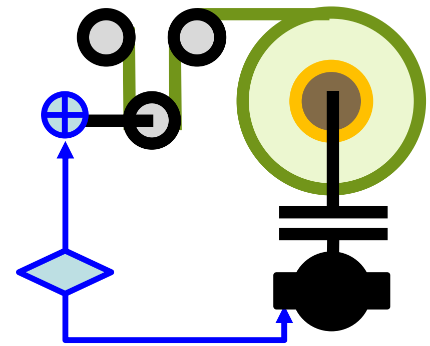
Open-Loop Clutch



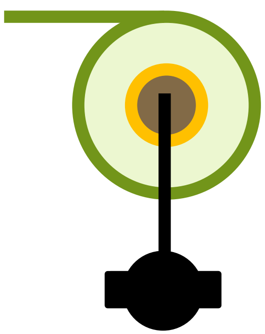
Clutch with Diameter Feedback



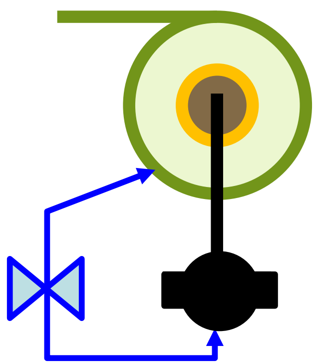
Clutch with Load Cell Feedback



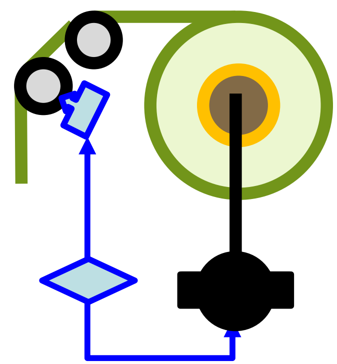
Clutch with Dancer Feedback



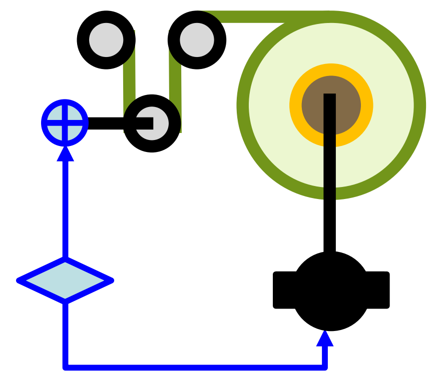
Open-Loop Torque Motor



Motor with Diameter Feedback

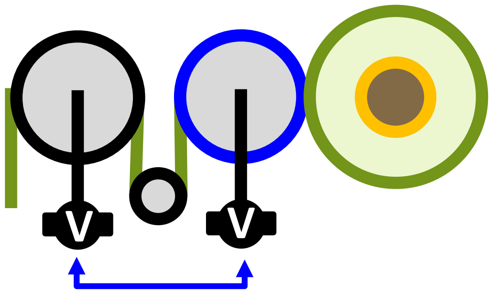


Motor with Load Cell Feedback

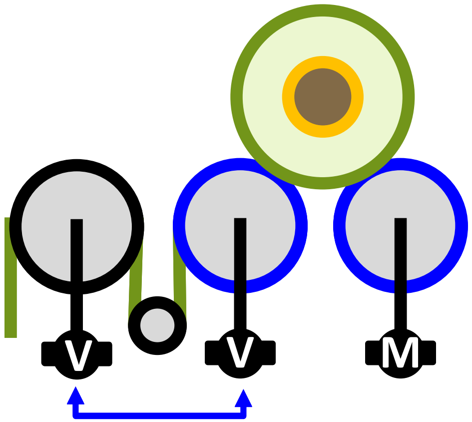


Motor with Dancer Feedback

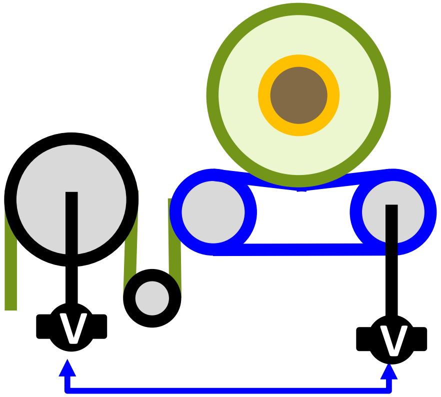
Winding Options: Surface Driven



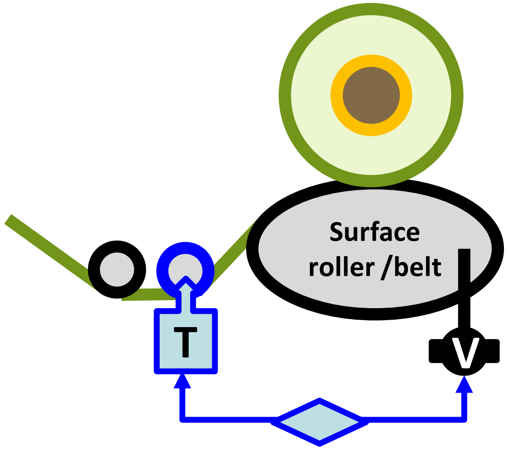
Surface Roller Driven Winder



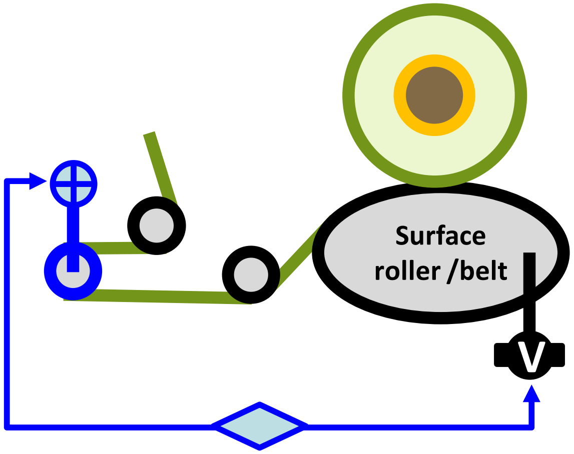
2-Drum Surface Roller Driven Winder



Surface Belt Driven Winder



Motor with Load Cell Feedback



Motor with Dancer Feedback

Winding Options: Surface-Center Driven

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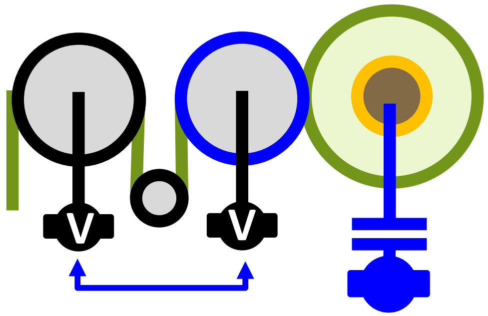
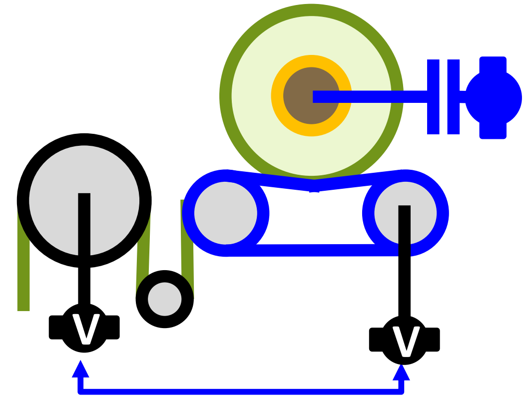
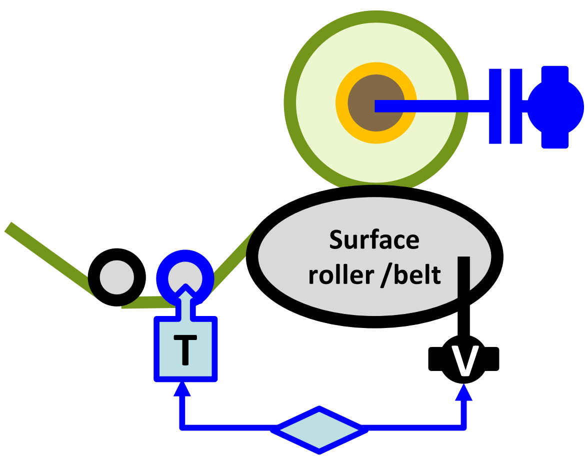
Add torque assist to center of any surface winder.



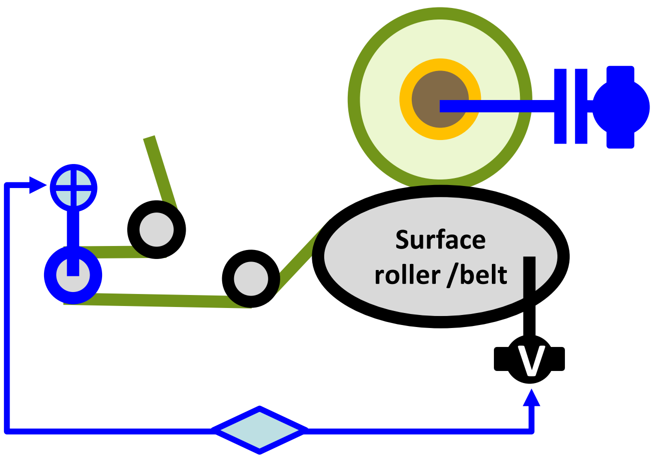
Surface Roller Driven Winder



Surface Belt Driven Winder



Motor with Load Cell Feedback

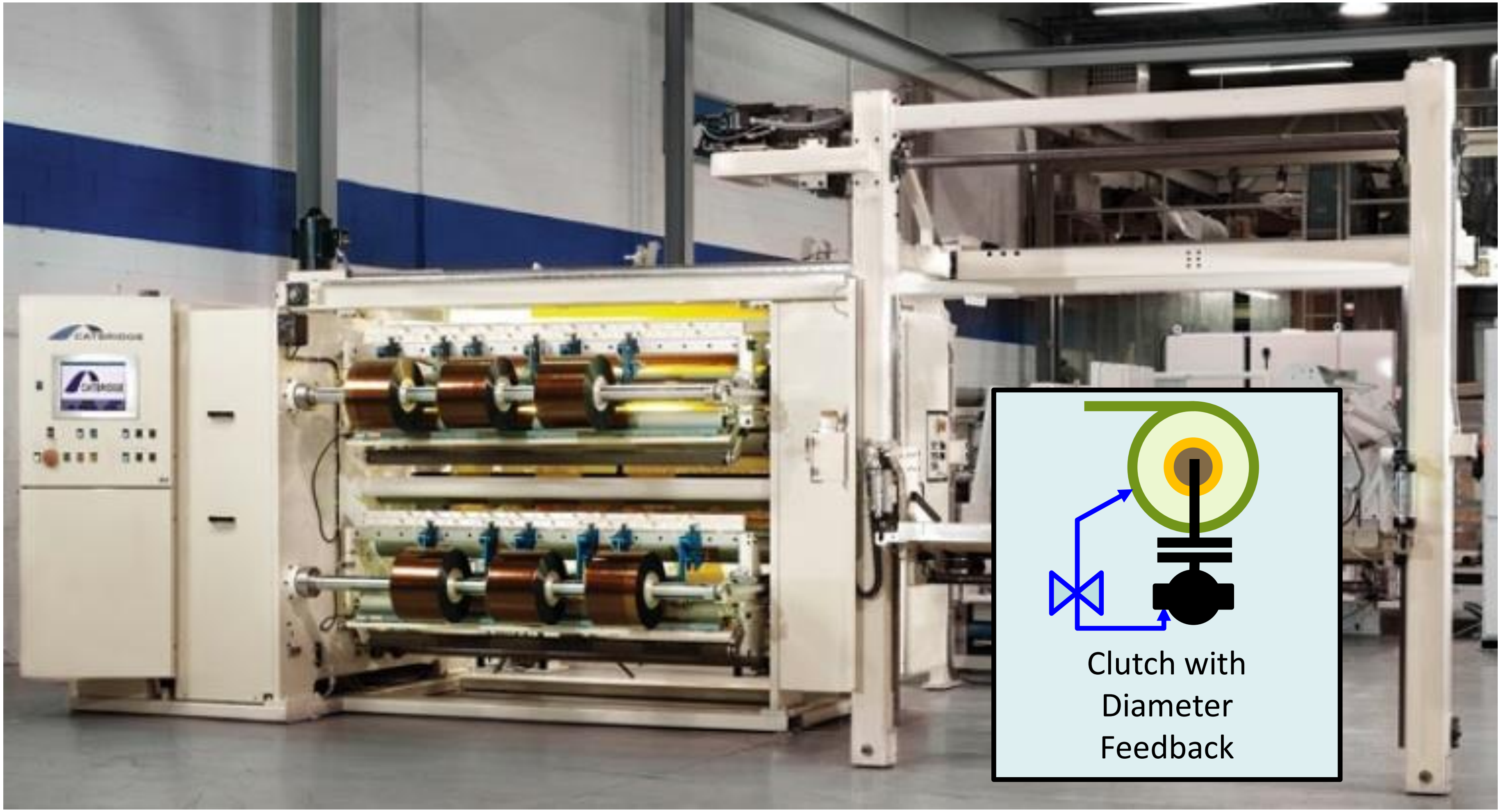


Motor with Dancer Feedback

One Big Turreted Center Winder

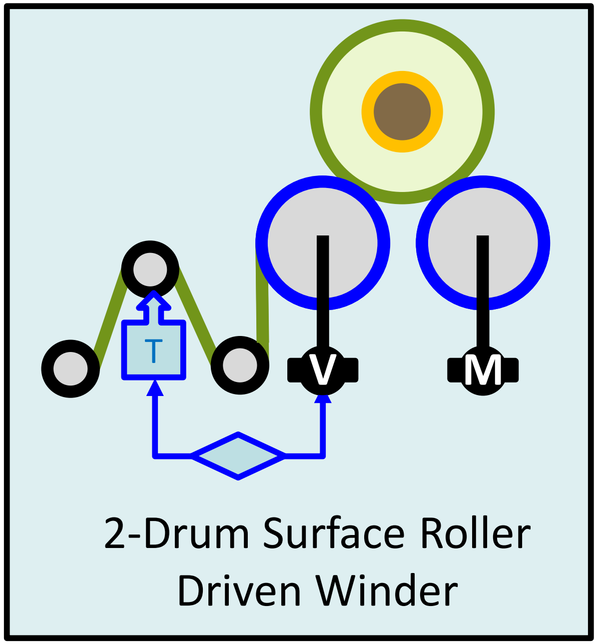
Wide Film Slitter-Rewinder

Duplex Center Slitter-Rewinder



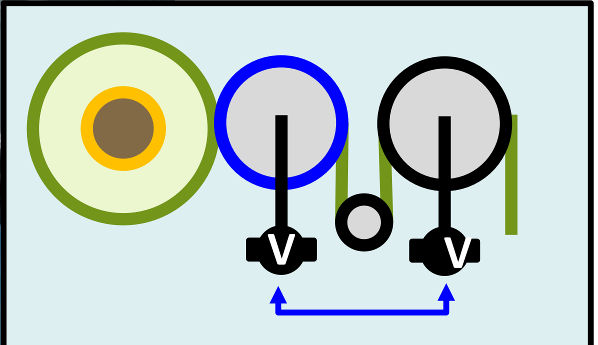
Two-Drum Slitter Rewinder

Catbridge 210



Valmet

One Big Surface Winder



Surface Roller
Driven Winder

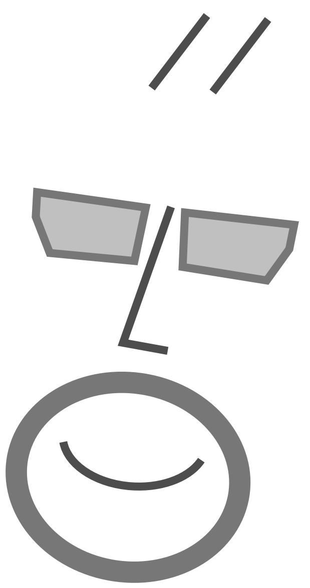
Image courtesy of
Valmet

Winding Equipment Qs

When is center winding used?

When is surface winding used?

When is center-surface winding used?



Product by Winder Type

Center

- Most films and foils
- Thin films, high speed (nipped)
- Coated papers (nipped)
- Rolls with small diameter buildup
- Tacky and sticky webs (gapped)

Surface

- Paper
- Newprint
- Thick webs wound to large diameter
- Carpet
- Textiles
- Large diameter stretchy webs

Center-Surface

- Toilet tissue
- Thick, low density non-wovens
- Slippery webs

These list indicate what is most common by product type.

Application by Winder Type

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Center

- Winding soft rolls (i.e. webs with gauge bands).
- Winding film with high tack.
- Winding small-diameter rolls.
- Easily designed for dual-direction winding.
- Able to provide adhesive-less transfers

Surface

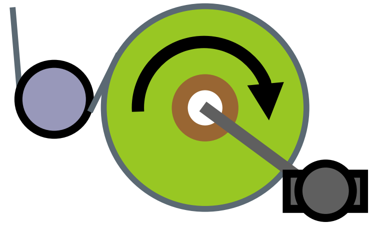
- Winding hard rolls (i.e., protective films).
- Least space and horsepower.
- Winding very large-diameter rolls.
- Less indexing waste during transfers.
- Less expensive.
- Mechanically simpler, with a single and smaller winding drive.

Center-Surface

- Winding low friction films to larger diameters.
- Slitting and winding extensible films to larger diameters.
- Maintaining width with large taper tensions.
- Supplies in-wound tension without stretching the web over thick bands.

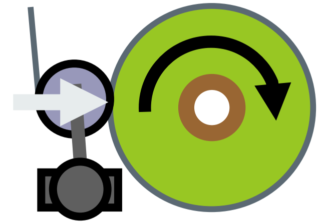
Winding Decisions

No Nip +



Center Winding

OR



Surface Winding

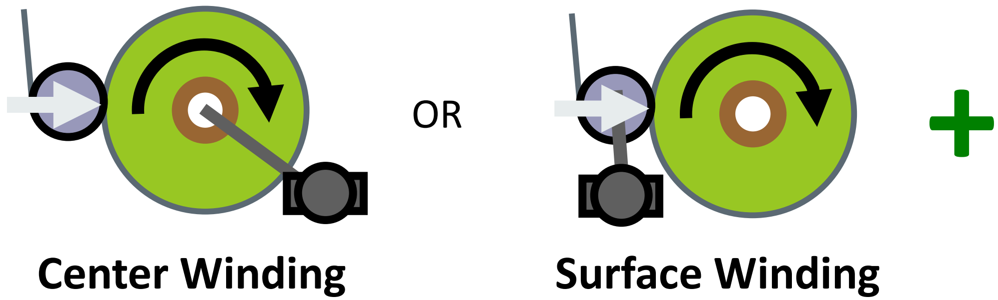
+ Porous

Key Deciding Factors:

Does nipping damage the winding roll?
Is the product able to absorb entrained air?

Pressure sensitive products may be damaged by a winding nip. If nipping damages the winding roll, center winding is the only option.

Winding Decisions

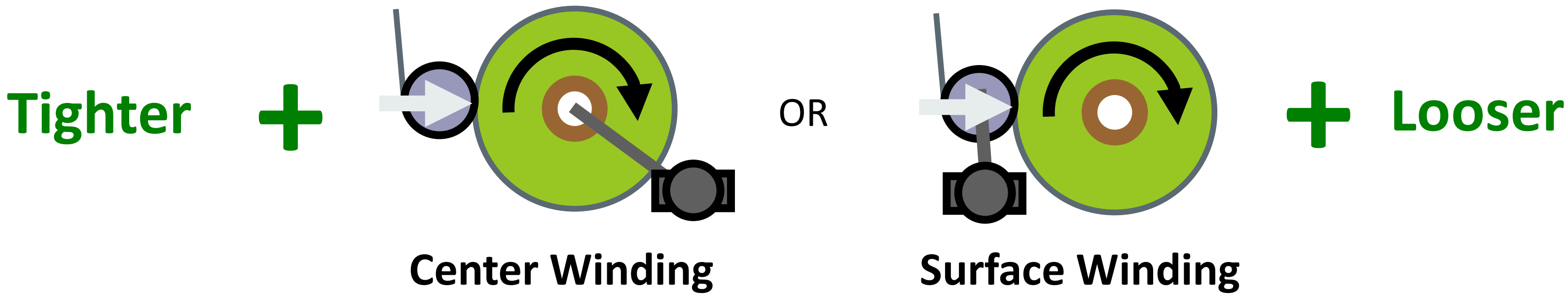


Key Deciding Factors:

Do you want to (and your product is able to) make rolls with a large buildup ratio?

Surface winders are better at winding large buildup rolls since the applied torque doesn't transmit through the roll's radius.

Winding Decisions



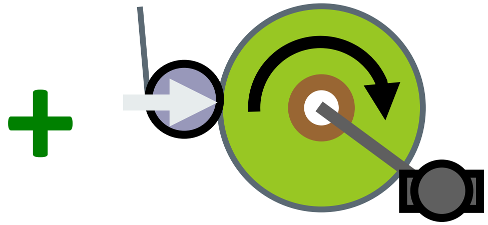
Key Deciding Factors:

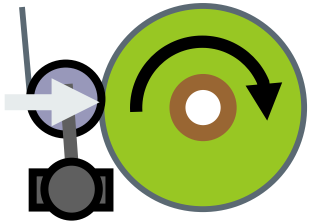
Do you have more problems with tight or loose wound rolls?

For the same tension and nip load, center winding will create a tighter roll.

Combination center-surface winding can wind looser or tighter than center or surface winding alone.

Winding Decisions





Center Winding

Surface Winding

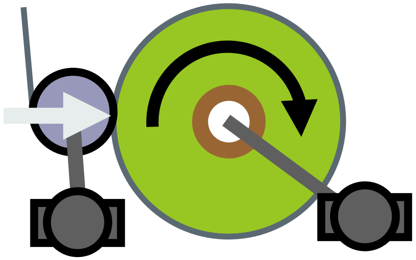
Key Deciding Factors:

Do you want automated roll transfers?

There are more commercially available designs for wrinkle-free roll transfer for turreted center winders.

Most automated surface winders create significant wrinkles and looseness at the layers near the core.

Winding Decisions



Center / Surface Combination Winding

$$T_{WH} = \frac{M_{CW}}{wr_{roll}} \pm \frac{M_{SW}}{wr_{nip}}$$

$$T_{WH,optimal} \neq \frac{M_{CW,optimal}}{rw}$$

Is center torque needed, but there is a conflict between best web handling tension and best winding torque?

Winding good rolls have an optimum torque vs. radius profile, but may create web handling problems if it creates tension that is too high, too low, or changing over time.

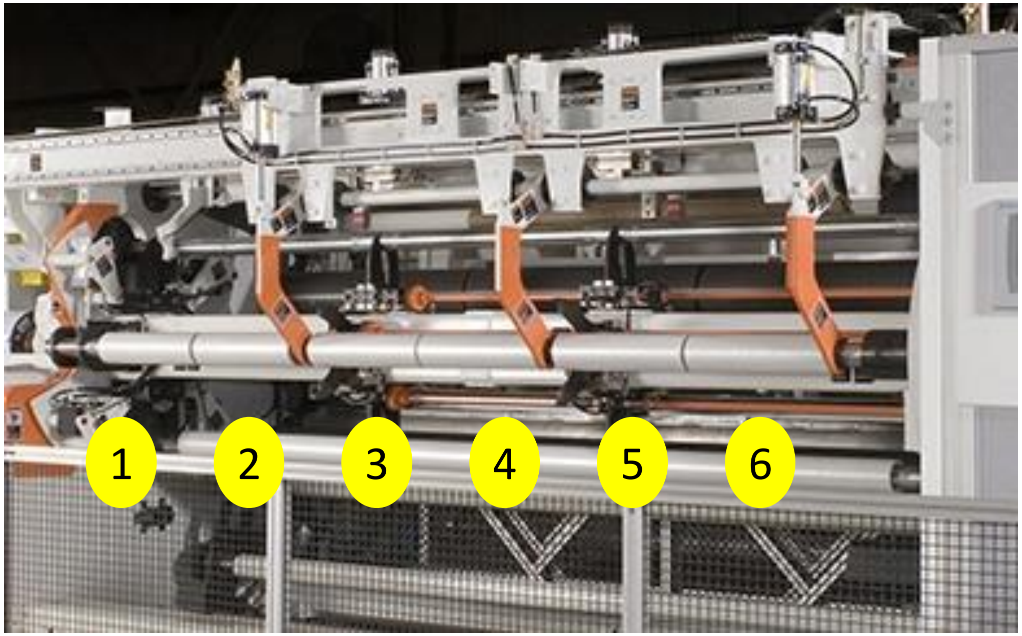
Adding a surface driven roller allows web handling tension to be optimized independently of winding torque.

3. Winding a Roll

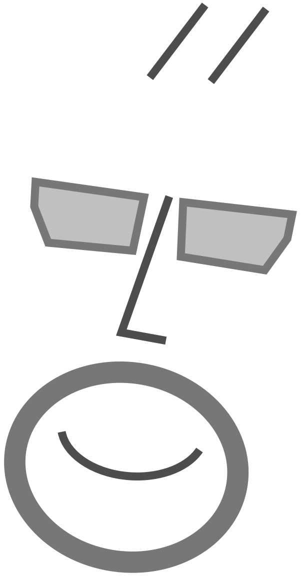
- Starting a Roll
- Managing Initial Contact
- Tension and Torque
- Winder Tensioning
- Co-Axial Winding

Winding Equipment Qs

How can multiple rolls be tensioned from one shaft?

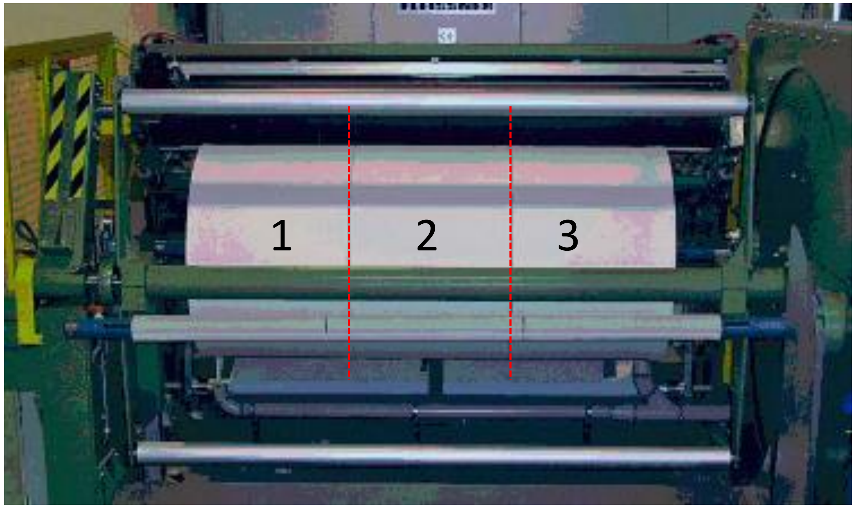


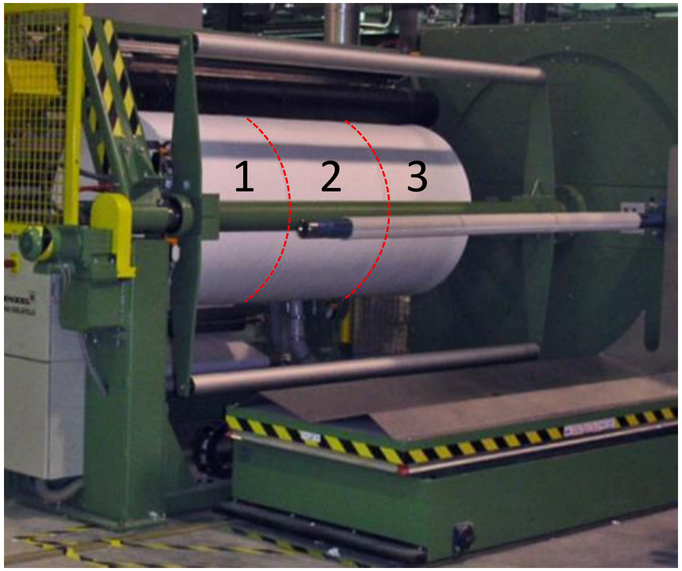
Courtesy of Davis-Standard Black Clawson



Winding >1 rolls on 1 Shaft

Courtesy of Macro Engineering

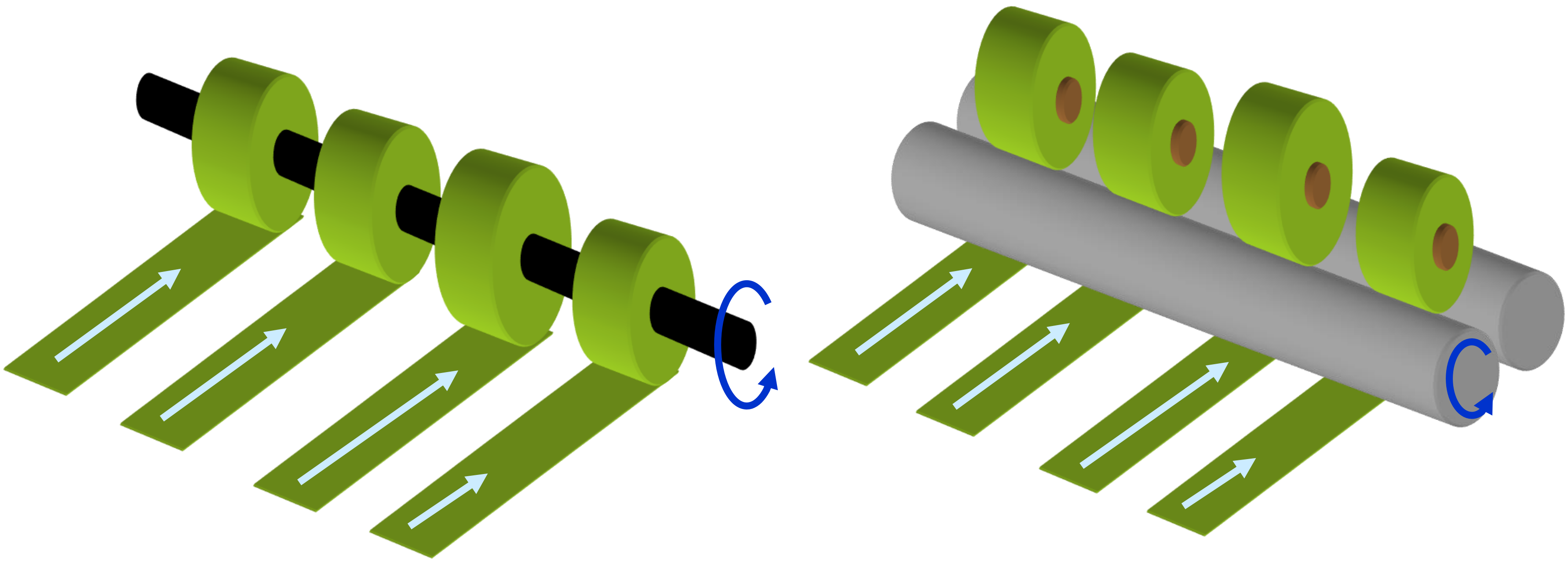




Courtesy of Menzel

Co-Axial Winding

Co-axial winding is the natural downstream process to slitting, where multiple rolls are wound on a common shaft or common surface drum.



Challenges of Co-Axial Winding

Thickness Profile and Independent Rotation

- Winding slit rolls from web with thickness variations will create rolls of differing diameters.
- Without independent rotation, winding tensions will increase where thick lanes create above average diameter rolls and decrease (potentially to slackness) in thin lanes forming smaller than average rolls.

Co-Axial Center Winding

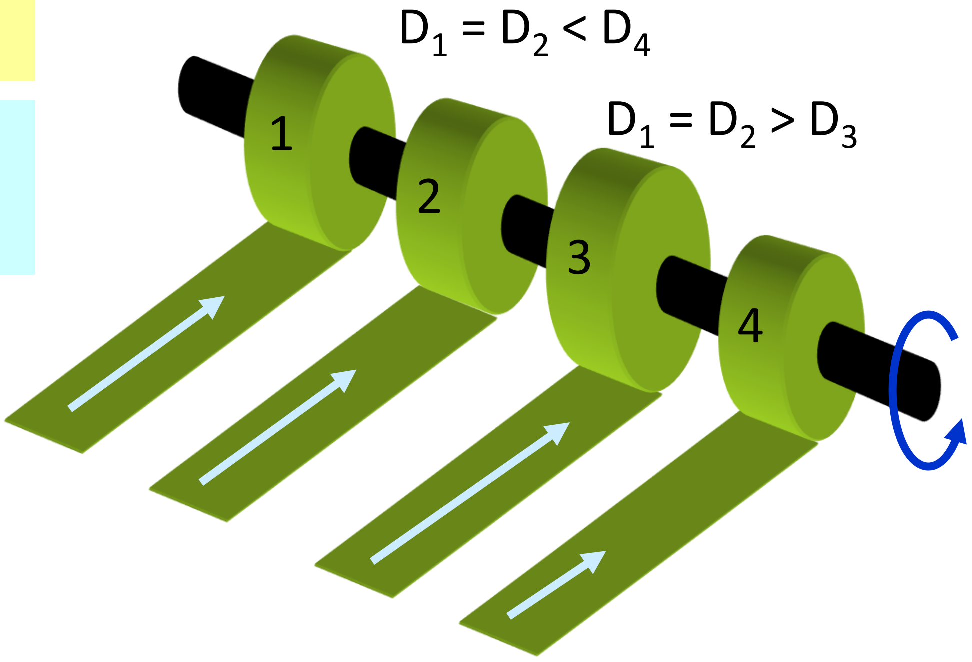
Locked shaft and slip shaft winding are used to wind multiple rolls on a common axis.

Locked shaft winding is used when $\Delta D + \Delta L <$ web strain.

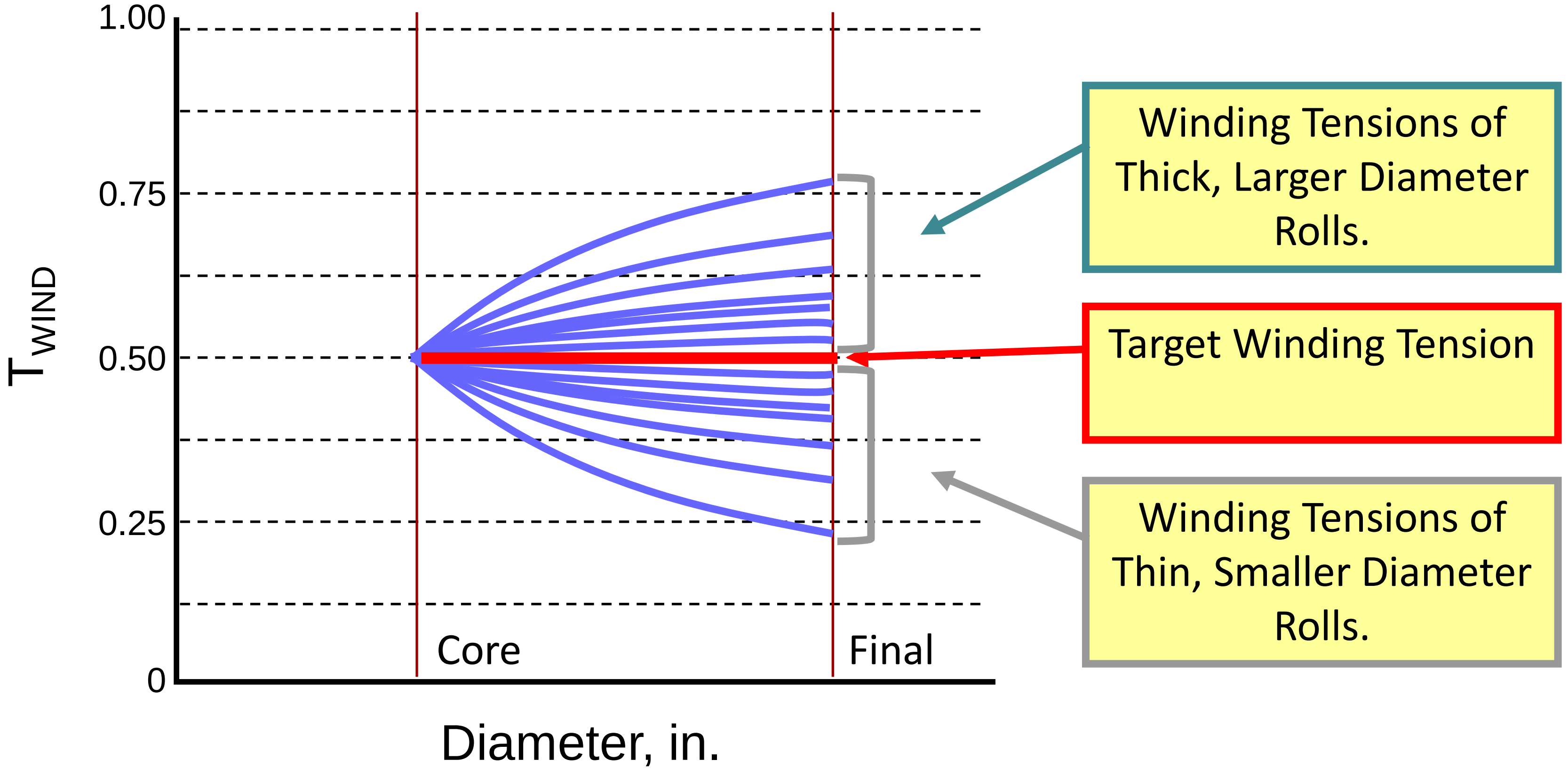
Slip shaft winding is used when $\Delta D + \Delta L >$ web strain.

ΔD is a function of:

- roll compressibility
- web thickness variations
- roll diameter
- web bagginess



Lock Bar Variations



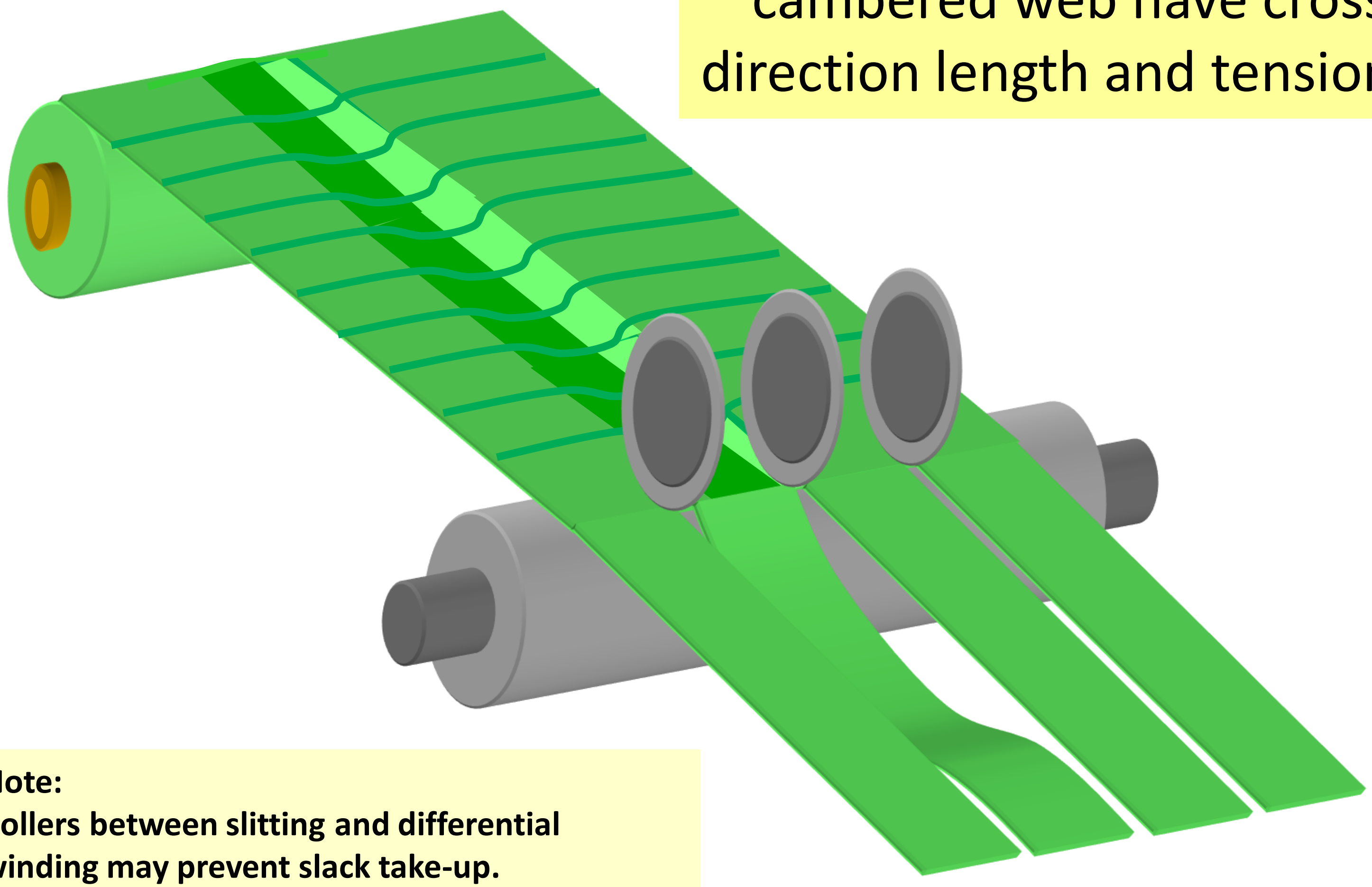
Challenges of Co-Axial Winding

Baggy Web

- Winding slit rolls from baggy web creates rolls of differing length.
- Without independent rotation, winding tension will increase in rolls formed from short lanes and decrease (or become slack) in rolls created from longer, baggy lanes.
- Bagginess is a problem when the percent length changes are similar to the winding strain of the web.
- Slip shaft or independent winding allows independent rotation rates across slit rolls, reducing winding tension variations of co-axial winding.

Slitting Baggy or Cambered Web

Narrow webs slit from baggy and cambered web have cross machine direction length and tension variations.



Note:
Rollers between slitting and differential winding may prevent slack take-up.

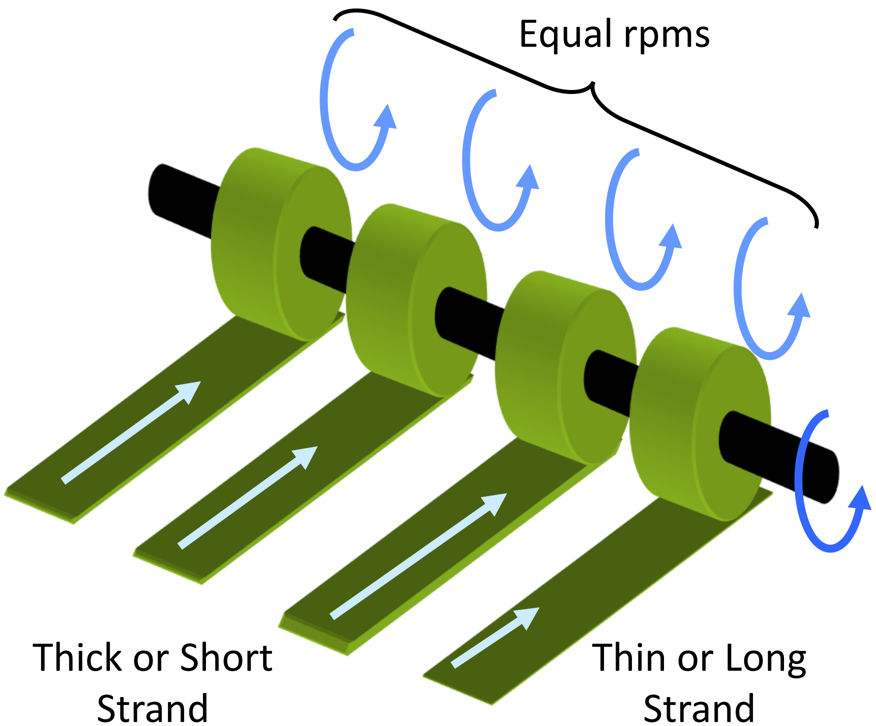
Locked vs. Slip Shaft Winding

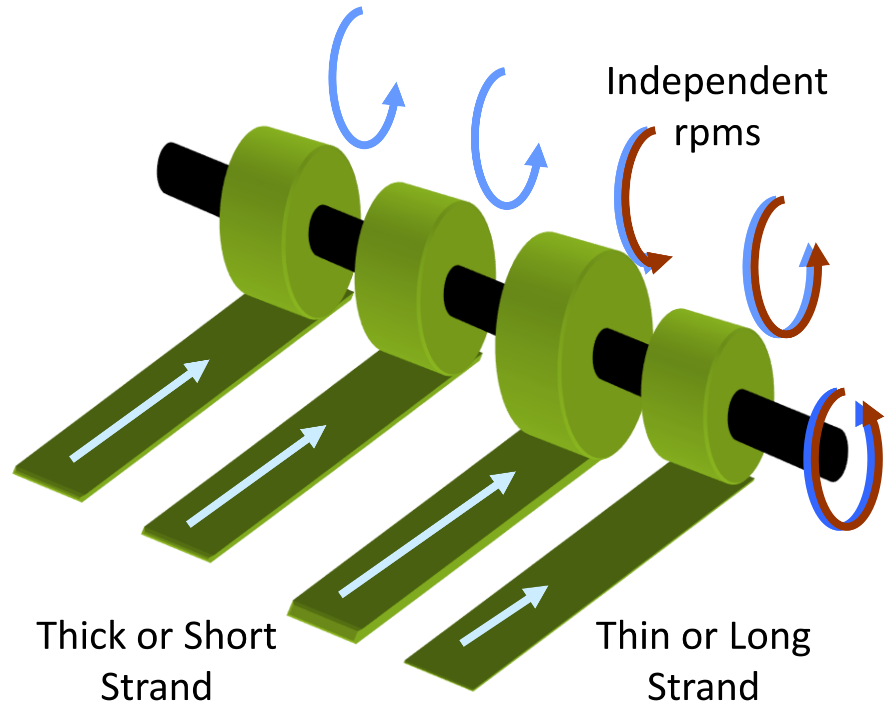
Locked Shaft:

Shaft and cores all rotate at same revolution rate.

Slip Shaft:

Shaft is oversped. All rolls slip relative to shaft and rotate at rate based on diameter and strand length.





Co-Axial Winding & Individual Nips

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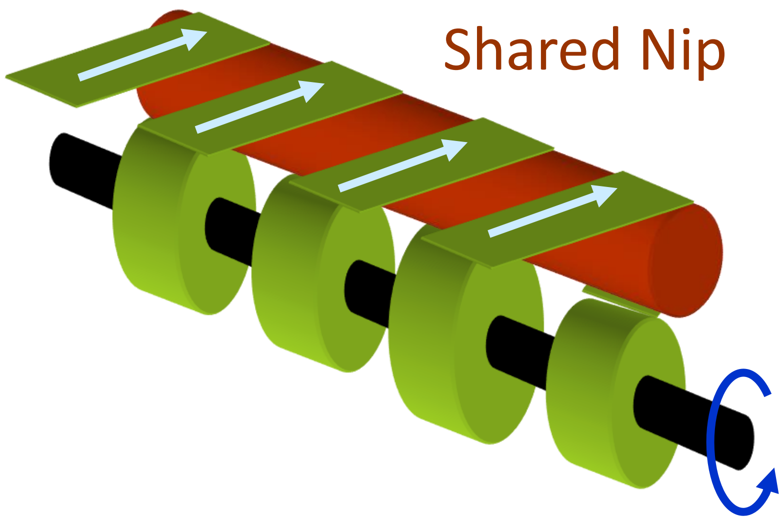
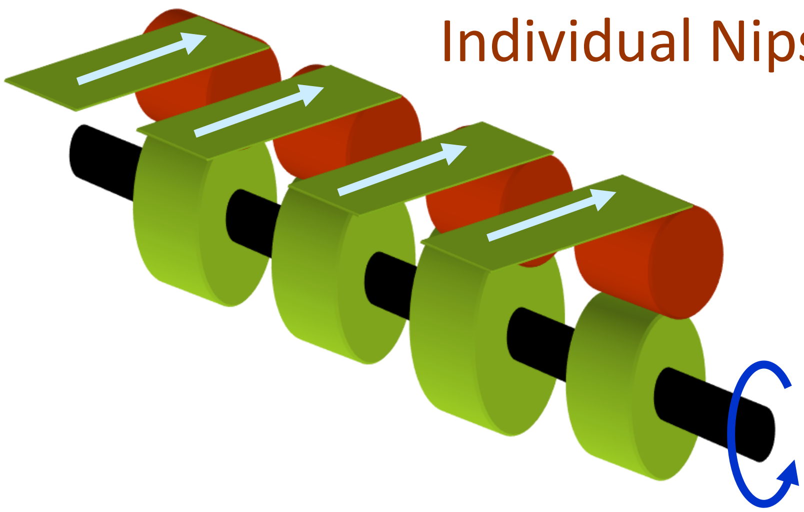
Compressible webs (papers, nonwovens) can share a common nip since individual rolls will can deform to maintain common diameter.

Incompressible webs (films, foils) require individual nips to maintain contact as diameter variations increase with thickness variations and roll diameter.


Shared Nip


Individual Nips

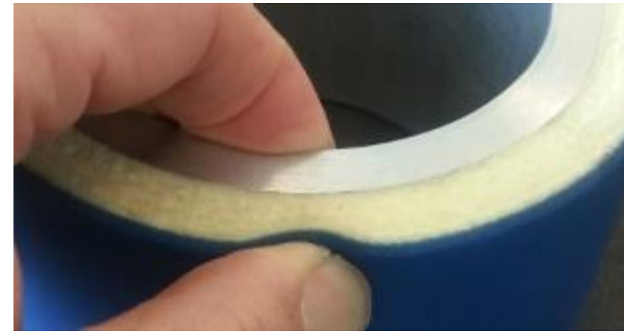
Challenges of Co-Axial Winding

Thickness Profile and Shared Nip Roller

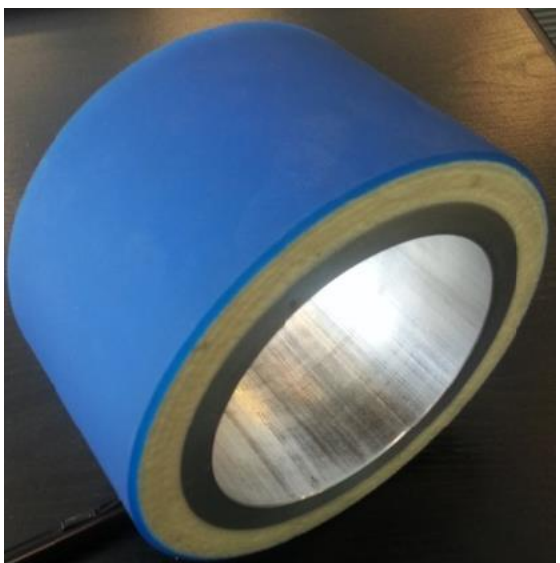
- In product where air lubrication is problematic (smooth webs and high speed, low tension, large diameters), a winding nip is required to prevent air lubrication or entrainment problems.
- A shared nip can work when roll-to-roll diameter variations are small (compressible webs, small roll buildups), but independent nips will be needed as diameter variations increase.
- Soft, thick rubber covered rollers can extend the diameter variations that can be contacted with a shared nip roller.

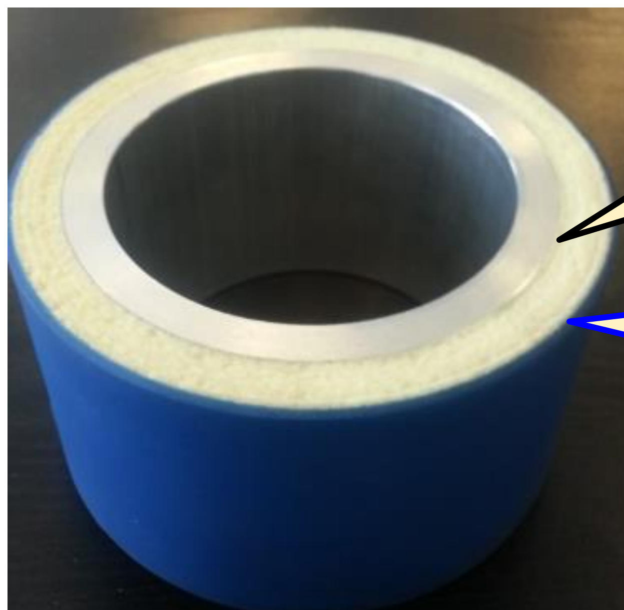
Challenges of Co-Axial Winding

Thickness Profile and Shared Nip Roller



- The dual Durometer roller covering provide extended compliance around roll diameters with better abrasion resistance than single low Durometer rollers.





Inner thick covering of soft foam rubber.

Outer thin covering of high durability rubber.

Challenges of Co-Axial Winding

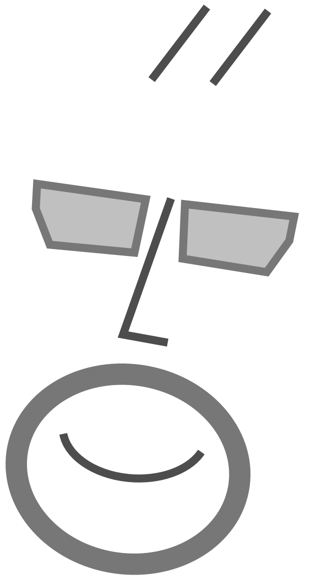
Individual Rolls Needing Independent Nip Rollers

- Thickness Profile

  Winding slit rolls from web with thickness variations will create rolls of differing diameters if the rolls have high radial modulus.

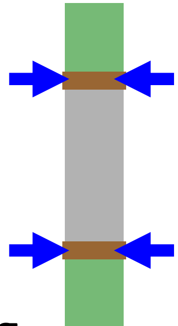
Winding Equipment Qs

How does a slip shaft create winding tension?



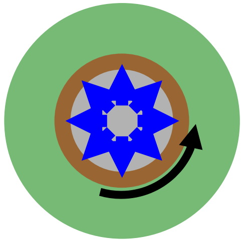
Tension and Torque of Slip Shaft Winding

Side Load:

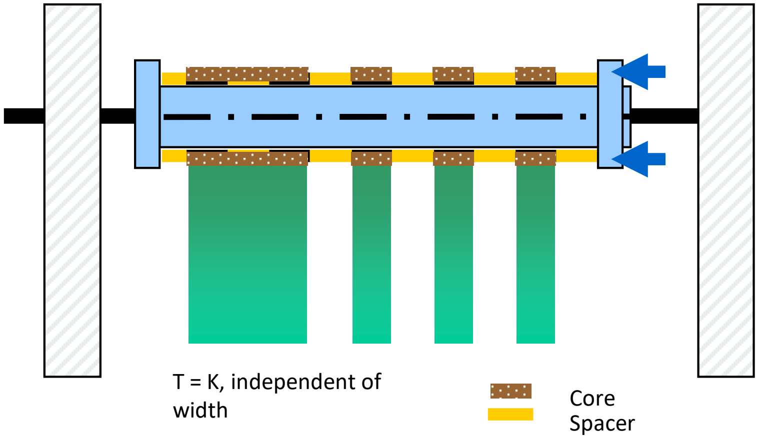


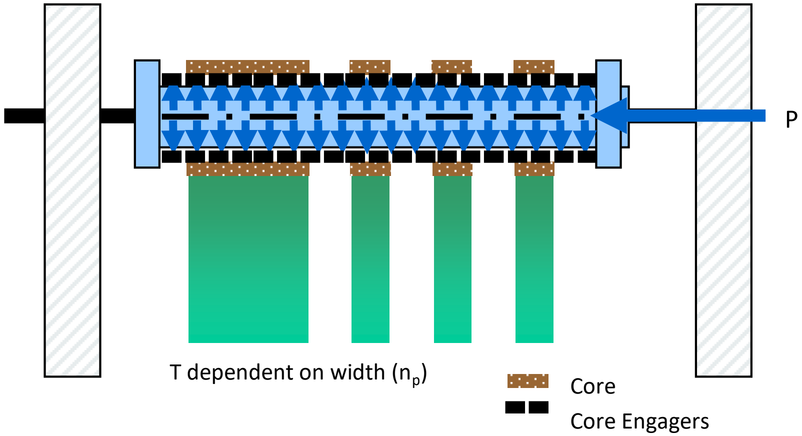
- Torque independent of width, making it best if all widths are equal.
- Allow narrowest rewinding
- Core-spacer system requires more roll change time.

Radial Load:



- Torque proportional to width, allowing different widths on same shaft.
- Slit width should be greater than segment widths (>25mm)
- Fastest roll change time.





Differential Bar: Axially Loaded

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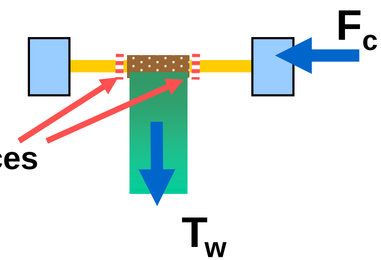
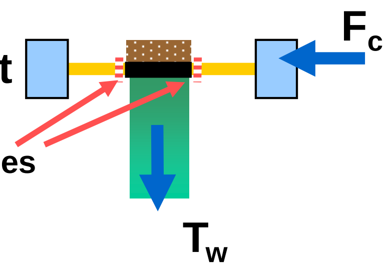
Pro: A simple, popular design. Handles narrowest widths.
Con: Unable to tension proportional to width.

Core Dependent

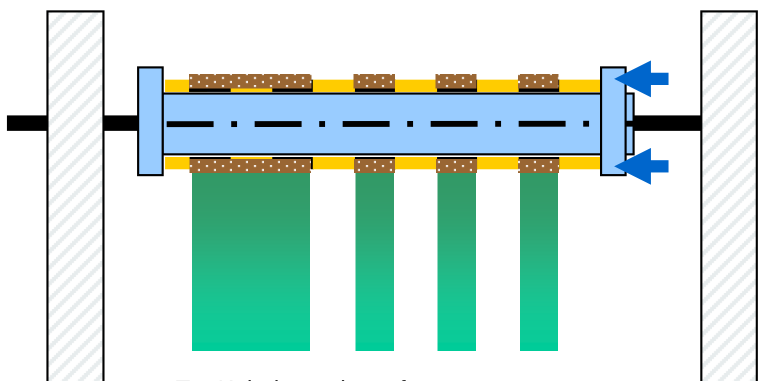


$$M = T_w R_w = 2\mu_{cs} F_c R_c$$

Core Independent

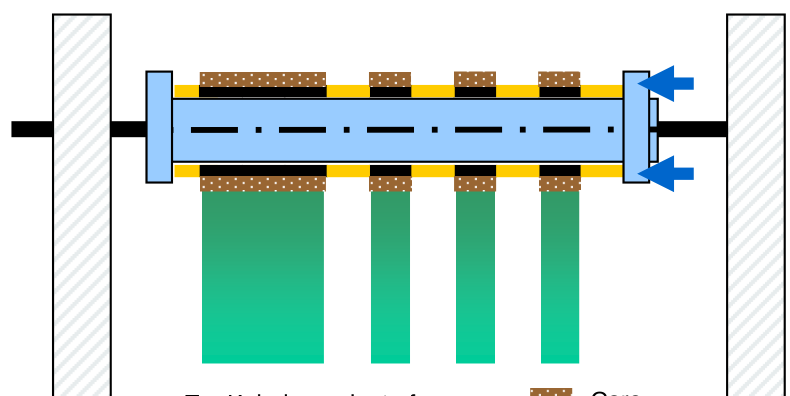


$$M = T_w R_w = 2\mu_{gs} F_g R_g$$



T = K, independent of width

Core
Spacer

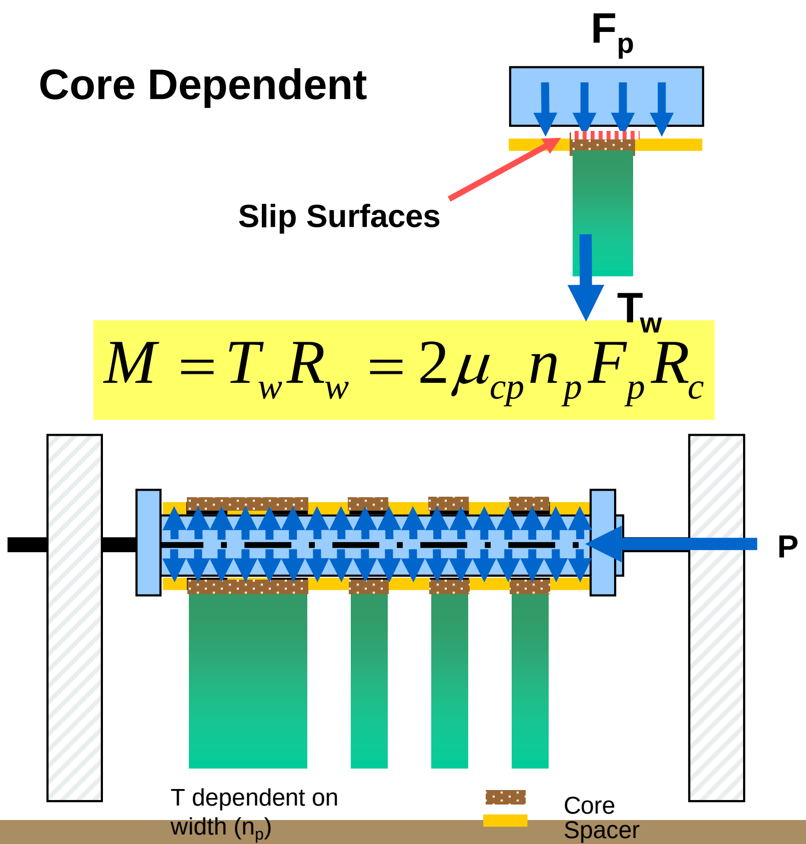


T = K, independent of width

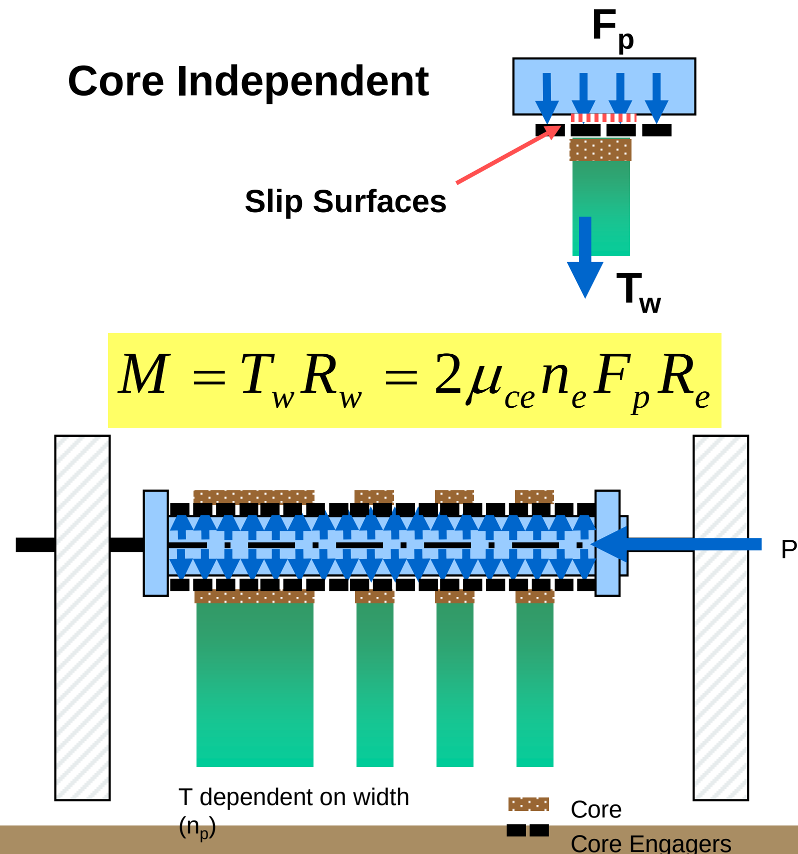
Core
Core Grabber
Spacer

Differential Bar: Radially Loaded

Pro: Tensions proportional to width.
Con: More complex and expensive,
Slit width must be greater than segment width.



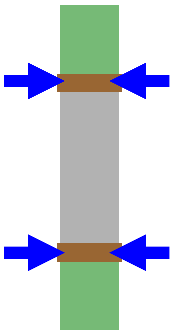
$$M = T_w R_w = 2\mu_{cp} n_p F_p R_c$$

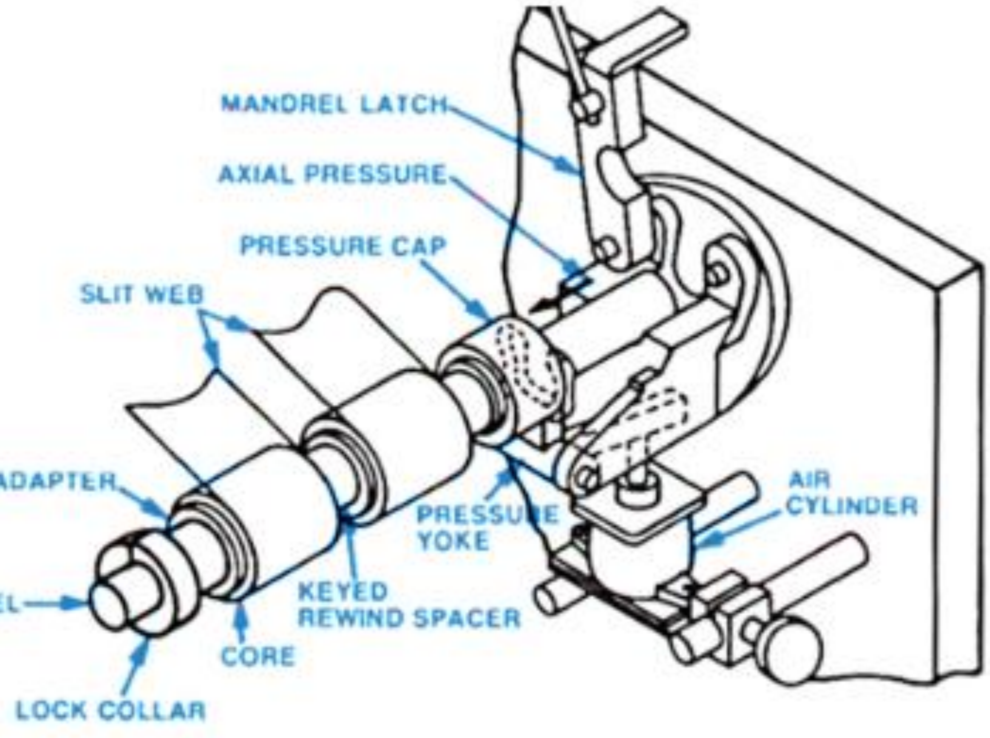


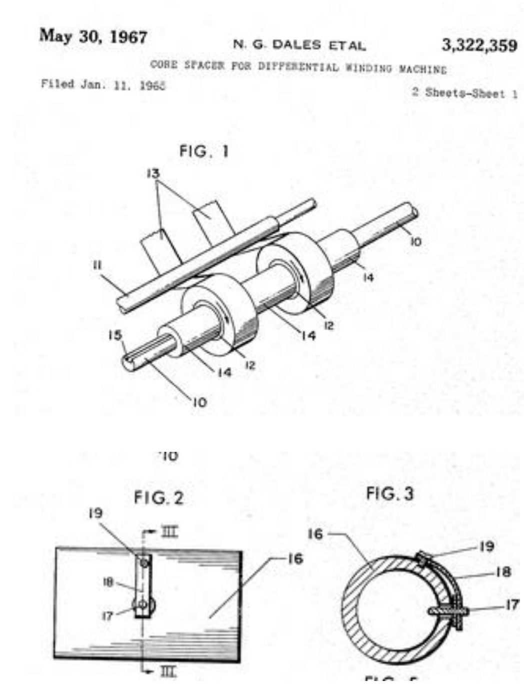
$$M = T_w R_w = 2\mu_{ce} n_e F_p R_e$$

Side Load Slip Shaft Winding

Side load slip shafts create a friction on the two sides of the winding core (or core adaptor) with keyed spacers between each slit roll.



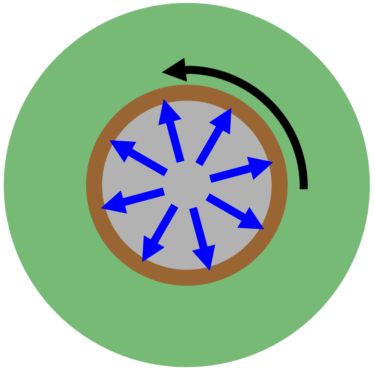




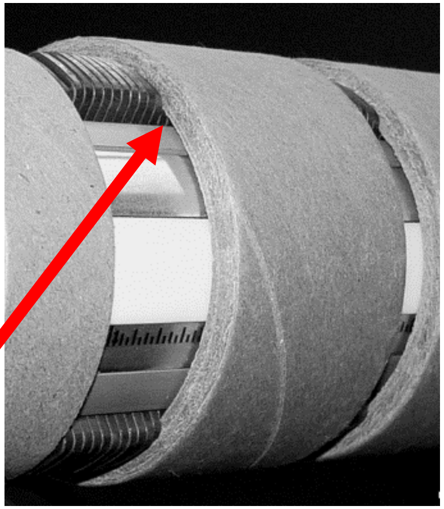
Courtesy of Parkinson (Dusenbery)

Radially Loaded Slip Shaft Winding

Radially loaded system create friction on the inner surface of the core (or core adaptor segments).



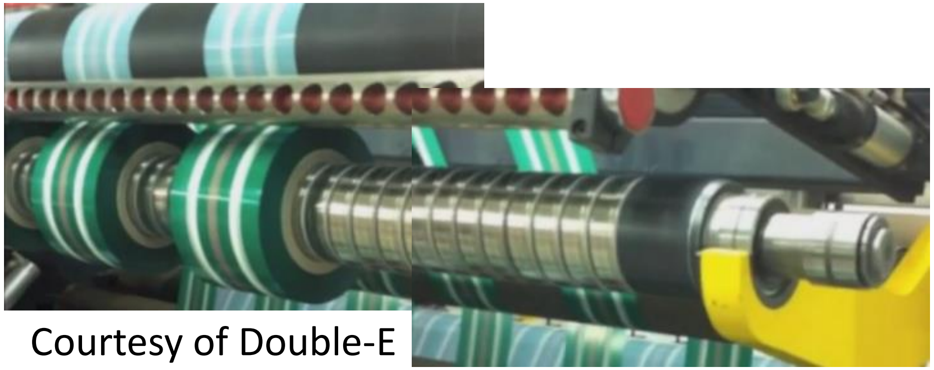
Core-Dependent Friction



Courtesy of Tidland-Maxcess

Core-Independent Friction

Courtesy of Convertech



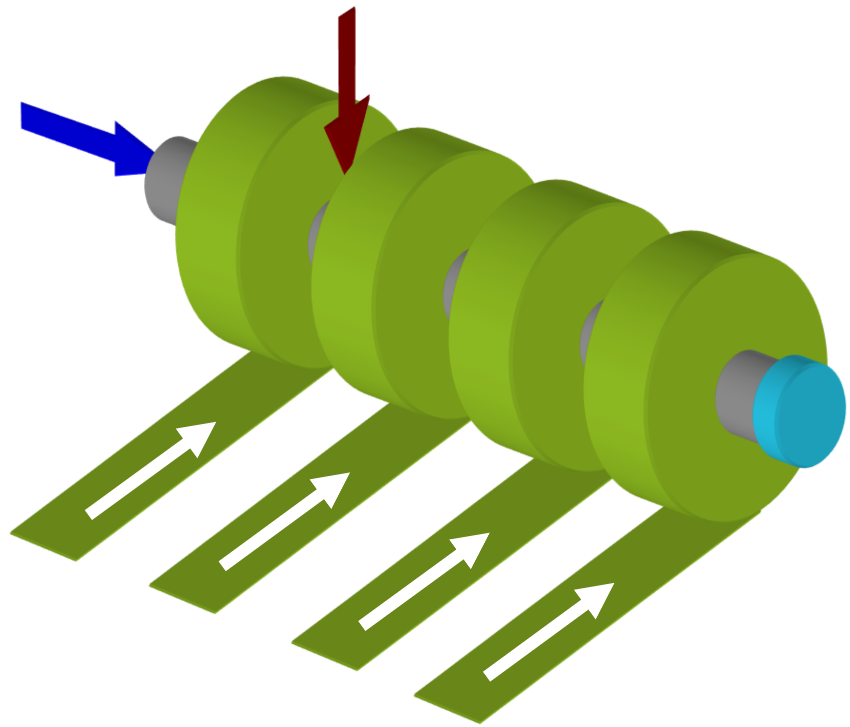
Courtesy of Double-E

What Create Tension in Slip Shaft Winding?

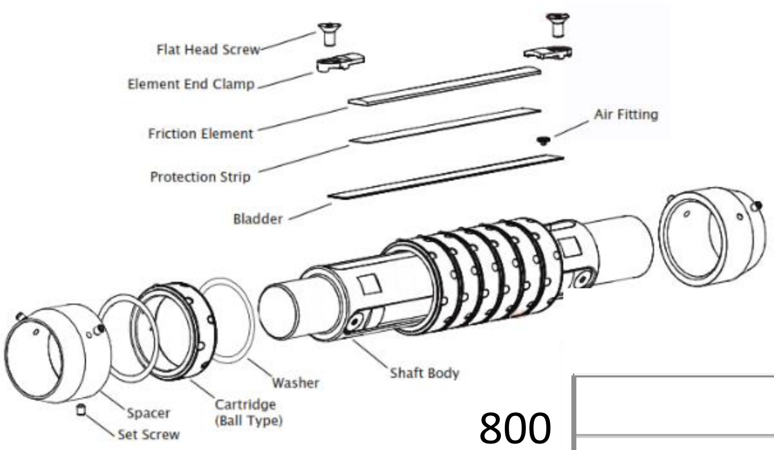
1. Pneumatic pressure apply a force to create friction (force) at a specific radius (distance) through radial load or side load. This apply a torque to each winding roll.

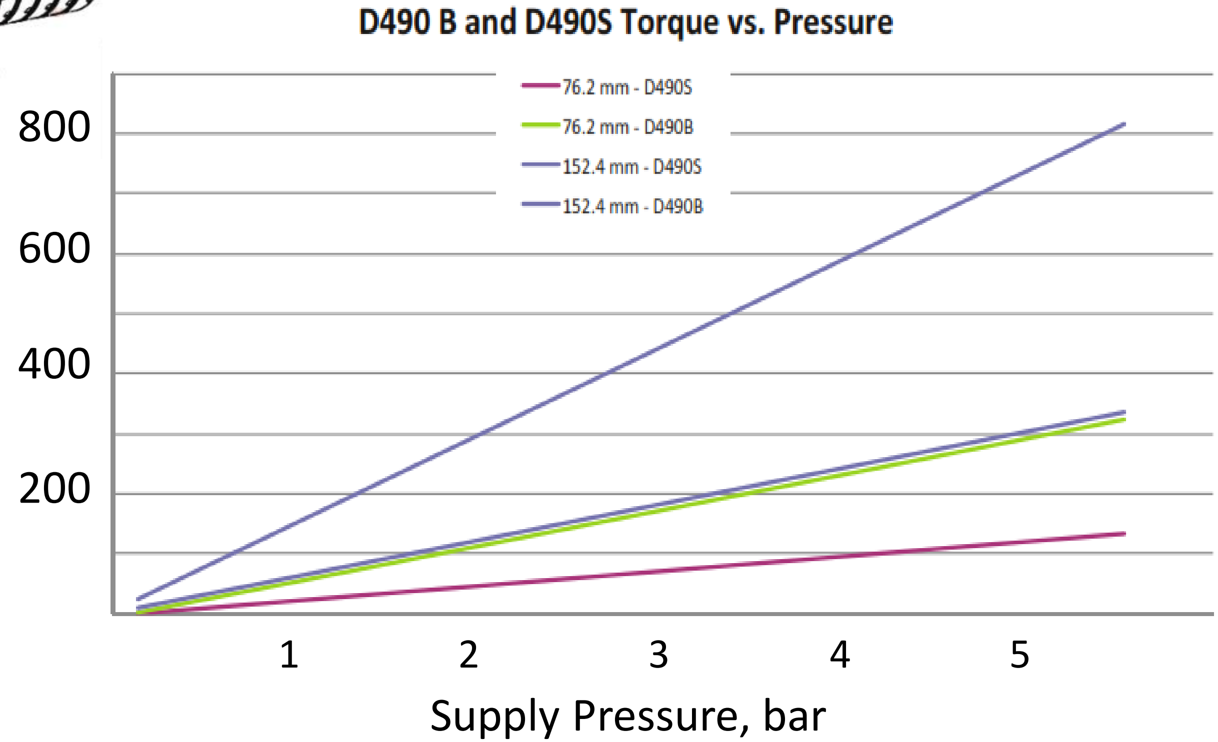
   Winding tension will be torque divided by roll radius.

$$\text{Tension} = \frac{\text{Torques (A)}}{\text{Radius}}$$



Example Radial Load Pneumatic Slip Shaft






Spring Cartridge D490S


Ball Cartridge D490B

Courtesy of Tidland Maxcess Int’l

Core Adaptor Design Friction

Core grabbing adaptors must be able to transmit applied torque to the cores without slippage.

Some core adaptors may slip at high applied torque with plastic or metal cores.

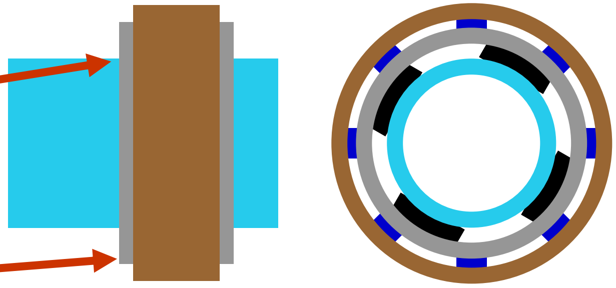
Frictional Torque Transmission

M_1, Shaft to Core Grabber

– Slipping, creates frictional torque.

M_2, Core Grabber to Core

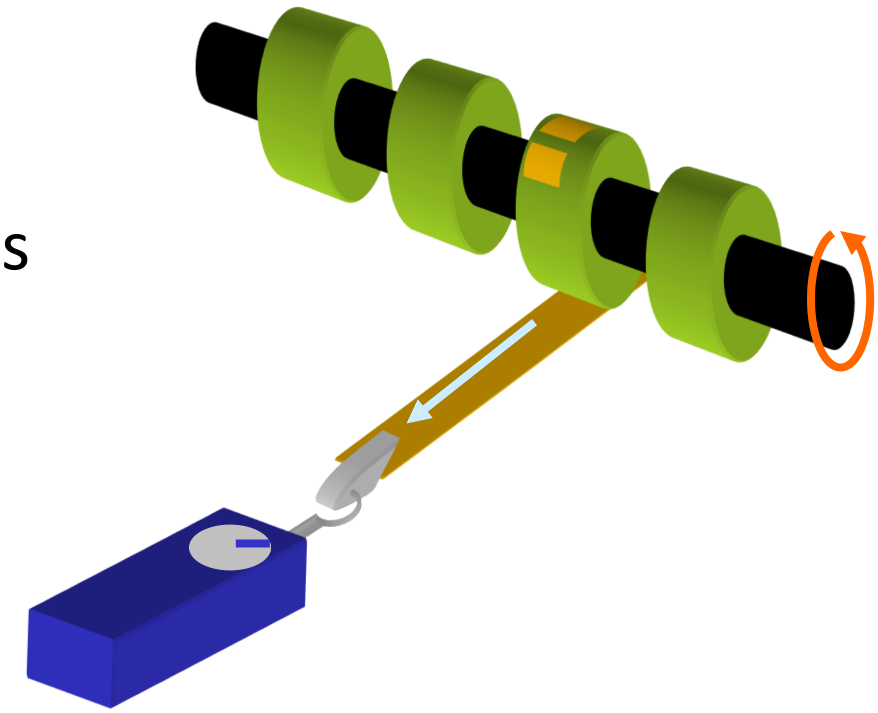
– No Slip, holds core, often mechanical only



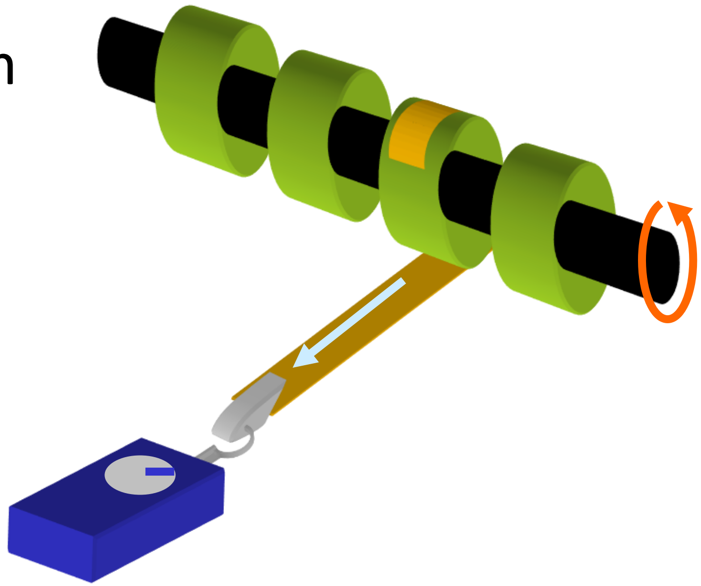
Calibrate Slip Shaft Winding Tension

Slip shaft winding is commonly used in open-loop torque control (without tension feedback). Calibrating the winding shafts helps to estimate tension magnitude and variations across shaft positions.

Contamination, lubrication, and wear can all create variations as large as 2:1 across a single shaft.



Calibrate Slip Shaft Winding Tension

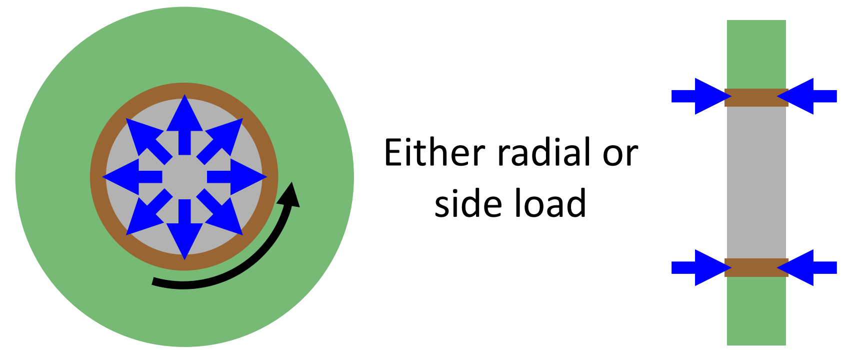


- Attach a strip of strong tape or rope from the core to a force gauge.
- Set the slip shaft air pressure clutch to a low setting (0.5 bar).
- Measure the force to make the core slip relative to the shaft.
- Repeat at higher pressures.
- Multiply measured force by core radius to calculate torque. Plot torque vs pressure.
- Repeat measurements at various core positions.
- Repeat test with full diameter roll to see effect of roll weight on slip torque.

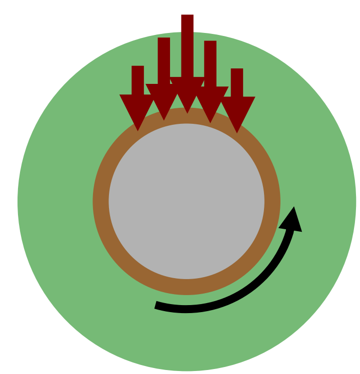
What Create Tension in Slip Shaft Winding?

1. Pneumatic pressure
2. Gravity will create a force between the roll's mass and the shaft, creating additional winding torque.
   At small diameters, gravity can be easily ignored, but this become significant for large diameter and high density products.

Torque A: Pneumatic Load

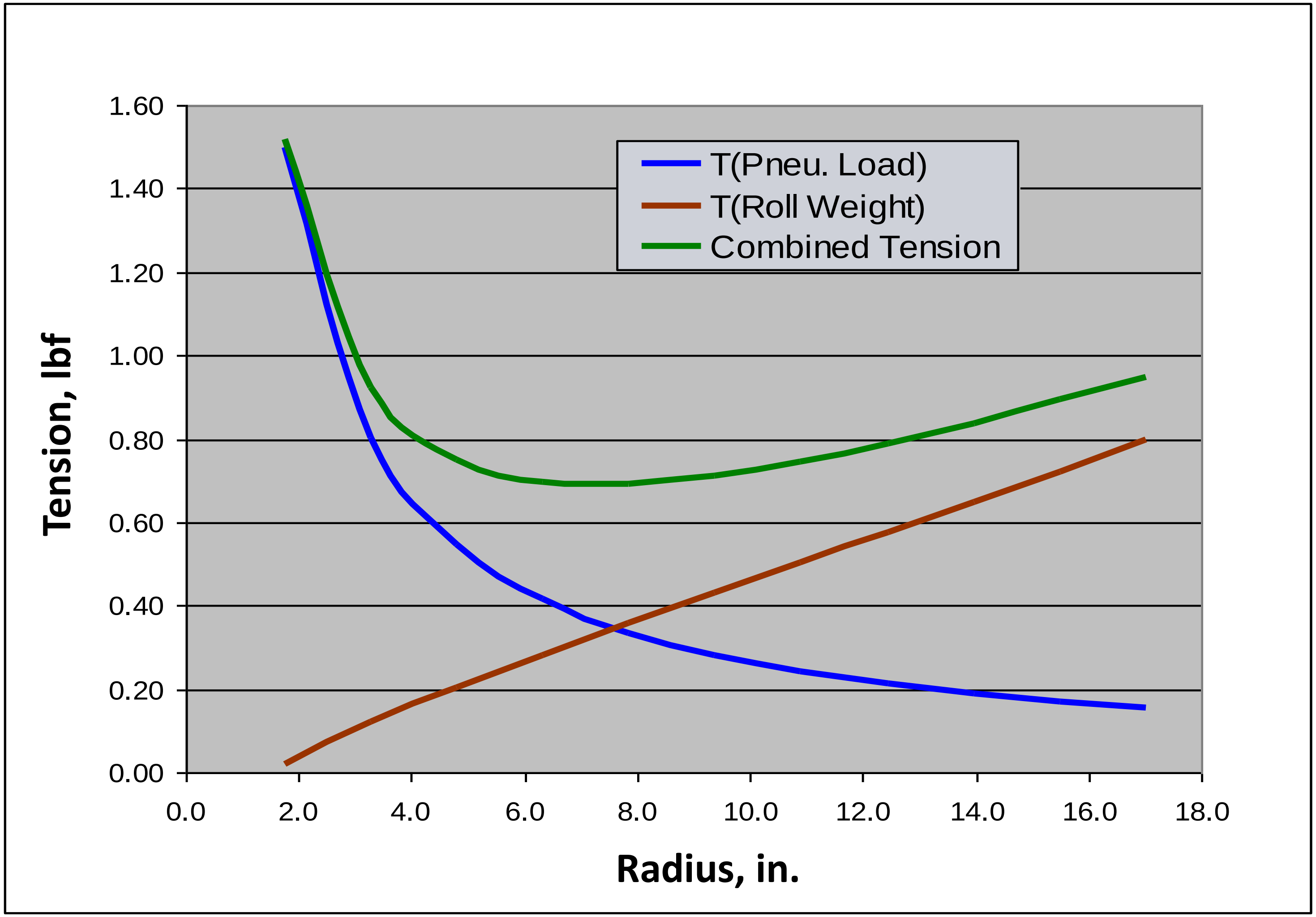


Torque B: Gravity Load



$$\text{Tension} = \frac{\text{Torques } (A+B)}{\text{Radius}}$$

Differential Winding Tension vs. Roll Radius



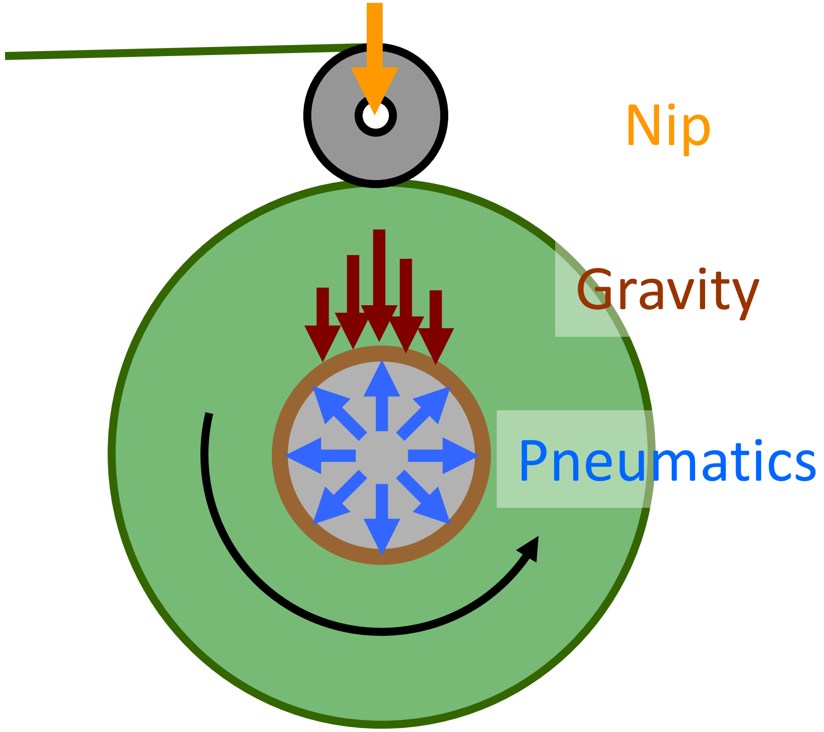
Three Torques of Slip Shaft Winding

1. Pneumatic pressure
2. Gravity
3. Nip loading will add to the force pushing the core against the shaft, creating another source of friction-induced torque.
   Nip loads usual increase slip shaft torques, but if applied from below, nip load could relieve gravity effects.

Torque C: Center Slip Shaft Load

Torque G: Gravity Load

Torque N: Nip Load



$$\text{Tension} = \frac{\text{Torque}(C+G+/-N)}{\text{Radius}*\text{Width}}$$